V.S.B ENGINEERING COLLEGE, KARUR
DEPARTMENT OF CIVIL ENGINEERING
CE 8302 - FLUID MECHANICS
PART-B
UNIT -1

FLUID PROPERTIES AND FILUID STATICS

1. One liter of crude oil weighs 9.6 N; calculate its specific weight, density, specific volume and specific gravity.

2. In a stream of glycerin in motion, at a certain point the velocity gradient is 0.25 meter per sec per meter. The mass density of fluid is 1268.4 kg per cubic meter and kinematics viscosity is 6.30×10^{-5} square meter per second. Calculate the shear force at that point.

3. What do you understand by the terms isothermal process and adiabatic process?

4. Determine the minimum size of glass tubing that can be used to measure water level, if the capillary rise in the tube is not to exceed 0.25 mm. take surface tension of water in contact with air as 0.0732N/m.

5. Determine the bulk modulus of elasticity of a liquid, if as the pressure of the liquid is increased from 7 MN/m^2 to 13MN/m^2 the volume of liquid decreased by 0.15%.

6. Determine the mass density, specific volume and specific volume and specific weight of a liquid whose specific gravity is 0.85.

7. A cylindrical shaft of 90mm rotates about a vertical axis inside a cylindrical tube of length 50 cm and 95 cm internal diameter. If the space between them is filled with oil of viscosity 2 poise find the power lost in friction for a shaft speed of 200rpm.

8. Obtain an expression for capillary rise of a liquid.

9. Define surface tension and obtain the relationship between surface tension and pressure inside a droplet of liquid in excess of outside pressure.

10. Find the surface tension in a soap bubble of 30mm diameter when the inside pressure is 1.962 N/m^2 above atmosphere.

UNIT -2

FLUID KINEMATICS AND DYNAMICS

1. A solid wood cylinder is with a diameter of 0.666 m and a height of 1.3 m. the specific gravity of the wood is 0.61. Would the cylinder be stable if places vertically in oil of specific gravity 0.85?

2. Give the definitions of metacentre and metacentirc height.

3. Derive the equation of continuity for three dimensional incompressible fluid flows and reduce it to one dimensional form.

4. A hollow cylinder closed at both ends has an outside diameter of 1.25m, length 3.5m and specific weight 75KN/m^3. it the cylinder is to float just in stable equilibrium in sea water (specific weight 10 KN/m^3) , find its minimum permissible thickness.

5. What is a flow - net"? Enumerate the methods of drawing flow nets. What are the uses and limitations of flow nets?

6. State and prove the hydrostatic law.

7. Determine the gauge and absolute pressure at a point which is 2m below the free surface of dewater. Take atmospheric pressure as 10.1093n/m^2.

8. A rectangular tank 4m long, 1.5m wide contains water up to a height of 2m. Calculate the force due to water pressure on the base of the tank. Find also the depth of centre of pressure from free surface.

9. A block of wood of specific gravity 0.7. Floats in water. Determine the meta-centric height of the block it its' size is 2mx1mx0.8m.

10. A 25cm diameter pipe carries oil specific gravity 0.9 at a velocity of 3m/s. at another section the diameter is 20cm. find the velocity at this section and also mass rate of flow of oil

UNIT -3

DIMENSIONAL ANALYSIS AND MODEL STUDIES

1. Briefly explain Rayleigh's method.

2. Briefly explain Buckingham's π-theorem.

3. The resisting force of a supersonic aircraft during its flight can be assured to depend on following variables such as length, velocity, viscosity, and density and bulk modulus. With the help of Buckingham π theorem derive an expression showing the relationship between resisting force and these variables.

4. Describe Buckingham's π theorem to formulate a dimensionally homogenous equation between the various physical quantities effecting a certain phenomenon.

5. By dimensional analysis, show that the power P developed by a hydraulic turbine is given by $P = \rho N^3 D^5 f(\frac{N2D2}{gH})$ where ρ- mass density of liquid , N- rotational speed , D- diameter of runner , H – working head , g- acceleration due to gravity.

6. The resistance 'R' is experienced by a partially submerged body depends upon the velocity 'V' , length of the body 'l', viscosity of fluid 'μ',density of the fluid 'ρ' and gravitational acceleration 'g'; Obtain the dimensionless expression for R.

7. State the reasons for constructing distorted models. What are the merits and demerits of distorted models as compared to undistorted models?

8. The resisting force (R)of the supersonic flight can be considered as dependent upon the length of the aircraft 'l', velocity 'v', air velocity 'μ', air density 'ρ' and the bulk modulus of air 'k'. express the functional relationship between these variables and the resisting force.

9. Explain the Reynolds's law of similitude and Froude's law of similitude.

10. In an aero plane model of size (1/10) of its prototype, the pressure drop is 7.5 $kN/m^{2.}$ The model is tested in water. Find the corresponding pressure drop in the prototype. Assume density of air = 1.24 kg/m^3, density of water = 1000 $kg/m^{3,}$ viscosity of air = 0.00018poise and viscosity of water = 0.01poise.

UNIT -4

FLOW THROUGH PIPES

1. The water is flowing through a pipe having diameters 200 mm and 100 mm at sections 1 & 2 respectively. The rate of slow through the pipe is 35 l/s. the section 1 is 2 m above datum and section 2 is 4 m above datum. If the pressure at section 1 is KN/m^2, find the intensity of pressure at section 2.

2. Derive Euler's equation of motion along a stream line and hence derive the Bernoulli's theorem.

3. Derive the Hagen - Poiseuille equation and state the assumptions made.

4. Give the equation for difference of pressure head for laminar flow between two fixed parallel plates with details.

5. Explain in detail about drag and lift coefficients.

6. Briefly explain about Moody's diagram, pipe roughness.

7. A pipe of diameter 400mm carries water at a velocity of 25m/s .the pressure at the points and a and b are given as 29.43n/$m^2$and 22.563n/m^2 respectively while the datum head at a& b are 28m & 30m. Find the loss of head between a & b.

8. A nozzle of diameter 20mm is fitted to a pipe of diameter 40mm. find the force exerted by the nozzle on the water which is flowing through the pipe at the rate of 1.2m^3/minute.

9. State Bernoulli's theorem for steady flow of an incompressible fluid.

10. Assumption's of Bernoulli's theorem.

UNIT -5

BOUNDARY LAYER

1. Derive the expression for momentum thickness of boundary layer.

2. Derive the expression for head loss due to friction for a pipe flow.

3. A 146.3 m long 45.7 cm diameter concrete pipe and a 179.8 m long 30.48 cm diameter concrete pipe are connected in series. Calculate the length of an equivalent of 25.4 cm diameter.

4. Explain briefly pipe network.

5. A crude oil of kinematic viscosity 0.4 stoke is flowing through a pipe of diameter 300mm at the rate of 300 liters per sec. find the head lost due to friction for a length of 50m of the pipe.

6. Find the loss of head when a pipe of diameter 200mm is suddenly enlarged to a diameter of 400mm. the rate of flow of water through the pipe is 250 l/s.

7. At a sudden enlargement of a water main from 240mm to 480mm diameter, the hydraulic gradient rises by 10mm. estimate the rate of flow.

8. A 150mm diameter pipe reduces in diameter abruptly to 100mm.if the pipe carries water at 30l/s. calculates the pressure loss across the contraction. Take the co-efficient of contraction as 0.6 .

9. Explain what you understand by boundary layer thickness and displacement thickness. Determine the relationship between the two for a boundary layer which is

(i) laminar throughout and

(ii) Turbulent throughout

Assume

(i) In the laminar boundary layer, the flow obeys the law, shear stress $\tau = \mu \frac{du}{dy}$ where μ is the viscosity , which leads to velocity profile $(U- u) = k(\delta – y)^2$ where U is the free stream , u is the velocity at distance y above the plate and k is constant.

(ii) The Velocity distribution in the turbulent boundaryis given by $\frac{u}{U} = (\frac{y}{\delta})^{1/7}$

10. Derive an expression for the calculation of loss of head due to

(i) Sudden enlargement (ii) Sudden contraction

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CE 8302 - FLUID MECHANICS
TWO MARKS QUESTION AND ANSWES
UNIT-1 - FLUID PROPERTIES AND FILUID STATICS

1. Define fluid?

A fluid is a substance having a property to flow easily.
Example: liquid, vapor, gas.

2. Define fluid mechanics?

Fluid mechanics is a branch of science which deals with property and behavior of fluids at rest and in motion.

3. Define specific volume?

It is the ratio of volume to the mass of a fluid. It is denoted by υ. Its unit is m^3/kg.
υ= volume of fluid
Mass of fluid
$\upsilon = V/m \ m^3/kg$

4. Define viscosity?

It is defined as the resisting property of liquid to its flow corresponding to its adjacent layers.

5. Which one of the following has high viscosity (i) water or (ii) lubricating oil?

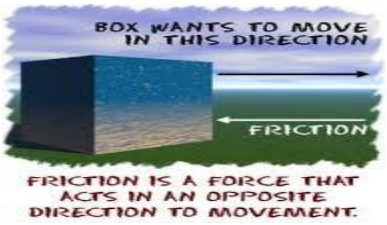
Lubricating oil has high viscosity because it reduces friction.

6. Define poise?

Poise is the other name of unit of viscosity in CGS system which equals dyne-sec/cm^2.

7. Define friction.

Friction is a force that acts in opposite direction to the movement.



8. What is Non Newtonian fluid?

A real fluid in which shear stress is not proportional to rate of shear strain.

$$\tau \neq \mu . \frac{du}{dy}$$

τ = Shear stress μ = viscosity of the fluid
du = change in velocity dy = change in perpendicular distance.

9. Define compressible fluid?

A liquid is considered to a compressible fluid only when there is a change in Volume of liquid that occurs under large pressure variation.

10. Define compressibility?

Compressibility is the property of fluid which undergoes change in volume under various pressure conditions.

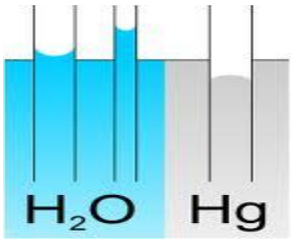
It is also defined of reciprocal of bulk modulus of elasticity (k).

i.e., compressibility = 1/k.

k= compressive stress / volumetric strain

11. Define capillarity?

It is the phenomenon of rise or fall of liquid surface relative to outside liquid surface.



12. Define specific gravity with respect to weight density?

It is the ratio of specific weight of fluid to specific weight of a standard fluid.

i.e., s = Specific weight of liquid (for liquids)
Specific weight of water

i.e., s = Specific weight of gas (for gasses)
Specific weight of air

13. Define dynamic viscosity?

The shear stress required to move one layer with unit velocity over another layer at unit distance. It is known as dynamic viscosity. It is denoted as μ.

14. What is cause for viscosity?

The causes for the viscosity are

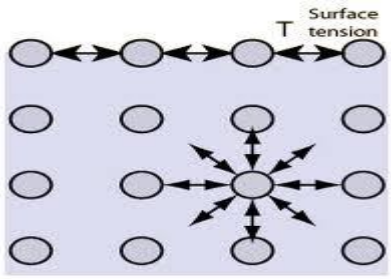
(i) Inter molecular force of cohesion and

(ii) Moment of molecules being exchanged.

15. Define Surface Tension.

Surface Tension is defined as the tensile force acting on the surface of a liquid in contact with a gas or on the surface between two immiscible liquids such that the contact surface behaves like a membrane under tension.

Unit: N/m.



16. Give the types of gas laws?

The types of gas laws are,

(i) Boyles law

(ii) Charles law

17. Define ideal fluid?

A fluid which is incompressible and has no viscosity is called as an ideal fluid. It is an imaginary fluid.

18. Define Newtonian fluid.

A real fluid in which shear stress is directly proportional to the rate of shear strain.

$\tau = \mu . du/dy$

19. Define ideal plastic fluid.

A fluid whose shear is more than yield value and its shear stress is directly proportional to shear strain is called as ideal plastic fluid.

20. What is an incompressible fluid?

A liquid is considered to be incompressible only when there is a change in volume of a liquid that occurs under smaller pressure variation.

UNIT-2 - FLUID KINEMATICS AND DYNAMICS

1. **Define "Pascal's Law".**

   It states that the pressure or intensity at a point in a static fluid is equal in all directions.

2. **What is meant by Absolute pressure and gauge pressure?**

   Absolute pressure :

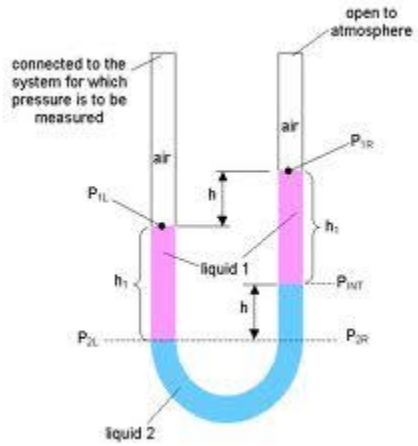
   It is defined as the pressure which is measured with the reference to absolute vaccum pressure.

   Gauge pressure:

   It is defined as the pressure which is measured with the help of a pressure measuring instrument, in which the atmospheric pressure is taken as datum. The atmospheric pressure on the scale is marked as zero.

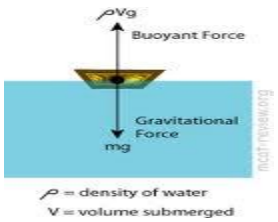
3. **Define Manometers.**

Manometers are defined as the devices used for measuring the pressure at a point in a fluid by balancing measuring the column of fluid by the same or another column of fluid.



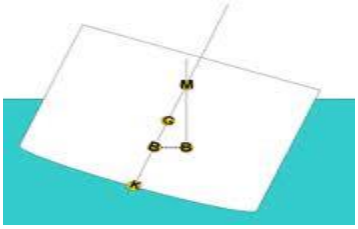
4. **Define Buoyancy.**

When a body is immersed in a fluid, an upward force is exerted by the fluid on the body. This upward force is equal to the weight of the fluid displaced by the body and is called the force of buoyancy or simply buoyancy.



5. Define META-CENTRE.

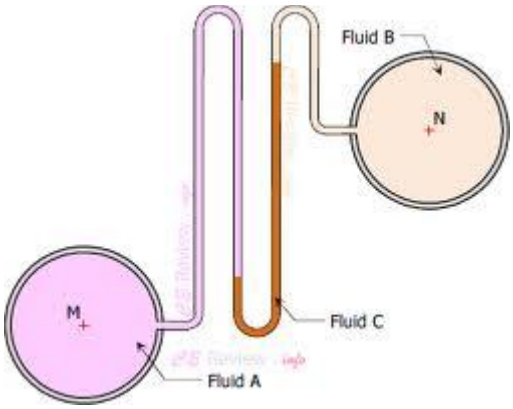
It is defined as the point about which a body starts oscillating when the body is fitted by a small angle. The meta-centre may also be defined as the point at which the line of action of the force of buoyancy will meet the normal axis of the body when the body is given a small angular displacement.



6. Write short notes on "Differential Manometers".

Differential manometers are the devices used for measuring the difference of pressure between two points in a pipe or in two different pipes , a differential manometer consists of a U tube containing a heavy liquid , whose two ends are connected to the points , whose difference of pressure is to be measured . Most commonly types of differential manometers are:

1. U-tube differential manometer
2. Inverted U-tube manometer.



7. Define Centre of pressure.

It is defined as the point of application of the total pressure on the surface.

The submerged surface may be:

1. Vertical plane surface
2. Horizontal plane surface
3. Inclined plane surface
4. Curved surface.

8. Write down the types of fluid flow.

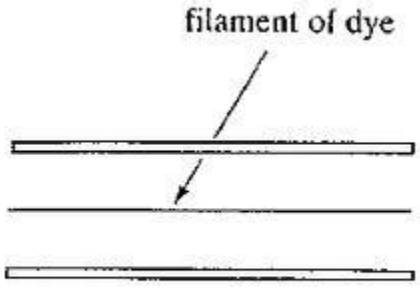
The fluid flow is classified as :

1. Steady flow and unsteady flows
2. Uniform and non-uniform flows
3. Laminar and turbulent flows
4. Compressible and incompressible flows

5. Rotational and irrotational flows
6. One , two and three dimensional flows

9. Write short notes on "laminar flow".

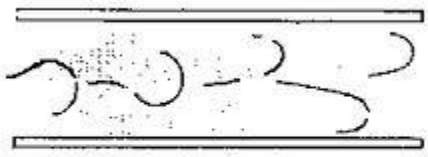
Laminar flow is defined as that type of flow in which the fluid particles move along well defined paths or stream line and all the stream lines are straight and parallel. Thus the particles move in laminas or layers gliding over the adjacent layer. This type of flow is also called stream line flow or viscous flow.



Laminar (viscous)

10. Define "turbulent flow".

Turbulent flow is that type of flow in which the fluid particles move in a zig-zag way. Due to the movement of fluid particles in a zig-zag way.



Turbulent

11. What is mean by Rate flow or Discharge?

It is defined as the quantity of a fluid flowing per second through a section of a pipe or channel. For an incompressible fluid (or liquid) the rate of flow or discharge is expressed as volume of fluid flowing across the section per section. For compressible fluids, the rate of flow is usually expressed as the weight of fluid flowing across the section.

Where , A = Cross – sectional area of pipe
V= Average velocity of fluid across the section.

12. What do you understand by continuity equation?

The equation based on the principle of conservation of mass is called continuity equation. Thus for a fluid flowing through the pipe at all the cross – section, the quantity of fluid per second is constant.

13. Differentiate steady and unsteady flow?

Steady flow	Unsteady flow
1. Steady flow is defined as that type of flow in which the fluid characteristics like velocity, pressure etc at a point do not change with time. 2. (dv/dt)(0,0,0)=0	1. Unsteady flow is that type of flow in which the velocity. pressure at a point like velocity, pressure etc at a point do changes with time. 2. (dv/dt)(0,0,0)≠0

13. Differentiate uniform and non-uniform flow?

Uniform flow	Non-uniform flow.
1. It is defined as that type of flow in which the velocity at any given time does not change with respect to space. 2. (dv/dt)t=constant=0	1. It is defined as that type of flow in which, the velocity at any given time changes with respect to time. 2. (dv/dt)t=constant≠0

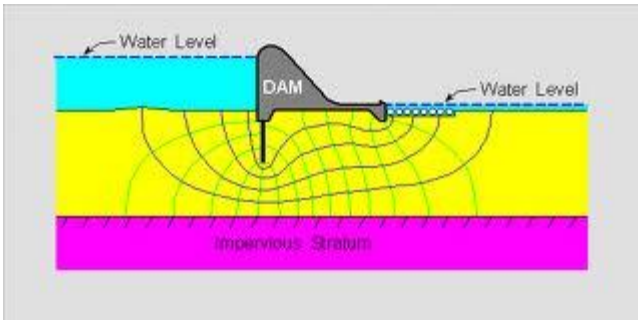
14. Define angular deformation.

It is defined as the average change in the angle contained by two adjacent sides. Let s & s is the change in angle between two adjacent sides of a fluid element.

The angular deformation=$1/2*(S\theta_1 + S\theta_2)$.

15. What is mean by Flow net?

A grid obtained by drawing a series of equipments lines and stream lines is called a flow net. The flow net is an important tool in analyzing the two –dimensional irrotational flow problems.



16. What is linear translation?

It is defined as the movement of a fluid element in such a way that it moves bodily from one position to represents in new position by a'b'& c'd' are parallel.

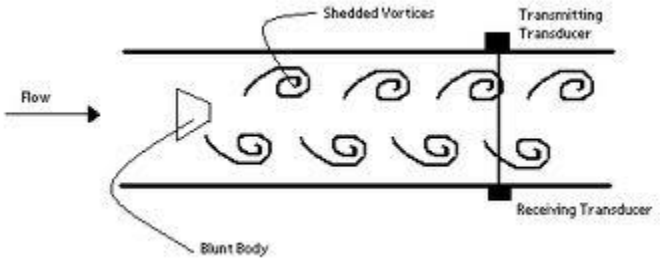
17. What are the types of Motion?

1. Linear Translation or pure Translation.
2. Linear deformation.
3. Angular deformation.
4. Rotation.

18. Define "Vortex flow".

Vortex flow is defined as the flow of a fluid along a curved path or the flow of a rotating mass of a fluid is "Vortex Flow". The vortex flow is of two types namely:

1. Forced vortex floe and
2. Free vortex flow.



19. What is linear deformation?

It is defined as the deformation of a fluid element in linear direction when the element moves the axes of the element in the deformation position and undeformation position are parallel but their lengths changes.

UNIT-3
DIMENSIONAL ANALYSIS AND MODEL STUDIES

1. **Define dimensional analysis.**

Dimensional analysis is defined as a mathematical technique used in research work for design and conducting model tests.

2. **What are the fundamental dimensions?**

The fundamental units quantities such as length (L), mass (M) and time (T) are fixed dimensions known as fundamental dimensions.

3. **Define Units.**

Unit is defined as a yardstick to measure physical quantities like distance, area, volume, mass etc.

4. **Derive the dimensions for velocity.**

Velocity is the distance (L) travelled per unit time (T)

Velocity = distance / time = [L/T] = LT^{-1}.

5. **Define Model.**

Model is nothing but small-scale repetition of the actual structure or machine.

6. **List out the advantage of model analysis.**

The advantages of model analysis are:

1. The performance of hydraulic structure or machine can be easily predicted in advance from its model.
2. The merits of alternative design can be predicted with the help of model testing and the most economical and safe design may be finally adopted.

7. **Define similitude.**

The three types of similarities are,

1. Geometric similarity
2. Kinematic similarity
3. Dynamic similarity

8. **Define Scale ratio.**

Scale ratio is the ratio of linear dimension in the model and prototype which are equal in a geometric similarity. It is denoted by L_r.

$L_r = L_p/L_m = b_p/b_m = D_p/D_m$.

9. **Define dynamic similarity.**

It means the similarity of forces at corresponding points in the model and prototype is equal.

10. Give the types of forces in a moving fluid.

The types of forces in a moving fluid are,

1. Inertia force (f_i)
2. Viscous force (f_v)
3. Gravity force (F_g)
4. Pressure force (F_p)
5. Surface tension (F_s)
6. Elastic force (F_e)
7.

11. Define dimensionless numbers.

Dimensionless numbers are the numbers obtained by dividing inertia force or gravity force or pressure force or elastic force or surface tension. They are called as non-dimensional parameters.

12. Define pressure force.

Pressure force is the product of pressure intensity and cross-sectional area of the flowing fluid in case of pipe flow.

13. Define elastic force.

Elastic force is defined as the product of elastic stress and the area of flowing fluid.

14. Give the types of dimensionless numbers.

The types of dimensionless numbers are :

1. Reynold's number
2. Froude's number
3. Euler's number
4. Weber' number
5. Mach's number

15. Define Reynolds's number.

Reynold's number is defined as the ratio of inertia force of flowing fluid and viscous force of the fluid. It is denoted by (Re).

$Re = V \times d/v = \rho Vd/\mu$

16. Define Froude's number.

Froude's number is defined as square root of ratio of inertia force of flowing fluid to gravity. It is denoted as $Fe = \sqrt{F_i/f_g} = V/\sqrt{Lg}$

17. Give the classification of models.

i. Undistorted models
ii. Distorted models

18. What is an undistorted model?

If the scale ratio for the linear dimensions of the model and prototype is same, then the model is said to be undistorted model.

19. What is distorted model?

A distorted model is said to be a distorted model only when it is not geometrically similar to prototype.

20. Define Weber's model laws.

When surface tensile forces alone are predominant a model may be taken to be dynamically similar to the prototype when ratio of inertial to the surface tensile forces is the same in the model and prototype.

UNIT-4
FLOW THROUGH PIPES

1. What are the assumptions made in the derivation of Bernoulli's equation?

(i) The fluid is ideal, ie., Viscosity is zero .
(ii) The flow is steady.
(iii) The flow is incompressible.
(iv) The flow is irrotational.

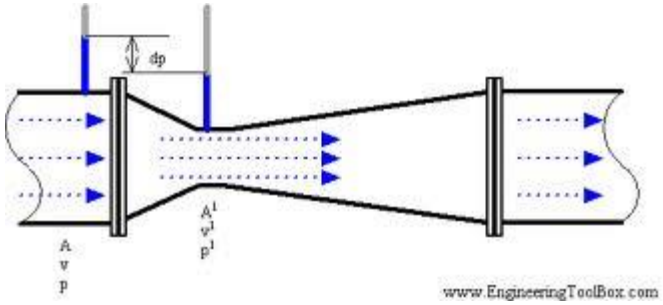
2. State the Bernoulli's theorem for steady flow of an incompressible fluid.

It states that in a steady, ideal flow of an incompressible fluid, the total energy at any point of the fluid is constant. The total energy consists of pressure energy, kinetic energy and potential energy or datum energy. These energies per unit weight of the fluid are:

Pressure energy = $p/\rho g$
Kinetic energy = $v^2/ 2g$
Datum energy = z
The mathematically, Bernoulli's theorem is written as
$(p/w)+(v^2/2g)+z$ = Constant.

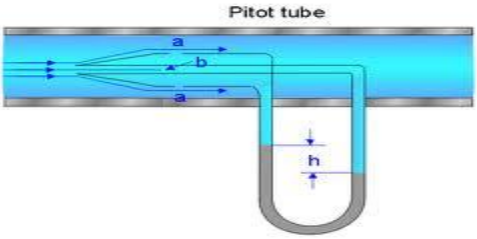
3. Define Venturimeter.

Venturimeter is a device used for measuring the rate of flow of a fluid flowing through a pipe. It consists of three (i) A short converging part
(ii) Throat
(iii) Diverging part.



4. Define pitot tube.

Pitot tube is a device used for measuring the velocity of flow at any point in a pipe or channel. It is based on the principle that if the velocity of flow at a point becomes zero.



5. What is meant by free liquid jet?

Free of liquid jet is defined as the jet of water coming out from the nozzle in atmosphere. The path travelled by the free jet is parabolic.

6. What is meant by dynamic of fluid flow?

The study of fluid motion with the forces causing flow is called dynamics of fluid flow. The dynamic behavior of the fluid flow is analyzed by the Newton's second law of motion, which relates the acceleration with the forces.

7. Write down the formulae for finding the discharge in venturimeter.

$$\mathbf{Q = C_d \frac{a1a2}{\sqrt{a12 - a22}} \times \sqrt{2gh}}$$

Where a_1= area of the inlet venturimeter

a_2 = area at the throat

C_a = co efficient of venturimeter

h = difference of pressure head in terms of fluid head flowing through venturimeter.

8. Write the equations of motion.

The net force $F_x$acting on a fluid element in the direction of x is equal to mass m of the fluid element multiplied by the acceleration $a_x$in the x-direction.

$F_x = ma_x$.

9. What are the forces present in the fluid flow?

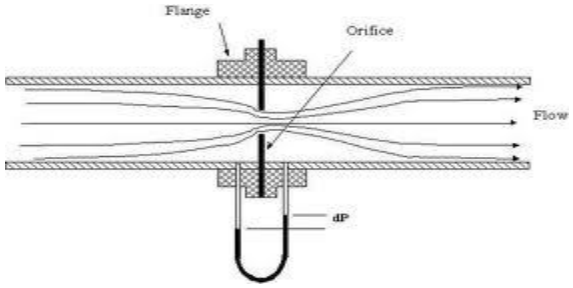
(i) Gravity force ,F_g
(ii) The pressure force ,F_p
(iii) Force due to viscosity ,F_v
(iv) Force due to turbulence ,F_t
(v) Force due to compressibility, F_c.

10. What are the devices used for measuring rate of flow?

(i) Venturimeter
(ii) Pitot tube
(iii) Orifice meter.

11. Define orifice meter.

It is the device used for measuring the rate of flow of a fluid through a pipe. It also works on the same principle as that of venturimeter.



12. State the momentum principle.

It states that the net force acting on a fluid mass is equal to the change in momentum of flow per unit time in that direction.

$$\mathbf{F = \frac{d(mv)}{dt}}$$

13. Define moody diagram.

The Moody chart or Moody diagram is a graph in non-dimensional form that relates the Darcy friction factor, Reynolds number and relative roughness for fully developed flow in a circular pipe. It can be used for working out pressure drop or flow rate down such a pipe.

14. Give the expression for Darcy weisbach equation.

$$h_f = f\frac{l}{d}\frac{V^2}{2g};$$

Where

hf = loss of head due to friction

f = friction factor

l = length of pipe, d = diameter of pipe

V = velocity of flow, g = acceleration due to gravity.

15. Define

(i) Stream line

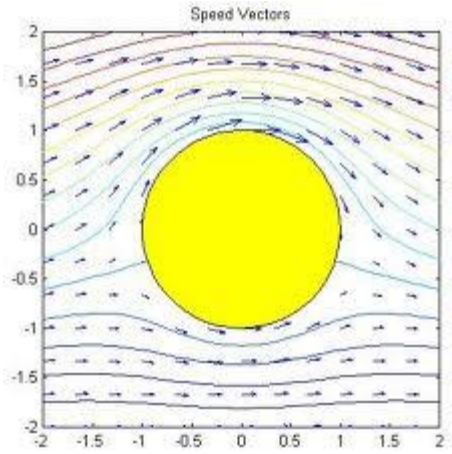
(ii) Path line

(iii) Streak line

(iv) Time line

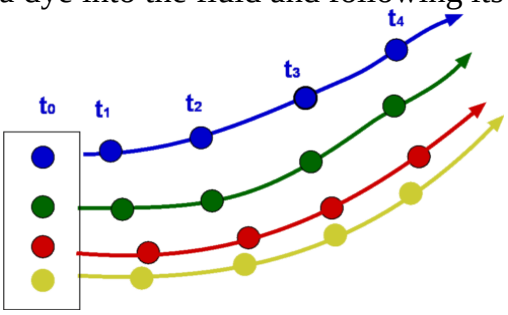
(i) Stream line:

Stream line is the path traced out by a mass less particle moving with the flow. Velocity is tangent to stream line at every point.



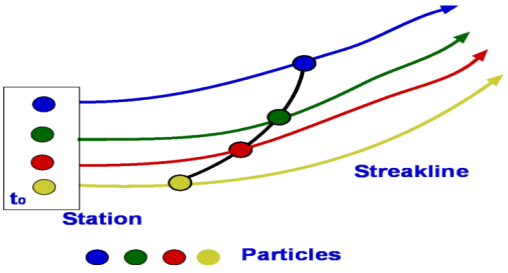
(ii) Path line:

Pathline is the line traced by a given particle. This is generated by injecting a dye into the fluid and following its path by photography or other means.



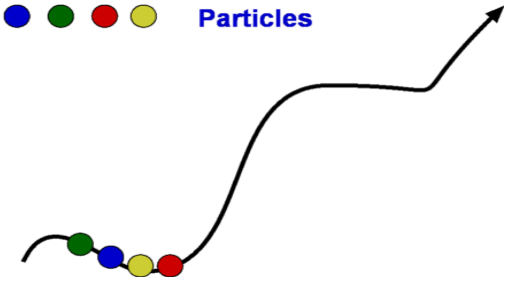
(iii) Streak line:

Streakline concentrates on fluid particles that have gone through a fixed station or point. At some instant of time the position of all these particles are marked and a line is drawn through them. Such a line is called a streakline.



(iv) Time line:

Timeline is generated by drawing a line through adjacent particles in flow at any instant of time.



16. Distinguish between stream line and path line.

Stream line	Path line
A stream line is an imaginary line drawn in a flow field such that a tangent drawn at any point on this line represents the direction of the velocity vector. Path line is the line traced by a single fluid particle as it moves over a period of time.	Stream line shows the direction of velocity of a number of fluid particles at the same instant of time. Path line shows the direction of velocity of the same fluid particle at successive instants of time.

17. Can the path line and a streamline cross each other at right angles?

A fluid particle always moves tangent to the streamline. In steady flow, the path lines and streamlines are identical. In unsteady flow, a fluid particle follows one stream line at one instant and another at the next instant and so on, so that the path line have no resemblance to any given instantaneous streamline.

18. Define convective and local acceleration.

Convective acceleration:

It is defined as the rate of change of velocity due to the change of position of fluid particles in a fluid flow.

Local acceleration:

It is defined as the rate of increase of velocity with respect to time at a given point in a flow field.

19. Define stream function.

It is defined as the scalar function of space and time, such that its partial derivative with respect to any direction gives the velocity component at right angles to that direction.

20. Define boundary layer thickness.

It is defined as the distance from the boundary of the solid body measured in the y-direction to the point, where the velocity of the fluid is approximately equal to 0.99 times the free stream velocity of the fluid.

UNIT-5

BOUNDARY LAYER

1. What do you meant by viscous flow?

A flow is said to be viscous if the Renold's number is less than 2000 (or) the flows in layers ie. Re <2000.

2. Define kinetic energy correction factor?

Kinetic energy factor is defined as the ratio of the kinetic energy of the flow per secbased on actual velocity across a section to the kinetic energy of the flow per sec based onaverage velocity across the same section. It is denoted by (σ).

K. E factor (σ) = K.E per sec based on actual velocity / K.E per sec based on Average velocity.

3. **Define momentum correction factor .**

It is defined as the ratio of momentum of the flow per sec based on actual velocity to themomentum of the flow per sec based on average velocity across the section.

It is denoted by(β) = $\frac{Momentum\ per\ \sec\ based\ on\ actual\ velocity}{Momentum\ per\ \sec\ based\ on\ average\ velocity}$

4. Give the expression for the loss of head due to friction in viscous flow.

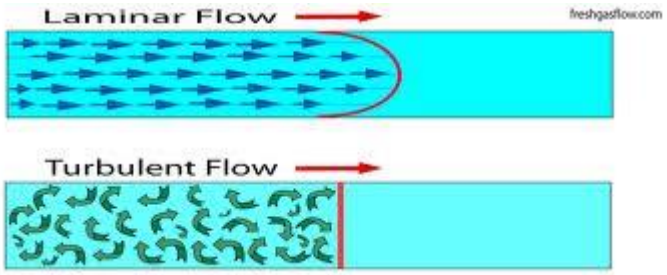
Loss of head due to friction in viscous flow f = 16/Re

Where f= coefficient of friction between the pipe and fluid

Re= Renolds number.

5. Differentiate between laminar flow and turbulent flow.

Laminar flow	Turbulent flow
A flow is said to be laminar if Reynolds number is less than 2000 is known as laminar flow.	A flow is said to be turbulent if Renolds number is greater than 4000 is known as Turbulent flow .
Laminar flow is possible only at low velocities and high viscous fluids .	Is the flow is possible at both velocities and low viscous fluid .
In such type of flow fluid particle moves in laminas or layers gliding smoothly over the adjacent layer .	In that type of flow fluid particle move in a zig – zag manner .



6. What is the expression for head loss due to friction in Darcy formula?

$h_f = 4fLV^2 / 2gD$

Where,

f = Coefficient of friction in pipe

L = Length of the pipe

D = Diameter of pipe

V = velocity of the fluid

7. What are the factors influencing friction loss in pipe flow?

Frictional resistance for the turbulent flow is

i. Proportional to vn where v varies from 1.5 to 2.0.

ii. Proportional to the density of fluid.

iii. Proportional to the area of surface in contact.

iv. Independent of pressure.

v. Depend on the nature of the surface in contact.

8. Give the equation for average velocity.

The equation for average velocity is given as

U = μ (-ðp/ðx) R2

Where

R = Radius of the pipe

9. Give the formula for velocity distribution.

The formula for velocity distribution is given as

U = ¼ μ (-ðp/ðx) (R2-r2)

Where

R = Radius of the pipe

r = Radius of the fluid element

10. What do you understand by the terms a) major energy losses , b) minor energy losses

Major energy losses : -

This loss due to friction and it is calculated by Darcy weis bach formula and chezy's formula .

Minor energy losses :-

This is due to

i. Sudden expansion in pipe .

ii. Sudden contraction in pipe .

iii. Bend in pipe .
iv. Due to obstruction in pipe .

11. How will you determine the loss of head due to friction in pipes?

Darcy weis-bach equation

$h_f = 4fLV^2 / 2gD$

Where, hf = Loss of head due to friction .
f = Coefficient of friction in pipe .
D = Diameter of pipe .
L = Length of the pipe
V = Mean velocity of flow .

Chezy's formula

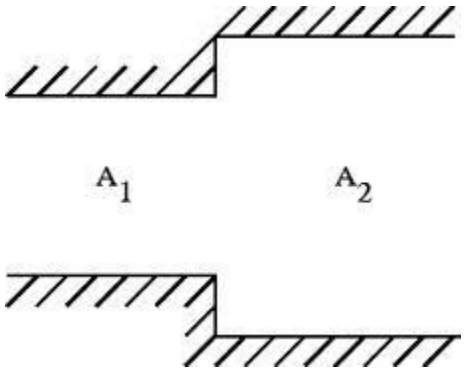
$V = C\sqrt{mi}$

Where
i = hf / l

12. Give an expression for loss of head due to sudden enlargement of the pipe.

$h_e = (V_1-V_2)^2 /2g$

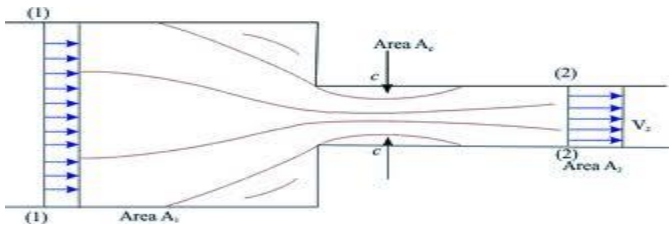
Where
he = Loss of head due to sudden enlargement of pipe .
V_1 = Velocity of flow at section 1-1
V_2 = Velocity of flow at section 2-2



13. Give an expression for loss of head due to sudden contraction.

$h_c = 0.5\ V^2/2g$

Where
hc = Loss of head due to sudden contraction .
V = Velocity at outlet of pipe.



14. Give an expression for loss of head at the entrance of the pipe.

$hi = 0.5V^2/2g$

where

hi = Loss of head at entrance of pipe .

V = Velocity of liquid at inlet and outlet of the pipe .

15. Derive the expression for drop of pressure for a given length of a pipe.

$P_1-P_2 = 32\ \mu UL / gD^2$

Where

P_1-P_2 is drop of pressure.

16. Give expression for co efficient of friction in terms of shear stress.

$f = 2i_o / \rho v^2$

Where

i_o= Shear stress

v = volume of pipes

f = coefficient of friction

17. Give an expression for loss of head due to an obstruction in pipe.

Loss of head due to an obstruction

$= V^2 / 2g\ (A/ Cc\ (A-a\ ) -1\ )^2$

Where

A = area of pipe

a = Max area of obstruction

V = Velocity of liquid in pipe

A-a = Area of flow of liquid at section 1-1.

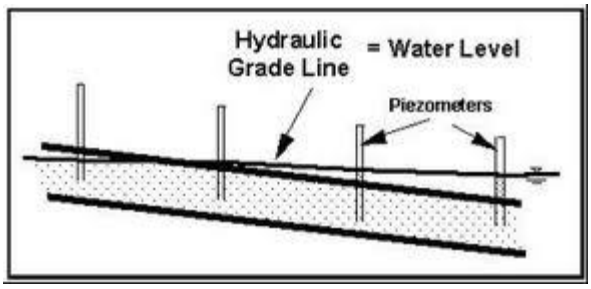
19. Define the terms

a) Hydraulic gradient line [HGL]

b) Total Energy line [TEL]

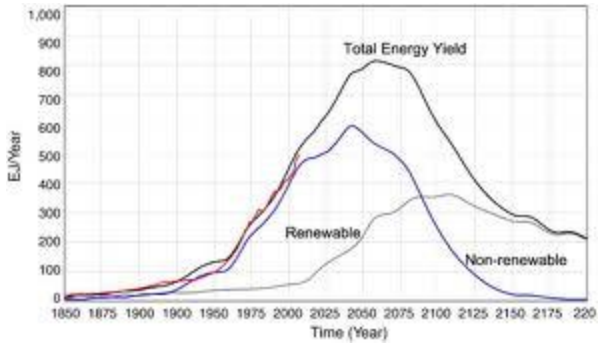
a) Hydraulic gradient line :-

Hydraulic gradient line is defined as the line which gives the sum of pressure head and datum head of a flowing fluid in a pipe with respect the reference line .



b) Total energy line :-

Total energy line is defined as the line which gives the sum of pressure head , datumhead and kinetic head of a flowing fluid in a pipe with respect to some reference line .



20. What is syphon? Where it is used?

Syphon is along bend pipe which is used to transfer liquid from a reservoir at a higher elevation to another reservoir at a lower level.

Uses of syphon : -

1. To carry water from one reservoir to another reservoir separated by a hill ridge .
2. To empty a channel not provided with any outlet sluice.

V.S.B ENGINEERING COLLEGE, KARUR

DEPARTMENT OF CIVIL ENGINEERING

CE 8391 –CONSTRUCTION MATERIALS– 16 MARKS

PART – B & PART-C

UNIT – I - STONE-BRICKS-CONCRETE BLOCKS

1. Explain in detail about the manufacturing process of bricks.
2. Describe the various Qualities of a good brick.
3. Briefly discuss the defects and preservation of stones.
4. Explain the various methods for the preservation of stone work.
5. List the various tests to be performed on stones.
6. Describe the characteristics to be considered for selection of stones for various civil Engineering works.
7. Explain the various types of common building stones and their uses.
8. Explain the properties of good bricks.
9. Explain in detail about the manufacture process of conventional bricks with neat sketches.
10. Describe the sketches the manufacture of bricks by Hoffman's kiln.
11. Describe the various tests to be conducted for testing of conventional bricks.
12. List out the types of special bricks. Briefly explain any four of them.
13. Discuss the reasons for the causes of defects in bricks.
14. Explain the recent advancements (special use) refractory bricks.
15. What are concrete blocks? Explain about the manufacturing of concrete blocks.
16. Explain in detail about the manufacturing and applications of light weight concrete blocks.

UNIT- 2 - LIME-CEMENT-AGGREGATES-MORTAR

1. Explain in detail about the manufacturing process of lime.
2. Briefly describe the manufacturing process of cement.
3. Explain with a flow chart the step by step dry process of manufacturing of cement.
4. Describe the wet and dry process of mixing of raw material for cement.
5. What are the various ingredients required for manufacturing cement? State their function.
6. Give a detailed account of types and grades of cement.
7. Explain the preparation, properties and uses of cement mortar.
8. List the various field and laboratory tests for cement.
9. Explain field tests on river sand and cement.
10. How do you perform the tensile strength of cement? Also explain the effect of fineness of cement.
11. How do you perform the compression strength of cement mortar cubes? Also explain the procedure for fineness of cement.
12. Explain the hydration process of cement.
13. Compare and contrast the advantages and disadvantages of using lime and cement in engineering works.
14. Explain detail about
    a) Consistency test as per IS specification on cement. b) Soundness of cement. c) Crushing strength of aggregate. d) Impact strength of aggregate
15. Explain various properties of coarse aggregates.
16. Explain the following test with neat sketches
    a) Abrasion resistance test on coarse aggregate. b) Soundness of cement.
17. Enumerate the various methods for physical testing of coarse aggregate. Discuss any two tests in detail.
18. Explain with codal provision for testing of conventional coarse aggregates.

UNIT 3 - CONCRETE

1. Explain in detail about preparation of concrete.
2. Discuss the functions of cement, fine aggregate and water in the concrete.
3. Explain any two tests for testing of conventional fresh concrete.
4. With sketches explain in detail about different flow properties of concrete.
5. What is compaction of concrete? Explain the different compaction process of concrete.
6. What is meant by curing of concrete? Explain the different curing process of concrete.
7. Explain the test on workability of concrete.
8. List the various tests to be performed on hardened concrete with neat sketch.
9. Explain the procedure of compressive and split tensile strength of hardened concrete with neat sketch.
10. Write down the complete procedure of concrete mix design as per BIS. Assume necessary data.
11. With sketches explain the various specifications for self-compacting concrete.
12. With sketches explain any two testing of workability for self- compacting concrete.
13. Briefly explain about the high strength and high performance concrete.
14. Write in detail about
    a) Modulus of rupture. b) Self- compacting concrete. c) High performance concrete
15. Write in detail about
    a) Curing b) High Strength Concrete

UNIT 4- TIMBER AND OTHER MATERIALS

1. Define timber. Explain the classification and advantages of timber.
2. Write the properties and uses of timber.
3. Explain the preservatives treatments available for timber.
4. List out and explain the various methods of seasoning of timber.
5. Explain the defects and market forms of timber.
6. Explain in detail about a) Plywood b) Veneer c) Thermacole d) Panels of laminates
7. Explain the important advantages and disadvantages of timbers and timber construction.
8. Explain the different types of paints used in building construction.
9. Discuss the reasons for the causes of defects in painting work.
10. Describe the manufacturing process and civil Engineering applications of steel.
11. List the various mechanical and heat treatment of steel.
12. Explain in detail about aluminum composite panel and its advantages.
13. Explain the important properties and advantages of aluminum.
14. Investigate the various applications of aluminium.
15. Enumerate the advantages of using aluminium as building material.
16. What are varnishes? Describe the various types of varnishes.
17. Write in detail about
    a) Steel b) Forms of aluminum c) Preparation of paint d) Classification of timber
18. Write in detail about a) Paints b) Varnishes c) Distempers d) Bitumen

UNIT-5- MODERN MATERIALS

1. List few modern materials and explain the uses of any two materials.
2. Explain the various classification and manufacturing process of glass.
3. Explain the different types of glass with its application in buildings.
4. Discuss the various forms of commercial glass.
5. Explain the properties and uses of glass in construction industry.
6. What is fabrication? Explain the various methods for fabrication of glass.
7. Explain in detail about ceramics.
8. Write an explanatory note on Mechanical properties of ceramics.
9. Explain in detail about a) Sealants for joins b) Ceramics c) Refractories
10. Explain in detail about refractories.
11. Discuss the various factors you considered for choosing floor tiles.
12. Discuss the uses of clay products.
13. Explain the following a) Properties of plastics b) Composite materials
14. Describe the various applications of clay products in construction works.
15. Discuss about the manufacture and properties of various clay products.
16. Write short notes on a) Laminar Composites b) Fibre glass reinforced plastic
17. Explain in detail about the composite materials and its types.
18. Discuss about the types of fiber textile and properties of textile fiber.
19. Explain the classification of Geo Synthetics and its uses.
20. Describe the various applications of i) Fibre textile ii) Geotextile
21. Describe the various applications of i) Geomembrane ii) Geotextile

PART-C

1. Discuss the criteria to recommend stone as a good building material.
2. As a civil Engineer recommend various building materials involved in the construction of a building with their criteria with respect to quality.
3. Discuss in detail about the factors affecting the durability of concrete and state protective measures established for durable concrete.
4. Describe the various classifications of geosynthetics and also explain various applications of geosynthetics in construction works.
5. Determine the shape test on coarse aggregate with suitable sketches.
6. What is meant by bitumen 80/100 penetration and explain its various applications in construction.
7. Explain the application of Bitumen in road construction.
8. Explain the application of geotextiles for earth reinforcement.
9. Describe the various applications of geotextiles in geo technical engineering works. And also explain the properties of geotextiles.
10. Explain any three types of special concrete and their applications.
11. Explain the various applications of geosynthetics in civil engineering works.

V.S.B ENGINEERING COLLEGE, KARUR
DEPARTMENT OF CIVIL ENGINEERING
CE83911- CONSTRUCTION MATERIALS-TWO MARKS & ANSWERS

UNIT I- STONES – BRICKS – CONCRETE BLOCKS

1. What are the uses of stone in construction?

Stone is an important building material and use in the construction of stone masonry for foundation, stone work for dam, wing walls for bridges, culverts and other important structural works.

2. What are the uses of stone?

- For masonry work
- For lintels and vertical columns
- For covering floors and buildings
- For paving of roads and footpaths
- As an aggregate in cement as well as lime concretes.

3. What are the various tests for stones?

1. Smith's test 6. Microscopic examination
2. Freezing and thawing test 7. Impact test
3. Frost action test 8. Attrition test
4. Acid test 9. Crushing test
5. Water absorption test 10. Hardness test

4. What are the causes for deterioration of stones?

1. Temperature variation 6. Vegetable growth
2. Wetting and drying of stone 7. Rain water
3. Frost action 8. Wind
4. Polluted atmosphere 9. Water
5. Living organisms 10. Chemicals

5. What is an artificial stone?

Cast stone or artificial stone is nothing, but hardened plain cement concrete, moulded in suitable shape and size. Artificial stone consists of 1.5 parts of coarse aggregate ofsize 3mm to 6mm and 1.5 parts of coarse aggregate of size less than 3mm.

6. Explain the factors influencing the selection of preservatives.

- Type of the stone
- Durability of the preservative.
- Colour of the preservative.
- Environmental conditions.
- Cost factors etc.

7. What are Kilns? State its types.

Kilns are the permanent structures or arrangements, used for large scale unit in the manufacturing process of bricks, for burning process of bricks, for burning purposes. Kilns are classified as,1)Intermittent Kilns 2)Continuous Kiln

8. State any four characteristics of good building stone?

1. Strength
2. Durability
3. Hardness
4. Toughness

9. What is meant by dressing of stones?

The Dressing of stone is defined as "The process of giving a proper size, shape and finish to the roughly broken stones as obtained from the quarry. The stones are used in different types of masonry; therefore, it has to be cut and shaped to fit in the type of work needed.

10. Define "natural bed of stone".

The building stones are obtained from rocks. These rocks have a distinct plane of division along which stones can easily be split. This plane is known as the natural bed of stone and it thus indicates the plane or bed on which the sedimentary stone was originally deposited. The natural bed of stone need not necessarily be horizontal.

11. Define light weight clay bricks and its ingredients.

Bricks are lightweight rectangular blocks used as important building materials having uniform size, and obtained by burning and drying of clay in moulding. A good brick earth mainly consists of silica (sand), alumina (clay), lime, oxide of ironand magnesia.

12. List out the raw materials used for manufacture of fire bricks.

Principal raw materials for firebrick include fireclays, 1) aluminum silicates; 2) aluminum oxide 3) silica, 4) magnesite, 5) chromite, 6) solid solution of chromic oxide

13. Differentiate first and second class bricks.

First class bricks are 19 x 9 x 9 cm in size and made from good earth. They should be thoroughly burnt, good color, regular shape with square edges and parallel faces.

Second class bricks are also fully burnt and give a clear ringing sound when struck together. Slightly irregularities in shape, size or color are accepted.

14. List the tests for bricks.

1) Absorption test 2) Shape and size test 3) Crushing strength test 4) Soundness test 5) Hardness test .6) Test for presence of soluble salts.

15. Define the term Efflorescence. How it can be removed?

If soluble present in the bricks, it will cause efflorescence on the surface of the bricks. In this test, the brick is immersed in water for 24 hours. It is taken out and allowed to dry in shade. The presence of grey and white deposits on bricks surface indicates the presence soluble salts.

16. List out the types of bricks for special use.

1) Facing bricks 2) Fireclay bricks 3) Silica bricks 4) Neutral bricks 5) Basic bricks 6) Acid bricks 7) Ordinary bricks.

17. What are refractory bricks and its types?

Refractory bricks (fire clay bricks) are manufactured from fire-clay and the process is same as that of ordinary bricks.Fireclay bricks are white or yellowish in colour. The weight of a fire-brick is about 30N to 35N.Following are the types of refractory bricks.1) Acidic bricks 2) Basic bricks 3) Neutral bricks.

18. Write down the uses of bricks.

- Construction of walls of any size.
- Construction of floors.
- Construction of arches and cornices.
- Construction of brick retaining wall.

19. What do you mean by lightweight concrete block and its advantages?

Lightweight concrete blocks are made by replacing the sand and gravel with expanded clay, shale or slate. Expanded clay, shale and slate are produced by crushing the raw materials and heating them to about 2000˚c.It has following advantages light in weight, high thermal insulation, high fire protection, high sound insulation and low water absorption.

20. Write the classification of concrete blocks.

1) Hollow concrete blocks. 2) Aerated Autoclaved Concrete Block (AAC) 3) Concrete Bricks 4) Solid concrete blocks. 5) Lintel blocks. 6) Paving Blocks. 7) Concrete stretcher block.

UNIT II - LIME – CEMENT – AGGREGATES – MORTAR

1. DefineLime.

Lime is a locally available binding material and it is used as an alternative for cementing material.

2. Define Calcination of lime.

Calcination is defined as the process of heating the lime to redness in contact with air. Calcination removes the moisture content and carbon-di-oxide from hydrated lime.

$$CaCO_3 \longrightarrow Cao \text{ (quick lime)} + CO_2$$

3. What is meant by slaking?

Slaking is defined as the process of chemical combination of quick lime with sufficient quantity of water for reducing the heat and, to get the powder form of lime.

$$CaO + H2O \rightarrow Ca(OH)_2 + Heat$$

4. Define slaked lime and Quick lime.

The product obtained by slaking of quick lime is known as slaked lime or hydrated lime.

The lime obtained from the calcinations of pure limestone is known as quick lime

5. What are the classifications of lime?

1. Fat lime
2. Hydraulic lime
3. Poor lime

6. What are called as bogue compounds?

When water is mixed with cement, the hydration process begins. The ingredients in concrete react each other with the help of water and forms more complex compounds .These compounds are called bogues compounds.

7. What is meant by cement and it types?

Cement is the binding material, obtained by burning and crushing of clay stones containing Calcium Carbonate and Magnesium Carbonate. ($CaCO_3$ and $MgCO_3$).Cement can be classified as 1.Natural cement 2.Artificial cement.

8. What are the normal ingredients of cement?

1. Lime 2.Silica 3.Alumina 4.Calcium Sulphate 5.Iron Oxide 6.Magnesia 7.Sulphur

9. What is the function of manufacturing of cement?

Three important processes are carried out in the manufacturing of ordinary cement.1).Mixing of raw materials 2).Burning 3).Grinding

10. What is meant by grade C-43 cement? What are the main active cementing compounds?

The 43 grade cement indicates that compression strength of the cement concrete after 28 days of setting, it gives a strength of 43 mpa.(or 43 N/mm2)

11. What is known as clinker?

Artificial cement is manufactured by burning approximately proportioned mixture of calcareous and argillaceous materials at a very high temperature and then grinding the burnt mixture to a fine powder. The burnt mixture of calcareous and argillaceous matter is known as clinker.

12. What is meant by hydration of cement? State its importance.

The chemical reaction takes place between cement and water is called hydration of cement. When water is added to the cement, various ingredients of cement react chemically with various complicated chemical compounds.

13. List the types and grades of cement?

Types of cement: 1. High alumina cement 2.Low heat cement 3.White cement 4.Coloured cement 5.Expanding cement 6.Quick setting cement.

Grades of cement: The gradesare classified depending upon the compressive strength of the cement.1) Grade 33 2) Grade 43 3) Grade 53.

14. Why do you understand by transition zone?

This **transition zone** is a plane of weakness and therefore has great influence on mechanical behavior of concrete.

15. Define Cement mortar.

The mortar is a paste like substance prepared by adding required amount of water to a dry mixture of sand or fine aggregate with some binding material like clay, lime or cement.

16. What do you understand by 53 grade cement?

The 53 grade cement indicates that compression strength of the cement concrete after 28 days of setting, it gives a strength of 53 mpa.(or 53 N/mm2)

17. What are the different types tests for cement?

1.Compressive strength test **2.**Tensile strength test **3.**Fineness test **4.**Soundness test **5.**Consistency test **6**.Setting time.

18. Define the term setting time of cement.

Setting time is used to detect the initial and final setting time of the cement under various climatic conditions. This test also is used to detect the deterioration of cement due to storage. The test is carried out to find out initial setting time and final setting time of cement.

19. Write down the chemical and physical characteristics of fly ash.

The chemical composition of fly ash depends upon the type of coal used and the methods used for combustion of coal. The physical properties of fly ash are Fineness of Fly Ash, Colour, Size and Shape of Fly Ash, Specific Gravity of Fly Ash.

20. State the functions of coarse aggregate in a concrete.

To fill, provide strength and form the consistency of the concrete. Also to provide physical interlocks thus raising shear strength. Fine aggregate mix in fills the voids created in coarser aggregate mix (fills the gaps between bigger stone particles)

21. **What do you mean by graded aggregates?**

1. Dense Graded Aggregate **2.**Gap-Graded Aggregate**3**.Uniformly Graded Aggregate **4**.Well Graded Aggregate **5**.Open Graded Aggregate.

22. List the test for coarse aggregates.

1. Crushing strength test **2.**Impact strength test **3.**Abrasion resistance test**4.**Flakiness Index and Elongation Index test.

23. State Flakiness Index test and Elongation Index.

Flakiness index is defined as the percentage by weight of particles whose least thickness is less than $3/5^{th}$ of their mean dimension. Thickness gauge is used to find out the flakiness index. **Elongation index** is the percentage by weight of particles whose greatest length is greater than $4/5^{th}$ 1.8 times their mean dimension.

24. What is known as bulking of sand?

Bulking of sand is defined as the volume increase of sand due to presence of moisture content. Fine sand bulks more as compared to coarse sand. Fine aggregate do not show any bulking when it is absolutely dry or completely saturated.

UNIT III CONCRETE

1. Define Concrete.

Concrete is defined as the solid-composite material and made up of suitable proportions of binding material, fine aggregate, coarse aggregate and water. Some special chemicals may be used to change the properties of concrete. The various ingredients of concrete are **1**.Binding Material **2**.Fine aggregate **3**.Coarse aggregate **4**.Water.

2. What precautions would you take in curing PPC?

After 24 hours of laying PCC, moist the concrete surface by flooding with water or by covering with moist gunny bags at least for seven days. **PPC** has initial setting time of 30 minutes and the ultimate setting time of 600 minutes. Typically, **PPC** also takes achieve **concrete** strength.

3. What are the various stages of preparation of concrete?

1. Batching of ingredients 2.Mixing of materials 3. Transporting 4.Placing 5.Compacting
6. Finishing 7.Curing.

4. Define Ready Mixed Concrete and list the ingredients of concrete.

Concrete prepared at plant or in trick mixers and transported and delivered to the construction site is called Ready Mix Concrete or Pre-Mixed Concrete. It has low cost and high durability. The various ingredients of concrete are **1**.Binding Material **2**.Fine aggregate **3**.Coarse aggregate **4**.Water.

5. Distinguish between bleeding and segregation.

Bleeding is defined as the type of segregation, in which sum of the water leaves out from the concrete and float on the surface of the concrete. It is sometimes referred as water gain.

The disintegration of ingredients of concrete mix, so that the mix is no longer in a homogeneous and stable condition is called segregations

6. Give the relationship between slump value and degree of workability.

Slump test is a field method used to roughly indicate the workability of concrete sample. Although there is no direct relationship between slump value and workability it gives a qualitative idea about workability of concrete. Higher is the slump value greater is the workability of concrete sample and vice versa.

7. List out the methods of transportation of concrete.

- Motor PAN
- Belt Conveyor
- Truck mixer and dumper
- Transit mixer

8. Mention the properties of fresh concrete.

- Workability.
- Slump loss.
- Temperature.
- Bleeding/segregation.
- Time of set.
- Plastic shrinkage.

9. Define workability.

Workability is that property of concrete which determines the amount of internal work necessary to produce full compaction. It is a measure with which concrete can be handled from the mixer stage to its final fully compacted stage.

10. What are the factors affecting workability.

1. Water-Cement ratio.
2. Mix proportions
3. Size of the aggregates.
4. Shape of the aggregates

11. List the types of tests for concrete.

1. Slump test 2.Compaction test 3.Flow test 4.Kelly ball test 5.Vee-Bee consistometer test

12. Define compaction Factor.

The Compaction factor is defined as the ratio of the weight of partially compacted concrete to the weight of fully compacted concrete.

13. List the properties of hardened concrete.

Following are the important properties of hardened concrete. 1).Strength 2).Durability 3).Permeability

14. Define Modulus of rupture.

The Flexural Tensile Strength is expressed as modulus of rupture (MR)in psi (Mpa)and is determined by standard test methods ASTM C78 or ASTM C 293. **Modulus of rupture** (MOR) is a measure of the maximum load-carrying capacity or strength of the crosstie and is **defined** as the stress at which the material breaks or ruptures

15. Define mixing of concrete.

The process of mixing cement, water, fine aggregate and coarse aggregate in suitableproportion is known as mixing of concrete.

16. What are the methods of consolidation or compaction of concrete?

1. Hand compaction
2. Machine compaction – i) Internal vibrators ii) Form vibrators iii) Surface vibrators

17. Define curing of concrete.

Curing is the operation by which moist conditions are maintained on finished concretesurface, to promote continued hydration of cement.

18. What are admixtures?

Admixtures are ingredients other than cement, fine aggregate and coarse aggregate toimprove the quality of concrete. The addition of an admixture may improve the concrete with respect to its strength, hardness, workability, water resisting power etc.

19. Write short notes on Mix proportioning and factors considered in mix proportioning.

Mix proportioning is the process of measuring the various ingredients of concrete of the required strength, quality and durability, and which is prepared by using the available materials.

1. Water cement ratio and consistency of fresh concrete
2. Cement paste or cement aggregate ratio
3. Gradation of aggregate

20. What is meant by grade of concrete? What is the lowest grade of concrete allowed for structural works in buildings?

Grade of concrete is defined as the minimum strength the concrete must posses after 28 days of construction with proper quality control. Min Grade of concrete for RCC is M20. Which is the minimum grade of concrete for RCC with ratio **1:1.5**:3 means 1 part of cement, 1.5 parts of fine aggregate and 3 parts of coarse aggregate?

21. Distinguish between nominal mix and design mix.

Nominal mix has volumetric batching. It is Performance based concrete. It is adopted for higher grade concrete. It has Quality control. However, design mix is a process where each and every ingredient of the concrete is first tested in the laboratory. . It is Prescriptive type concrete. It is used in ordinary concrete involving concrete grade not higher than M20. There is no quality control

22. Distinguish between HSC and HPC.

High Strength concrete having the compressive strength of more than 35 Mpa. It is used in structures that must resist high compressive loads, HSC is used in high-rise buildings and long bridges.

High-strength concrete is defined as having a specified compressive strength of 8000 psi (55 MPa) or greater. which possesses high strength, high workability, high modulus of elasticity, high density, high dimensional stability, high resistance to chemical attacks and low permeability.

23. Write the formula for flow percent.

$$\text{Flow}(\%) = \frac{\text{Spread dia.(cm)} - 25}{25} \times 100$$

24. Define Self Compacting Concrete.

Self-compacting concrete has evolved as an innovative method. In this, no vibration is required for the concrete, which can flow around obstructions, reinforcement and fill the formwork completely under its own self weight.

25. Why do we need compaction of concrete?

Compaction significantly increases the ultimate strength of concrete. It also increases the abrasion resistance and general durability of the concrete, decreases the permeability and helps to minimise its shrinkage and creep characteristics.

26. Why do you understand by consolidation of concrete?

Consolidation of concrete is the process of removing this excessive air and making the concrete into a uniformly solid mass with the needed properties. Consolidationalso levels the fresh concrete and improves the bonding with the reinforcement.

UNIT IV TIMBER AND OTHER MATERIALS

1. What is timber?

Timber is defined as the wood suitable for building purposes and it is applied to the trees measuring not less than 0.6 m in circumference or girth of the trunk.

2. List out the defects in timber.

- Defects due to abnormal growth. (check)
- Defects due to Conversion. (knot)
- Defects due to Seasoning.(twist)

3. Define seasoning of timber.

Seasoning is defined as the process of drying the timber to reduce the moisture content and thus to prevent the timber from possible fermentation and making it suitable for use.

4. State any four characteristics of good timber.

1) Hardness, 2) Strength, 3) Toughness 4) Elasticity, 5) Durability.

5. Define Industrial timber and its types.

Industrial Timber is defined as the timber prepared in a factory with specification and having shape, appearance, strength etc. 1. Veneers 2.Plywoods 3.Fibre Boards 4.Thermacole 5.Panels of laminates.

6. Enumerate the various market forms of timber.

1) Board, 2) Batten, 3) Plank, 4) Log, 5) Deal, 6) End, 7) Baulk, 8) Scantling, 9) Quartering.

7. Define veneers and its advantages.

Veneers are thin sheets or slices of wood of superior quality and its thickness varies from 0.4mm to 0.6mm.The maximum thickness of the veneer is limited to 1mm.Depending on the cutting process, the veneers are classified as 1.Rotary veneers 2.Sliced veneers. Advantages: It provide a natural tooth appearance. Gum tissue tolerates porcelain well. Porcelain **veneers** are stain resistant.

8. Define plywood.

An odd number of layers of wood or veneers are arranged and glued by using glues under pressure is called plywood.

9. What is meant by thermacole and its advantages.

Thermal insulation is the process of restricting the transfer of heat from warmer to cooler areas. It is a light and cellular plastic material used for sound and heat insulation. Advantages:- 1) Fire proof, 2) Impermeable to water, 3) Heat and sound insulation.

10. What are the panels of laminates?

Laminated boards have a core of strips, each not exceeding 7mm in thickness. The edges are glued together to form a solid sheet, which is then glued between two or more outer veneers.

11. What is meant by Steel and its defects?

The steel is a ferrous material containing a maximum of 1.5% of carbon in its composition.Steel is harder and tougher than iron, because its carbon chemically combines with complete iron, exist in its free state. Defects of steel:-1. Checks 2.Segregation 3.Blow Holes 4.Cold Shortness

12. Explain the process of manufacturing of steel.

1. Bessemer process 2.Open Hearth process 3.Crucible process 4.Duplex process 5.Cementationprocess 6.Electric smelting process 7.Lintz and Donawitz process.

13. What are the standard size of bars as per IS code?

Standard sizes (in nominal diameters): 10mm, 12mm, 16mm, 20mm, 25mm, 28mm, 32mm, and 36mm. Larger sizes such as 40mm and 50mm bars may be produced via special order. For steel bars, the terms "size" and "nominal diameter" are used interchangeably.

14. Name of the market forms of steel.

1. Angle section 2.Channel section 3.T-section 4.I-section 5.Plates 6.Tor steel.7.Expanded metal.

15. Define annealing.

Annealing is the process of heating and slow cooling of metal, glass or any other material, which has developed strain due to rapid cooling.

16. Define Aluminum.

Aluminum is a silvery white, soft, ductile metal and it is the third most important metal in the earth crust. It has low density and high ability to resist corrosion.

17. Define Aluminium composites and its applications.

Aluminium composite panel (ACP) also aluminum composite material, (ACM) is a type of flat panel that consists of two thin aluminum sheets bonded to a non-aluminium core. ACPs are frequently used for external cladding or facades of buildings, insulation, robots, high speed machinery, high-speed rotating shafts, and automotive engine and brake parts.

18. Define paint.

Paint is a surface coating liquid used to protect, color or to provide texture to objects. Paint can be made in many colors and in many different types, such as water colors, artificial etc

19. Write the various constituents of enamel paints.

Most enamel paints are alkyd resin based. Some enamel paints have been made by adding varnish to oil-based paint. They have since been superseded by new synthetic coatings like alkyd, acrylic and vinyl, due to toxicity, safety, and conservation concerns.

20. State the important consideration to be made in choosing paints.

1) Light source, 2) Room orientation,3) Surfaces, 4) Paint finishes, 5) Placement of color

21. What are the components of paint and types of paints?

Components: 1. Vehicle or binder 2.Pigments 3.Solvents or Thinners 4.Plasticizers 5.Adultrants.

Types: 1. Oil paints 2.Bituminous paints 3.Water paints 4.Cement Paints 5.Special paints

22. What is meant by varnish?

Varnish is a solution of some resinous substance in alcohol, oil or turpentine. The process of covering the surface with varnish is known as varnishing. Varnishing is done on wooden surface. **Constituents:** 1. Resinous material2. Driers3. Solvents. **Types:** 1.Oil varnish 2.Spar varnish 3.Flat varnish 4.Spirit varnish 5.Asphalt varnish

23. Write down the functions of solvent.

In paints, solvents dissolve or disperse different components used in the paintformulation (such as pigment and resin), making paint the desired consistency for application. Once paint is applied, the solvent evaporates, allowing resin and pigment to produce a film of paint (a coat) and dry rapidly.

24. What are the exterior paints? Explain their use in buildings?

Exterior paints have solvents and more pigments Exterior paint is exposed to all types of varying weather conditions. Exterior paint is therefore required to provide protection against Ultra Violet radiation of the sun as well as fungal growth and of course dampness during the rains and snow.

25. What is meant by distempering?

Distemper is the mixture of pigment, suspended in a liquid used as a protective or decorative coating of surface it dries to form a hard coating. It is made with base as white chalk and thinner as water.

26. What are the necessity of bituminous paint?

Bitumen is oil based, high viscous, semi-solid material produced by removing the lighter materials from heavy crude oil during the refining process.

27. List out the various test on bitumen.

1**)** Penetration **test., 2)** Ductility **test**, 3) Softening point **test**, 4) Specific gravity **test**.
5) Viscosity **test**., 6) Flash and Fire point **test**., 7) Float **test**., 8) Water content **test**.

UNIT V MODERN MATERIALS

1. What is Glass?

Glass is a hard, brittle, amorphous, transparent inorganic material, which is mad from the pure iron fine quartz sand or crushed quartzite rock.

2. State the Composition of Glass.

Glass is not a single material and which is formed by various constituents. The general formula of glass is as follows.

$aX_2O,bYO.6SiO2$

Where**1)a** and **b –** Number of molecules**.**

2) X-alkali metal atom (Na, K, etc),**3)Y-**bivalent metal atom (ca, Pb, etc).

3. State the types of glass.

Glasses are classified based on the properties and uses and are as follows.

1. Soda-Lime Glass
2. Potash-Lime Glass
3. Potash-Lime Glass
4. Boro-Silicate Glass
5. Common Glass.

4. Explain the procedure adopted in the manufacture of glass.

1. Raw materials collection
2. Batch Preparation
3. Melting
4. Fabrication 5.Annealing

5. Name the coloring agents (decolourisers) used in the manufacturing of glass.

1. Antimony oxide
2. Arsenic oxide
3. Manganese Dioxide
4. Cobalt Oxide
5. Nickel Oxide

6. Define Batch or Frit.

The raw materials, cullet and decolourises are weighed, and mixed in correct proportions and finely powdered in grinding units. The uniform mixture of the raw materials of glass is called **Batch** or **Frit**.

7. What is fabrication in glass manufacturing?

Fabrication is carried out to shape (to form) the glass from the molten glass, operated manually or mechanically. **Manual Fabrication** (Hand Fabrication) is employed for small scale production and **Mechanical Fabrication**(Machine Fabrication) is employed for large scale production,various methods of fabrication of glass1. Blowing 2. Casting 3.Drawing 4. Pressing 5. Rolling 6. Spinning 7.Annealing.

8. Define Sealant and its applications.

Sealant is a substance used to block the passage of fluids through the surface or joints or openings in materials, a type of mechanical seal. In building construction sealants is sometimes synonymous with caulking and also serve the purposes of blocking dust, sound and heat transmission, repair of large cracks, joints in exterior walls etc..

9. What are the characteristics of good sealant?

1. It should be always in shear in lap joints.
2. It should not be affected by temperature.
3. It should be cheap in cost.
4. It should be always in tension.

10. What are ceramics?

Ceramics are defined as the products or articles or materials, which are produced from the products of heating the earthly materials. Following are the general classification of ceramics as follows. **1. Clay products 2.Refractories 3. Glass**

11. What are clay tiles?

Clay tiles are thin slabs of low melting clays, used for various purposes in engineering constructions. These give a very pleasing appearance and good service properties. Roofing tiles, flooring tiles, wall tiles and partition tiles are some of the examples.

12. What are the characteristics of good floor tile?

1) Durability: Ceramic tile is more durable as compared to other types of tiles that are used as floor and wall tiles. **2)** Strength: It has a high strength at high temperature. **3)** Dirt Resistance**. 4)** Colour Permanence**. 5)** Slip Resistance: **6)** Fire Resistance: **7)** Heat Resistance

13. Define refractories.

Refractories are the materials that can with stand high temperatures and used in high temperature furnaces. In the use of refractory materials, there are two important principles involved, 1.Thermal **insulation 2.Heat conduction**

14. Write the properties of refractories.

1) Apparent Porosity. 2) Thermal Shock Resistance. 3) Reversible Thermal Expansion and Permanent Linear Change. 4) Thermal Conductivity. 5) Abrasion Resistance

15. What is meant FRP and its uses?

Fibre glass reinforced plastic (FRP) is defined as the composite material formed by two different materials having different properties. FRPs are used in prestressed concrete**,** high-performance hybrid structures.F RPs are employed for seismic retrofitting.

16. Mention the applications of Fibre glass reinforced plastic?

1) Fountains and aquariums:
2) Cooling towers:
3) Beverage industry
4) Automotive industry
5) Road bridges

17. Define Composite materials.

Composite materials are defined as the combination of two or more material having the different properties, differ from those of the individual materials of the composite. The properties of composite materials are always better than any one of the individual materials of the composite.

18. What is Fibre Reinforced Composite?

Fibre Reinforced Composites are high strength, low-density, fibre and matrix materials and in which the disperse phase is in the form of the fibre.

19. What are Laminar Composites?

Laminar Composites are defined as the composite materials. In which the layers of two dimensional sheets or panels cemented together, such that the orientation of the high strength direction varies with each successive layer.

20. Define Fibre textiles.

It is defined as the hair like substances that are very small in diameter in relation to their length. It is the important raw material to produce various types of finished textile products.

21. What is Geomembrane?

A **Geomembrane** is very low permeability-synthetic membrane liner which is used with any geotechnical engineering material, in order to control fluid migration in a man-made project, structure or system.

22. What are the applications of Geotextiles for earth reinforcement?

Geotextiles are employed in the following civil Engineering works. **1**. Road works **2**.Railway works **3**.Rivr canals and coastal works **4**.Drainage **5**.Sports field construction.

23. What are the forms of sealants?

1) Acrylic resins. 2) Butyl rubber.3) Epoxy thermosets. 4) Expanding foam.5) Hot wax.
6) Latex sealants. 7) Metal sealants. 8) Paint sealants.

24. What are the different forms of adhesives?

1) Epoxy resin. 2) PVA (polyvinyl acetate) 3) White Craft Glue. 4) Yellow Wood Glue.
5) Hot glue. 6) Spray adhesives. 7) Fabric adhesives.

UNIT 1 IMPORTANT 16 MARK QUESTION

<table>
<tr><th colspan="4">PART – B</th></tr>
<tr><td>1.</td><td>(i) What are the basic principles of surveying? Describe it.
(ii) Discuss about the different sources of error in chain surveying.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>2.</td><td>(i) Describe the field and office work in chain surveying?
(ii) Examine how you will conduct chain survey to measure a land in agriculture field.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>3.</td><td>(i) Describe the methods of ranging by using a line ranger.
(ii) Show the different methods of overcoming difficulties if there are obstacles in chaining and ranging both.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>4.</td><td>Explain the methods of chaining with neat sketches. While you do chaining to overcome obstacles for chaining and not for ranging?</td><td>BT-4</td><td>Analyse</td></tr>
<tr><td>5.</td><td>(i) Prepare a list of accessories required for a chain survey? Explain the functions of each.
(ii) With a simple sketch, state the construction and use of a cross staff.</td><td>BT-5</td><td>Evaluate</td></tr>
<tr><td>6.</td><td>On a closed compass traverse survey PQRST, following are the observation made with a suspicion of local attraction.
<table>
<tr><th>Line</th><th>Fore Bearing</th><th>Back Bearing</th></tr>
<tr><td>PQ</td><td>147° 3</td><td>26°45’</td></tr>
<tr><td>QR</td><td>74° 30’</td><td>253° 00’</td></tr>
<tr><td>RS</td><td>41° 30’</td><td>222° 45’</td></tr>
<tr><td>ST</td><td>312° 15’</td><td>132° 45’</td></tr>
<tr><td>TP</td><td>219° 15’</td><td>39° 15’</td></tr>
</table>
Identify the station affected with local attraction, included angles and the corrected bearings.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>7.</td><td>Evaluate the working principle of prismatic compass in detail.</td><td>BT-5</td><td>Evaluate</td></tr>
<tr><td>8.</td><td>Show which stations are affected by local attraction. Workout the correct bearing of closed traverse ABCDEA.
<table>
<tr><th>Line</th><th>Fore Bearing</th><th>Back Bearing</th></tr>
<tr><td>AB</td><td>190° 30’</td><td>17°0’</td></tr>
<tr><td>BC</td><td>73°30’</td><td>250° 30’</td></tr>
<tr><td>CD</td><td>36° 15’</td><td>214° 30’</td></tr>
<tr><td>DE</td><td>266° 45’</td><td>84° 45’</td></tr>
<tr><td>EA</td><td>234°15’</td><td>57°0’</td></tr>
</table></td><td>BT-6</td><td>Create</td></tr>
</table>

<table>
<tr><td>9.</td><td>The following staff readings were observed successively with a level, the instrument having been moved after third, sixth and eighth readings 2.228, 1.606, 0.988, 2.090, 2.864, 1.262, 0.602, 1.982, 1.044, 2.684 meters. Enter the above readings in a page of a level book and evaluate the R.L. of points if the first reading was taken with a staff held on a bench mark of 432.384 m.</td><td>BT-3</td><td>Applying</td></tr>
<tr><td>10.</td><td>The following bearings were observed with a compass. Workout the local attraction? Find the correct bearings.
<table>
<tr><td>Line</td><td>Fore Bearing</td><td>Back Bearing</td></tr>
<tr><td>AB</td><td>74° 0’</td><td>254°0’</td></tr>
<tr><td>BC</td><td>91°0’</td><td>271° 0’</td></tr>
<tr><td>CD</td><td>166° 0’</td><td>343° 0’</td></tr>
<tr><td>DE</td><td>177° 0’</td><td>0° 0’</td></tr>
<tr><td>EA</td><td>189°0’</td><td>9°0’</td></tr>
</table></td><td>BT- 3</td><td>Applying</td></tr>
<tr><td>11.</td><td>What are the different sources of error in leveling and explain them in detail.</td><td>BT-2</td><td>Understanding</td></tr>
<tr><td>12.</td><td>Describe the effects of curvature and refraction in leveling and their corrections.</td><td>BT-2</td><td>Understanding</td></tr>
<tr><td>13.</td><td>Describe the profile leveling and cross sectional leveling.</td><td>BT-4</td><td>Analyze</td></tr>
<tr><td>14.</td><td>i)Define Bench mark. Describe the different types of bench marks.
ii)Compare the rise and fall and line of collimation method in reducing leveling observation.</td><td>BT-2</td><td>Understanding</td></tr>
</table>

<table>
<tr><td colspan="4">PART-C</td></tr>
<tr><td>1.</td><td>Find out the station affected by the local attraction and work out the corrected bearing of the lines the following are the observed bearing of the lines of a traverse ABCDEA with a compass in a place where local attraction was suspected.
<table>
<tr><td>Line</td><td>AB</td><td>BC</td><td>CD</td><td>DE</td><td>EA</td></tr>
<tr><td>FB</td><td>191°45’</td><td>22°15’</td><td>22°15’</td><td>242°45’</td><td>330°15’</td></tr>
<tr><td>BB</td><td>39°30’</td><td>222°30’</td><td>200°30’</td><td>62°45’</td><td>150°45’</td></tr>
</table></td><td>BT-2</td><td>Understanding</td></tr>
<tr><td>2.</td><td>The following reading were taken with a dumpy level (a) when the instrument is midway between two pegs A and B, 100 mts apart. The staff reading on A= 3.345 m. The staff reading on B= 2.025 m. (b) when the instrument is kept very near A. The staff reading on A = 2.950 m. The staff reading on B= 2.000m. Is the instrument in adjustment or not? When the instrument is very near to A. What should be the correct reading on staff B?</td><td>BT 3</td><td>Applying</td></tr>
</table>

<table>
<tr>
<td>3.</td>
<td>Explain the method of direct and reciprocal ranging in detail. Determine the area for the following observation s by
(i) Mid Ordinate rule
(ii) Average Ordinate rule
(iii) Trapezoidal rule
<table>
<tr><td>Ordinate</td><td>O1</td><td>O2</td><td>O3</td><td>O4</td><td>O5</td><td>O6</td><td>O7</td><td>O8</td><td>O9</td></tr>
<tr><td>Distance(m)</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td></tr>
<tr><td>Offset(m)</td><td>3.25</td><td>5.60</td><td>4.20</td><td>6.65</td><td>8.75</td><td>6.20</td><td>3.25</td><td>4.20</td><td>5.65</td></tr>
</table>
</td>
<td>BT-6</td>
<td>Create</td>
</tr>
<tr>
<td>4.</td>
<td>Explain the working of a dumpy level with a sketch. What are the various temporary adjustments?</td>
<td>BT 2</td>
<td>Understanding</td>
</tr>
</table>

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UNIT 2 IMPORTANT 16 MARK QUESTION

PART-B

1.	Two observations were taken upon a vertical staff by means of a theodolite, the reduced level of its trunnion axis being 160.95. In the case of the first, the angle of elevation was 4°36' and the staff reading 0.75. In the case of second observation, the staff reading was 3.45 and the angle of elevation 5°48'. Calculate the reduced level of the staff station and its	BT - 1	Remembering
2.	<table><tr><th>Instrumentation Station</th><th>Reading in BM</th><th>Angle of elevation</th><th>Remarks</th></tr><tr><td>A</td><td>0.862</td><td>18º 36'</td><td>RL of BM=421.380m</td></tr><tr><td>B</td><td>1.222</td><td>10º 12'</td><td>Distance B=50m</td></tr></table> (i) Collect the elevation of the top of a chimney from the following data. Stations A and B and top of chimney are in the same vertical plane. ii) Describe the temporary adjustments of theodolite.	BT - 1	Remembering
3.	The lengths and bearings of lines of closed traverse ABCDE is given below. Examine the length and bearing of line EA. <table><tr><th>Line</th><th>Length, m</th><th>Bearing</th></tr><tr><td>AB</td><td>194.1</td><td>85 º 30'</td></tr><tr><td>BC</td><td>201.2</td><td>15 º 00'</td></tr><tr><td>CD</td><td>165.4</td><td>285 º 30</td></tr><tr><td>DE</td><td>172.6</td><td>195 º 30'</td></tr></table>	BT 2	Understanding
4.	Explain the different between tangential and stadia tachometry. How will you determine the stadia constants?	BT 2	Understanding
5.	What is meant by interpolation of contours? Describe the various methods used.	BT-1	Remembering
6.	Describe the indirect methods of locating contours.	BT-1	Remembering
7.	From a topograph map, the areas enclosed by contour lines for a proposed dam are given below. Estimate the volume of water that can be impounded in this location <table><tr><td>Contour:</td><td>300</td><td>305</td><td>310</td><td>315</td><td>320</td></tr><tr><td>Areas enclosed (hectares):</td><td>20</td><td>110</td><td>440</td><td>930</td><td>1120</td></tr></table>	BT-2	Understanding
8.	What considerations would you have while selecting the contour interval?	BT-4	Analysing
9.	Summarize on grade contour. How will you locate it (a) on the ground (b) on the map?	BT-5	Evaluate

<table>
<tr><td>10.</td><td>An embankment of width 8 m and side slopes 1:5:1 is required to be made on a ground which is leveled in a direction transverse to the centre line. The centre height at 24 m intervals is as follows 0.80, 1.42, 1.90, 2.20, 2.65, 2.20, 2.20, 1.65, 1.30 and 0.90. Calculate the volume of earthwork to be carried out informing the embankment.</td><td>BT-3</td><td>Applying</td></tr>
<tr><td>11.</td><td>The lengths and bearings of lines of closed traverse ABCDE is given below. Examine the length and bearing of line EA.
<table><tr><td>Line</td><td>Length, m</td><td>Bearing</td></tr><tr><td>AB</td><td>194.1</td><td>85 º 30’</td></tr><tr><td>BC</td><td>201.2</td><td>15 º 00’</td></tr><tr><td>CD</td><td>165.4</td><td>285 º 30</td></tr><tr><td>DE</td><td>172.6</td><td>195 º 30’</td></tr></table></td><td>BT-4</td><td>Analysing</td></tr>
<tr><td>12.</td><td>Explain the different between tangential and stadia tachometry. How will you determine the stadia constants?</td><td>BT-3</td><td>Applying</td></tr>
<tr><td>13.</td><td>Show the expressions for horizontal and vertical distances by stadia method when the line of sight is inclined, but staff is held vertically and considering the angle of elevation.</td><td>BT – 3</td><td>Applying</td></tr>
<tr><td>14.</td><td>Recommend the various methods of horizontal angle using a theodolite.</td><td>BT - 6</td><td>Create</td></tr>
</table>

<table>
<tr><td colspan="4">PART C</td></tr>
<tr><td>1.</td><td>(i) Explain the essential parts of theodolite?
(ii) Explain the permanent adjustment of theodolite?</td><td>BT 2</td><td>Understanding</td></tr>
<tr><td>2.</td><td>Calculate the horizontal and vertical distances using tangential tacheometry, when Both the observed angles are angle of elevation and angle of depression.</td><td>BT 4</td><td>Analysing</td></tr>
<tr><td>3.</td><td>Discuss the various characteristics and uses of contours with suitable examples.</td><td>BT 2</td><td>Understanding</td></tr>
<tr><td>4.</td><td>Describe a contour line? What is the importance of contour maps in civil engineering works?</td><td>BT 4</td><td>Analysing</td></tr>
</table>

UNIT 3 IMPORTANT 16 MARK QUESTION

<table>
<tr><th colspan="4">PART B</th></tr>
<tr><td>1.</td><td>The following observations were made on a satellite station S to determine angle BAC.
<table>
<tr><td>Line</td><td>Length</td><td>Line</td><td>Bearing</td></tr>
<tr><td>SA</td><td>9.500 m</td><td>SA</td><td>0º 00'00"</td></tr>
<tr><td>AB</td><td>2950 m</td><td>SB</td><td>78º 46'00"</td></tr>
<tr><td>AC</td><td>3525 m</td><td>SC</td><td>100º 15'00"</td></tr>
</table>
Identify the angle BAC.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>2.</td><td>(i) Describe the satellite station and reduction to centre?
(ii) Show the expression for reducing the angles measured at the satellite station to centre.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>3.</td><td>Two triangulation stations A and B are 60 km apart and have elevations 240m and 280m. Identify the minimum height of signal required at B so that the line of sight may not pass the ground than 2metres.The intervening ground may be assumed to have a uniform elevation of200metres.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>4.</td><td>Examine the most probable values of the angles A, B, C from the following observations at a station P.
<table>
<tr><td>A =38° 25’20”</td><td>Weight 1</td></tr>
<tr><td>B=32° 36’12”</td><td>Weight 1</td></tr>
</table></td><td>BT-1</td><td>Remembering</td></tr>
</table>

	A+B =71° 01’29” Weight 2 A+B+C =119° 10’43” Weight 1 B+C =80° 45’ 28” Weight 2		
5.	Describe the triangulation adjustment and explain the different conditions and cases with sketches.	BT-2	Understanding
6.	(i) Explain the general principles of least squares. (ii) Explain the various cases for the determination of most probable value.	BT-2	Understanding
7.	After measuring the length of a base line, the correct length of the line is computed by applying various applicable corrections. Discuss the following corrections and provide expressions for a.Correction for temperature. b.Correction for pull. c.Correction for sag d.Correction for absolute length e.Correction for slope	BT-2	Understanding
8.	Show the most probable value of the following. A= 28º 24’27.4” B= 32º 14’16.3” C= 51º 18’18.8” A+B= 60º 38’45.6” B+C= 83º 32’28.2”	BT-3	Applying
9.	(i) How the triangulation systems are classified? (ii) Calculate sag correction for 30 m steel under a pull of 100 N in three equal spans of 10 m each. Weight of one cubic cm of steel =0.078 N. Area of cross-section of tape =0.08 sq.cm	BT-3	Applying
10.	(i)Explain the inter visibility of triangular station? (ii)A tape 20m long of standard lengthat29˚C was used to measure a line, the mean temperature during measurement being19˚C.The measured distance was 882.10 meters, the following being the slopes : 2°20’ for 100m ;4°12’ for 150m; 1°6’ for 50m; 7°48’ for 200 m; 3°00’for 300 m ; 5°10’for82.10m;Examine the true length of the line if the coefficient of expansion is 6.5 X 10^{-4}per degree F.	BT-4	Analyzing
11.	A30m steel tape was standardized on the fiat and was found to be exactly 30m under no pull at 66°F.it was used in catenary to measure a base of 5 bays. The temperature during the measurement was 92°F and the pull exerted during measurement twas100N.The area of cross-section of the tape was 8mm^2.The specific weight of steel is78.6KN/m^2.α=0.63x10^{-5}F and E=2.1 x$10^5$$N/mm^2$.Invent the true length of the tape.	BT-5	Evaluating
12.	What is base line? Explain the accessories used to measure the baseline.	BT-6	Creating

13.	The following are the observed values of the angle A with the corresponding weights. (i) 51º20'30'' Weight 2 (ii) 51º20'28'' Weight 3 (iii) 51º20'29'' Weight Examinethe following terms: (1) the standard deviation (2) the standard error of the weighted mean (3) the probable error of single observation of weight 3 (4) The probable error of the weighted mean	BT-4	Analyzing
14.	(i)Formulate the normal equations for x,y and z in the following equation of equal weight: 3x+3y+z–4 =0 x+2y+2z–6 =0 5x+y+4z–21 =0 ii)If the weights of the above equation are2,3 and1 respectively form the normal equations for x,y and z.	BT-5	Evaluating

PART C			
1.	(i)Discuss the laws of accidental errors. (ii) The following are the three angles P, Q and R observed at a station O, Closing the horizon along with standard errors. Angle P =84°15'12"± 3" Angle Q =125°13'15"±4" Angle R= 150°31'18" ±5" Predict the corrected angles.	BT-2	Understanding
2.	Write down the various laws of weight. Explain it.	BT-6	Creating
3.	i)What are the different layouts of primary triangulation for large countries? Explain it briefly. List out the criteria for selection of the layout of triangles. ii) what are signals? Classify them, enumerate the requirements to be fulfilled by signal.	BT-2	Understanding
4.	Invent the difference of levels of points P and Q and the R.L.of P from the following data:- (i) Horizontal distance between P and Q =7118 (ii) Angle of depression to P and Q =1°32'12" (iii) Height of signal at P =3.87 m (iv) Height of instrument at Q =1.27 m (v) Coefficient of Refraction = 0.07 (vi) R Sin1" =30.88m (vii) m = 0.07 (viii) R.L of Q = 417.860 m	BT-6	Creating

UNIT 4 IMPORTANT 16 MARK QUESTION

PART B			
1.	With the help of suitable sketches, describe the following methods of Locating soundings. (i) Location by range and one angle from the shore. (ii) Location by two angles from the shore.	BT-1	Remembering
2.	What is a three point problem in hydrographic surveying? List the Various solutions for the problem? Explain in detail.	BT-1	Remembering
3.	Describe briefly the different methods of prediction of tides.	BT-1	Remembering
4.	Enumerate and explain the relationships between the coordinates of celestial sphere.	BT-1	Remembering

5.	Estimate the hour angle and declination of a star from the following data. Altitude of the star = 21° 30' Azimuth of the star = 140° E Latitude of the observer = 48° N.	BT-2	Understanding
6.	(i)Summarize briefly the procedures for setting out compound curve. (ii) How reconnaissance survey is conducted for railway project?	BT-2	Understanding
7.	i) Explain the correction needed for observation to sun to determine the azimuth of a line.(5) ii) Explain about Nautical Almanac .(5)	BT-2	Understanding
8.	(i) Classify the different types of tides? Explain any two. (ii) Explain the various sounding methods.	BT-3	Applying
9.	A, B and C are three visible stations in a hydrographical survey. The computed sides of the triangle ABC are: AB, 1130 m ; BC, 1372 m ; CA, 1889 m. Outside this triangle (and nearer to AC), a station P is established and its position is to be found by three point intersection on A, B and C, the angles APB and BPC being respectively $42^o35'$and$54^o20'$.Calculate the distances PA and PC.	BT-3	Applying
10.	Explain the Tilt Distortion with neat sketch in Photographic method.	BT-4	Analyzing
11.	Calculate the Sun's Azimuth and Hour angle at sunset at a place in Latitude 52°N, When its Declination is (i) 20°N and (ii) 14°S	BT-4	Analyzing
12.	Interpret about the celestial coordinate system.	BT-5	Evaluating
13.	Conclude about the following i) Equation of time ii) Sidereal time	BT-5	Evaluating
14.	Infer about equation of time? Show that it vanishes four times a year.	BT-6	Creating

PART C			
1.	i) Explain clearly how would you determine the levels at river bed points and fix the position of sounding by use of sextant in a boat.(8) ii) Derive the expression for correction of refraction to be applied to the observed or apparent altitudes of celestial body.(8)	BT-2	Understanding
2.	Find the hour angle and declination of star from the following data : Altitude of the star 22º 30' Altitude of the star 145º E Latitude of the observer 49º N.	BT-5	Evaluating
3.	Enumerate the various investigations involved in hydrographic Surveying.	BT-6	Creating
4.	Briefly explain the applications of remote sensing.	BT-2	Understanding

UNIT 5 IMPORTANT 16 MARK QUESTION

PART B			
1.	What are the space, control and user segments of GPS and their functions?	BT-1	Remembering
2.	List out the various measurements of GPS. Explain them.	BT-1	Remembering
3.	Briefly explain the Characteristics of GPS Navigation and Satellite navigation?	BT-1	Remembering
4.	List out the features of total station and merits and demerits of total station.	BT-1	Remembering
5.	(i) What are the types of GPS receivers? (ii) Explain the task of control segment in GPS	BT-2	Understanding
6.	(i)Describe briefly about sources of errors in GPS (ii)Explain the hand held receiver and geodetic receiver of GPS.	BT-2	Understanding
7.	Discuss in detail the phase and pulse method.	BT-2	Understanding
8.	Classify the main components of GPS receiver and explain them briefly	BT-3	Applying

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9.	Illustrate the fundamental measurement system of total station.	BT-3	Applying
10.	Distinguish between single frequency receivers and Double frequency receivers.	BT-4	Analyzing
11.	Describe in detail about Anti-spoofing and Selective Availability	BT-4	Analyzing
12.	Summarize the Requirements of GPS Signals and explain in brief.	BT-5	Evaluating
13.	Write down the steps involved in GPS data processing.	BT-5	Evaluating
14.	Explain the orbit determination and representation.	BT-6	Creating

PART C			
1.	Explain the various types of GPS devices and their uses.	BT-2	Understanding
2.	Write an Essay on Future of GPS Tracking Systems.	BT-3	Applying
3.	Discuss the emerging Trends in GPS Technology.	BT-5	Evaluating
4.	Elaborate in detail the History of GPS and Technical Specifications of its Orbits.	BT-6	Creating

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VALLIAMMAI ENGINEERING COLLEGE

SRM Nagar, Kattankulathur – 603 203.

DEPARTMENT OF CIVIL ENGINEERING

QUESTION BANK

SUBJECT : CE8351-SURVEYING **SEM / YEAR: III/II**

UNIT 1- FUNDAMENTALS OF CONVENTIONAL SURVEYING AND LEVELLING

Classifications and basic principles of surveying - Equipment and accessories for ranging and chaining - Methods of ranging - Compass - Types of Compass - Basic Principles- Bearing – Types - True Bearing - Magnetic Bearing - Levelling- Principles and theory of Levelling – Datum- - Bench Marks – Temporary and Permanent Adjustments- Methods of Levelling- Booking – Reduction - Sources of errors in Levelling - Curvature and refraction.

PART - A

Q.No	Questions	BT	Competence
1.	What is meant by digital planimeter?	BT-1	Remembering
2.	List the different cumulative errors in chain surveying.	BT-1	Remembering
3.	Tell the length of one link in engineer's chain and Gunter's chain.	BT-1	Remembering
4.	Name the instruments/accessories needed to carry out a chain surveying.	BT-1	Remembering
5.	List the errors that will occur when the distance between the two points is measured using tape.	BT-1	Remembering
6.	Discuss ranging. Mention its types.	BT-2	Remembering
7.	What is the use of trough compass in plane table surveying?	BT-4	Remembering
8.	Distinguish between Magnetic Dip and Declination.	BT-4	Remembering
9.	Why Eastern and Western ends are interchanged in Surveyor's compass.	BT-5	Remembering
10.	Distinguish between true bearing and magnetic bearing.	BT-2	Understanding
11.	Discuss the term bearings and its types.	BT-3	Applying
12.	Calculate the reduced bearing values for the whole circle bearings 150° and 270°.	BT-3	Applying
13.	Change the following whole circle bearing to reduced bearing: (a) 151° 20' (b) 332° 40'.	BT–3	Applying
14.	What is meant by sensitivity of level tube?	BT-2	Remembering
15.	List the different types of leveling staves.	BT-5	Remembering
16.	List out the types of levelling instruments.	BT-2	Remembering
17.	Write about datum.	BT-6	Remembering

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18.	Define Benchmark and give its types.	BT-4	Remembering
19.	Evaluate is reduction in leveling?Name the methods.	BT-6	Remembering
20.	Calculate the combined correction for value 6000m &9000m.	BT-3	Applying

PART – B

1.	(i) What are the basic principles of surveying? Describe it. (ii) Discuss about the different sources of error in chain surveying.	BT-1	Remembering
2.	(i) Describe the field and office work in chain surveying? (ii) Examine how you will conduct chain survey to measure a land in agriculture field.	BT-1	Remembering
3.	(i) Describe the methods of ranging by using a line ranger. (ii) Show the different methods of overcoming difficulties if there are obstacles in chaining and ranging both.	BT-1	Remembering
4.	Explain the methods of chaining with neat sketches. While you do chaining to overcome obstacles for chaining and not for ranging?	BT-4	Analyse
5.	(i) Prepare a list of accessories required for a chain survey? Explain the functions of each. (ii) With a simple sketch, state the construction and use of a cross staff.	BT-5	Evaluate
6.	On a closed compass traverse survey PQRST, following are the observation made with a suspicion of local attraction. Line / Fore Bearing / Back Bearing PQ / 147° 3 / 26°45’ QR / 74° 30’ / 253° 00’ RS / 41° 30’ / 222° 45’ ST / 312° 15’ / 132° 45’ TP / 219° 15’ / 39° 15’ Identify the station affected with local attraction, included angles and the corrected bearings.	BT-1	Remembering
7.	Evaluate the working principle of prismatic compass in detail.	BT-5	Evaluate
8.	Show which stations are affected by local attraction. Workout the correct bearing of closed traverse ABCDEA. Line / Fore Bearing / Back Bearing AB / 190° 30’ / 17°0’ BC / 73°30’ / 250° 30’ CD / 36° 15’ / 214° 30’ DE / 266° 45’ / 84° 45’ EA / 234°15’ / 57°0’	BT-6	Create

<table>
<tr><td>9.</td><td>The following staff readings were observed successively with a level, the instrument having been moved after third, sixth and eighth readings 2.228, 1.606, 0.988, 2.090, 2.864, 1.262, 0.602, 1.982, 1.044, 2.684 meters. Enter the above readings in a page of a level book and evaluate the R.L. of points if the first reading was taken with a staff held on a bench mark of 432.384 m.</td><td>BT-3</td><td>Applying</td></tr>
<tr><td>10.</td><td>The following bearings were observed with a compass. Workout the local attraction? Find the correct bearings.
<table>
<tr><td>Line</td><td>Fore Bearing</td><td>Back Bearing</td></tr>
<tr><td>AB</td><td>74° 0’</td><td>254°0’</td></tr>
<tr><td>BC</td><td>91°0’</td><td>271° 0’</td></tr>
<tr><td>CD</td><td>166° 0’</td><td>343° 0’</td></tr>
<tr><td>DE</td><td>177° 0’</td><td>0° 0’</td></tr>
<tr><td>EA</td><td>189°0’</td><td>9°0’</td></tr>
</table></td><td>BT- 3</td><td>Applying</td></tr>
<tr><td>11.</td><td>What are the different sources of error in leveling and explain them in detail.</td><td>BT-2</td><td>Understanding</td></tr>
<tr><td>12.</td><td>Describe the effects of curvature and refraction in leveling and their corrections.</td><td>BT-2</td><td>Understanding</td></tr>
<tr><td>13.</td><td>Describe the profile leveling and cross sectional leveling.</td><td>BT-4</td><td>Analyze</td></tr>
<tr><td>14.</td><td>i)Define Bench mark. Describe the different types of bench marks.
ii)Compare the rise and fall and line of collimation method in reducing leveling observation.</td><td>BT-2</td><td>Understanding</td></tr>
</table>

<table>
<tr><td colspan="4">PART-C</td></tr>
<tr><td>1.</td><td>Find out the station affected by the local attraction and work out the corrected bearing of the lines the following are the observed bearing of the lines of a traverse ABCDEA with a compass in a place where local attraction was suspected.
<table>
<tr><td>Line</td><td>AB</td><td>BC</td><td>CD</td><td>DE</td><td>EA</td></tr>
<tr><td>FB</td><td>191°45’</td><td>22°15’</td><td>22°15’</td><td>242°45’</td><td>330°15’</td></tr>
<tr><td>BB</td><td>39°30’</td><td>222°30’</td><td>200°30’</td><td>62°45’</td><td>150°45’</td></tr>
</table></td><td>BT-2</td><td>Understanding</td></tr>
<tr><td>2.</td><td>The following reading were taken with a dumpy level (a) when the instrument is midway between two pegs A and B, 100 mts apart. The staff reading on A= 3.345 m. The staff reading on B= 2.025 m. (b) when the instrument is kept very near A. The staff reading on A = 2.950 m. The staff reading on B= 2.000m. Is the instrument in adjustment or not? When the instrument is very near to A. What should be the correct reading on staff B?</td><td>BT 3</td><td>Applying</td></tr>
</table>

<table>
<tr><td>3.</td><td>Explain the method of direct and reciprocal ranging in detail. Determine the area for the following observation s by
(i) Mid Ordinate rule
(ii) Average Ordinate rule
(iii) Trapezoidal rule
<table>
<tr><td>Ordinate</td><td>O1</td><td>O2</td><td>O3</td><td>O4</td><td>O5</td><td>O6</td><td>O7</td><td>O8</td><td>O9</td></tr>
<tr><td>Distance(m)</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td></tr>
<tr><td>Offset(m)</td><td>3.25</td><td>5.60</td><td>4.20</td><td>6.65</td><td>8.75</td><td>6.20</td><td>3.25</td><td>4.20</td><td>5.65</td></tr>
</table></td><td>BT-6</td><td>Create</td></tr>
<tr><td>4.</td><td>Explain the working of a dumpy level with a sketch. What are the various temporary adjustments?</td><td>BT 2</td><td>Understanding</td></tr>
</table>

UNIT 2- THEODOLITE AND TACHEOMETRIC SURVEYING

Horizontal and vertical angle measurements - Temporary and permanent adjustments - Heights and distances - Tacheometer - Stadia Constants - Analytic Lens -Tangential and Stadia Tacheometry surveying - Contour – Contouring – Characteristics of contours – Methods of contouring – Tacheometric contouring - Contour gradient – Uses of contour plan and map

Q.No	Questions	BT	Competence
1.	What is meant by balancing in theodolite surveying?	BT-1	Remembering
2.	Define transit.	BT-1	Remembering
3.	List out the steps in temporary adjustments of theodolite surveying.	BT-1	Remembering
4.	What is contour? State the uses of contour.	BT-1	Remembering
5.	Label the fundamental axis of Theodolite.	BT-1	Remembering
6.	List the different field works to be carried out in theodolite traversing.	BT-1	Remembering
7.	Differentiate between latitude and departure.	BT-2	Understanding
8.	Describe about anallactic lens	BT-2	Understanding
9.	Summarize the methods of interpolating contours.	BT-2	Understanding
10.	Discuss few points on contour gradient.	BT-2	Understanding
11.	Discover the reason for taking face left and face right observations.	BT-3	Applying
12.	Illustrate the location and function of plate bubble in a theodolite.	BT-3	Applying
13.	Illustrate the contour lines for hill and a depression	BT-3	Applying
14.	What do you infer about centring of a theodolite?	BT-4	Analysing
15.	Compare theodolite and tacheometer.	BT-4	Analysing
16.	How will you analyze the capacity of the reservoir.	BT-4	Analysing
17.	What if the contour line crosses a valley at right angles or a stream?	BT-5	Evaluate
18.	List the errors which are eliminated in measurement of horizontal angles by repetition method.	BT-5	Evaluate
19.	Summarize the principle of stadia method.	BT-6	Create
20.	Summarize the methods of locating contouring	BT-6	Create

<table>
<tr><th colspan="4">PART-B</th></tr>
<tr><td>1.</td><td>Two observations were taken upon a vertical staff by means of a theodolite, the reduced level of its trunnion axis being 160.95. In the case of the first, the angle of elevation was 4°36’ and the staff reading 0.75. In the case of second observation, the staff reading was 3.45 and the angle of elevation 5°48’. Calculate the reduced level of the staff station and its</td><td>BT - 1</td><td>Remembering</td></tr>
<tr><td>2.</td><td>
<table>
<tr><td>Instrumentation Station</td><td>Reading in BM</td><td>Angle of elevation</td><td>Remarks</td></tr>
<tr><td>A</td><td>0.862</td><td>18º 36’</td><td>RL of BM=421.380m</td></tr>
<tr><td>B</td><td>1.222</td><td>10º 12’</td><td>Distance B=50m</td></tr>
</table>
(i) Collect the elevation of the top of a chimney from the following data.

Stations A and B and top of chimney are in the same vertical plane.
ii) Describe the temporary adjustments of theodolite.</td><td>BT - 1</td><td>Remembering</td></tr>
<tr><td>3.</td><td>The lengths and bearings of lines of closed traverse ABCDE is given below. Examine the length and bearing of line EA.
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<tr><td>Line</td><td>Length, m</td><td>Bearing</td></tr>
<tr><td>AB</td><td>194.1</td><td>85 º 30’</td></tr>
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<tr><td>6.</td><td>Describe the indirect methods of locating contours.</td><td>BT-1</td><td>Remembering</td></tr>
<tr><td>7.</td><td>From a topograph map, the areas enclosed by contour lines for a proposed dam are given below. Estimate the volume of water that can be impounded in this location
<table>
<tr><td>Contour:</td><td>300</td><td>305</td><td>310</td><td>315</td><td>320</td></tr>
<tr><td>Areas enclosed (hectares):</td><td>20</td><td>110</td><td>440</td><td>930</td><td>1120</td></tr>
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<tr><td>9.</td><td>Summarize on grade contour. How will you locate it
(a) on the ground
(b) on the map?</td><td>BT-5</td><td>Evaluate</td></tr>
</table>

10.	An embankment of width 8 m and side slopes 1:5:1 is required to be made on a ground which is leveled in a direction transverse to the centre line. The centre height at 24 m intervals is as follows 0.80, 1.42, 1.90, 2.20, 2.65, 2.20, 2.20, 1.65, 1.30 and 0.90. Calculate the volume of earthwork to be carried out informing the embankment.	BT-3	Applying
11.	The lengths and bearings of lines of closed traverse ABCDE is given below. Examine the length and bearing of line EA. Line / Length, m / Bearing AB / 194.1 / 85 º 30' BC / 201.2 / 15 º 00' CD / 165.4 / 285 º 30 DE / 172.6 / 195 º 30'	BT-4	Analysing
12.	Explain the different between tangential and stadia tachometry. How will you determine the stadia constants?	BT-3	Applying
13.	Show the expressions for horizontal and vertical distances by stadia method when the line of sight is inclined, but staff is held vertically and considering the angle of elevation.	BT – 3	Applying
14.	Recommend the various methods of horizontal angle using a theodolite.	BT - 6	Create

PART C			
1.	(i) Explain the essential parts of theodolite? (ii) Explain the permanent adjustment of theodolite?	BT 2	Understanding
2.	Calculate the horizontal and vertical distances using tangential tacheometry, when Both the observed angles are angle of elevation and angle of depression.	BT 4	Analysing
3.	Discuss the various characteristics and uses of contours with suitable examples.	BT 2	Understanding
4.	Describe a contour line? What is the importance of contour maps in civil engineering works?	BT 4	Analysing

UNIT 3- CONTROL SURVEYING AND ADJUSTMENT

Horizontal and vertical control – Methods – specifications – triangulation- baseline – satellite stations – reduction to centre- trigonometrical levelling – single and reciprocal observations – traversing – Gale's table. -Errors Sources- precautions and corrections – classification of errors – true and most probable values - weighed observations – method of equal shifts –principle of least squares - normal equation – correlates- level nets- adjustment of simple triangulation networks.

PART A

1.	Name the equipments used for base line measurement.	BT-1	Remembering
2.	Quote the applications of Gale's table.	BT-1	Remembering
3.	Describe the satellite station and reduction to centre	BT-1	Remembering

4.	Define most probable value	BT-1	Remembering
5.	State the principle of least square.	BT-1	Remembering
6.	Define correlates	BT-1	Remembering
7.	Distinguish the sag correction and temperature correction	BT-2	Understanding
8.	Summarize the specifications of first order triangulation.	BT-2	Understanding
9.	Distinguish between the observed value and the most probable value of a quantity.	BT-2	Understanding
10.	Discuss the conditioned quantity.	BT-2	Understanding
11.	How to apply the figure adjustment in triangulation?	BT-3	Applying
12.	How to apply the figure adjustment in triangulation?	BT-3	Applying
13.	When do you apply the method of equal shift?	BT-3	Applying
14.	Explain in detail about the weight of an observation	BT-4	Analyzing
15.	Explain normal equations.	BT-4	Analyzing
16.	Compare the Horizontal and Vertical controls in hydrographic surveying.	BT-4	Analyzing
17.	Design a well conditioned triangle.	BT-5	Evaluating
18.	How do you prepare the figure adjustments in triangulation?	BT-5	Evaluating
19.	Compare the systematic and accidental errors.	BT-6	Creating
20.	When would you recommend the triangulation system in surveying projects?	BT-6	Creating

PART B

1.	The following observations were made on a satellite station S to determine angle BAC. Line | Length | Line | Bearing SA | 9.500 m | SA | 0º 00'00" AB | 2950 m | SB | 78º 46'00" AC | 3525 m | SC | 100º 15'00" Identify the angle BAC.	BT-1	Remembering
2.	(i) Describe the satellite station and reduction to centre? (ii) Show the expression for reducing the angles measured at the satellite station to centre.	BT-1	Remembering
3.	Two triangulation stations A and B are 60 km apart and have elevations 240m and 280m. Identify the minimum height of signal required at B so that the line of sight may not pass the ground than 2metres.The intervening ground may be assumed to have a uniform elevation of200metres.	BT-1	Remembering
4.	Examine the most probable values of the angles A, B, C from the following observations at a station P. A =38° 25'20" | Weight 1 B=32° 36'12" | Weight 1	BT-1	Remembering

	A+B =71° 01’29” Weight 2 A+B+C =119° 10’43” Weight 1 B+C =80° 45’ 28” Weight 2		
5.	Describe the triangulation adjustment and explain the different conditions and cases with sketches.	BT-2	Understanding
6.	(i) Explain the general principles of least squares. (ii) Explain the various cases for the determination of most probable value.	BT-2	Understanding
7.	After measuring the length of a base line, the correct length of the line is computed by applying various applicable corrections. Discuss the following corrections and provide expressions for a.Correction for temperature. b.Correction for pull. c.Correction for sag d.Correction for absolute length e.Correction for slope	BT-2	Understanding
8.	Show the most probable value of the following. A= 28º 24’27.4” B= 32º 14’16.3” C= 51º 18’18.8” A+B= 60º 38’45.6” B+C= 83º 32’28.2”	BT-3	Applying
9.	(i) How the triangulation systems are classified? (ii) Calculate sag correction for 30 m steel under a pull of 100 N in three equal spans of 10 m each. Weight of one cubic cm of steel =0.078 N. Area of cross-section of tape =0.08 sq.cm	BT-3	Applying
10.	(i)Explain the inter visibility of triangular station? (ii)A tape 20m long of standard lengthat29˚C was used to measure a line, the mean temperature during measurement being19˚C.The measured distance was 882.10 meters, the following being the slopes : 2°20’ for 100m ;4°12’ for 150m; 1°6’ for 50m; 7°48’ for 200 m; 3°00’for 300 m ; 5°10’for82.10m;Examine the true length of the line if the coefficient of expansion is 6.5 X 10^{-4}per degree F.	BT-4	Analyzing
11.	A30m steel tape was standardized on the fiat and was found to be exactly 30m under no pull at 66°F.it was used in catenary to measure a base of 5 bays. The temperature during the measurement was 92°F and the pull exerted during measurement twas100N.The area of cross-section of the tape was 8mm^2.The specific weight of steel is78.6KN/m^2.α=0.63x10^{-5}F and E=2.1 x$10^5$$N/mm^2$.Invent the true length of the tape.	BT-5	Evaluating
12.	What is base line? Explain the accessories used to measure the baseline.	BT-6	Creating

13.	The following are the observed values of the angle A with the corresponding weights. (i) 51º20'30'' Weight 2 (ii) 51º20'28'' Weight 3 (iii) 51º20'29'' Weight Examinethe following terms: (1) the standard deviation (2) the standard error of the weighted mean (3) the probable error of single observation of weight 3 (4) The probable error of the weighted mean	BT-4	Analyzing
14.	(i)Formulate the normal equations for x,y and z in the following equation of equal weight: 3x+3y+z–4 =0 x+2y+2z–6 =0 5x+y+4z–21 =0 ii)If the weights of the above equation are2,3 and1 respectively form the normal equations for x,y and z.	BT-5	Evaluating

	PART C		
1.	(i)Discuss the laws of accidental errors. (ii) The following are the three angles P, Q and R observed at a station O, Closing the horizon along with standard errors. Angle P =84°15'12"± 3" Angle Q =125°13'15"±4" Angle R= 150°31'18" ±5" Predict the corrected angles.	BT-2	Understanding
2.	Write down the various laws of weight. Explain it.	BT-6	Creating
3.	i)What are the different layouts of primary triangulation for large countries? Explain it briefly. List out the criteria for selection of the layout of triangles. ii) what are signals? Classify them, enumerate the requirements to be fulfilled by signal.	BT-2	Understanding
4.	Invent the difference of levels of points P and Q and the R.L.of P from the following data:- (i) Horizontal distance between P and Q =7118 (ii) Angle of depression to P and Q =1°32'12" (iii) Height of signal at P =3.87 m (iv) Height of instrument at Q =1.27 m (v) Coefficient of Refraction = 0.07 (vi) R Sin1" =30.88m (vii) m = 0.07 (viii) R.L of Q = 417.860 m	BT-6	Creating

UNIT 4- ADVANCED TOPICS IN SURVEYING

Hydrographic Surveying – Tides – MSL – Sounding methods – Three point problem – Strength of fix astronomical Surveying – Field observations and determination of Azimuth by altitude and hour angle methods – .Astronomical terms and definitions - Motion of sun and stars - Celestial coordinate systems - different time systems - Nautical Almanac - Apparent altitude and corrections - Field observations and determination of time, longitude, latitude and azimuth by altitude and hour angle method

PART A

1.	What is Hydrographic Survey?	BT-1	Remembering
2.	Define Sounding.	BT-1	Remembering
3.	List the functions of transition curves.	BT-1	Remembering
4.	What is Mean sea level?	BT-1	Remembering
5.	Define the right ascension.(R.A)	BT-1	Remembering
6.	Enumerate the properties of spherical triangle.	BT-1	Remembering
7.	Distinguish between compound and reverse curves.	BT-2	Understanding
8.	Describe the Azimuth.	BT-2	Understanding
9.	Explain Nautical Almanac .	BT-2	Understanding
10.	Discuss about Celestial equator.	BT-2	Understanding
11.	Classify the different equipment's needed for soundings.	BT-3	Applying
12.	Illustrate the methods for determining the latitude of a place.	BT-3	Applying
13.	Write the corrections applied to the observed altitude of sun.	BT-3	Applying
14.	Write the equation of time	BT-4	Analyzing
15.	Differentiate between latitude and co latitude.	BT-4	Analyzing
16.	Compare the relation between right ascension and hour angle?	BT-4	Analyzing
17.	Write a short note on echo-sounding.	BT-5	Evaluating
18.	Summarize about sidereal time.	BT-5	Evaluating
19.	State three point problem in hydrographic surveying.	BT-6	Creating
20.	Conclude about lunar and solar tides.	BT-6	Creating

PART B

1.	With the help of suitable sketches, describe the following methods of Locating soundings. (i) Location by range and one angle from the shore. (ii) Location by two angles from the shore.	BT-1	Remembering
2.	What is a three point problem in hydrographic surveying? List the Various solutions for the problem? Explain in detail.	BT-1	Remembering
3.	Describe briefly the different methods of prediction of tides.	BT-1	Remembering
4.	Enumerate and explain the relationships between the coordinates of celestial sphere.	BT-1	Remembering

5.	Estimate the hour angle and declination of a star from the following data. Altitude of the star = 21° 30' Azimuth of the star = 140° E Latitude of the observer = 48° N.	BT-2	Understanding
6.	(i)Summarize briefly the procedures for setting out compound curve. (ii) How reconnaissance survey is conducted for railway project?	BT-2	Understanding
7.	i) Explain the correction needed for observation to sun to determine the azimuth of a line.(5) ii) Explain about Nautical Almanac .(5)	BT-2	Understanding
8.	(i) Classify the different types of tides? Explain any two. (ii) Explain the various sounding methods.	BT-3	Applying
9.	A, B and C are three visible stations in a hydrographical survey. The computed sides of the triangle ABC are: AB, 1130 m ; BC, 1372 m ; CA, 1889 m. Outside this triangle (and nearer to AC), a station P is established and its position is to be found by three point intersection on A, B and C, the angles APB and BPC being respectively $42^o35'$and$54^o20'$.Calculate the distances PA and PC.	BT-3	Applying
10.	Explain the Tilt Distortion with neat sketch in Photographic method.	BT-4	Analyzing
11.	Calculate the Sun's Azimuth and Hour angle at sunset at a place in Latitude 52°N, When its Declination is (i) 20°N and (ii) 14°S	BT-4	Analyzing
12.	Interpret about the celestial coordinate system.	BT-5	Evaluating
13.	Conclude about the following i) Equation of time ii) Sidereal time	BT-5	Evaluating
14.	Infer about equation of time? Show that it vanishes four times a year.	BT-6	Creating

PART C			
1.	i) Explain clearly how would you determine the levels at river bed points and fix the position of sounding by use of sextant in a boat.(8) ii) Derive the expression for correction of refraction to be applied to the observed or apparent altitudes of celestial body.(8)	BT-2	Understanding
2.	Find the hour angle and declination of star from the following data : Altitude of the star 22º 30' Altitude of the star 145º E Latitude of the observer 49º N.	BT-5	Evaluating
3.	Enumerate the various investigations involved in hydrographic Surveying.	BT-6	Creating
4.	Briefly explain the applications of remote sensing.	BT-2	Understanding

UNIT 5- MODERN SURVEYING

Total Station : Advantages - Fundamental quantities measured - Parts and accessories - working principle - On board calculations - Field procedure - Errors and Good practices in using Total Station GPS Surveying : Different segments - space, control and user segments - satellite configuration - signal structure - Orbit determination and representation - Anti Spoofing and Selective Availability - Task of control segment - Hand Held and Geodetic receivers - data processing - Traversing and triangulation

PART A			
1.	Define total station?	BT-1	Remembering
2.	List out the errors in total station.	BT-1	Remembering
3.	Define GPS	BT-1	Remembering
4.	State the basics of GPS	BT-1	Remembering
5.	List the different segments of GPS	BT-1	Remembering
6.	Define GPS data processing.	BT-1	Remembering
7.	Discuss about microwave and electro optical system adopted in total station.	BT-2	Understanding
8.	Distinguish between Space and User segment?	BT-2	Understanding
9.	Discuss few points on GPS navigation.	BT-2	Understanding
10.	Explain shortly about Orbit Representation.	BT-2	Understanding
11.	Demonstrate the components of satellite signals.	BT-3	Applying
12.	Write about anti-spoofing.	BT-3	Applying
13.	Write about selective availability?	BT-3	Applying
14.	Explain in short about Satellite Configuration.	BT-4	Analyzing
15.	State the applications of GPS.	BT-4	Analyzing
16.	Explain the term satellite configuration?	BT-4	Analyzing
17.	Write the principles of GPS.	BT-5	Evaluating
18.	How are personal GPS receivers different from survey-grade GPS Receivers? Justify.	BT-5	Evaluating
19.	Compose a short note on Task of control segment	BT-6	Creating
20.	Formulate steps for GPS data processing for establishment of control points .	BT-6	Creating

PART B			
1.	What are the space, control and user segments of GPS and their functions?	BT-1	Remembering
2.	List out the various measurements of GPS. Explain them.	BT-1	Remembering
3.	Briefly explain the Characteristics of GPS Navigation and Satellite navigation?	BT-1	Remembering
4.	List out the features of total station and merits and demerits of total station.	BT-1	Remembering
5.	(i) What are the types of GPS receivers? (ii) Explain the task of control segment in GPS	BT-2	Understanding
6.	(i)Describe briefly about sources of errors in GPS (ii)Explain the hand held receiver and geodetic receiver of GPS.	BT-2	Understanding
7.	Discuss in detail the phase and pulse method.	BT-2	Understanding
8.	Classify the main components of GPS receiver and explain them briefly	BT-3	Applying

9.	Illustrate the fundamental measurement system of total station.	BT-3	Applying
10.	Distinguish between single frequency receivers and Double frequency receivers.	BT-4	Analyzing
11.	Describe in detail about Anti-spoofing and Selective Availability	BT-4	Analyzing
12.	Summarize the Requirements of GPS Signals and explain in brief.	BT-5	Evaluating
13.	Write down the steps involved in GPS data processing.	BT-5	Evaluating
14.	Explain the orbit determination and representation.	BT-6	Creating

PART C			
1.	Explain the various types of GPS devices and their uses.	BT-2	Understanding
2.	Write an Essay on Future of GPS Tracking Systems.	BT-3	Applying
3.	Discuss the emerging Trends in GPS Technology.	BT-5	Evaluating
4.	Elaborate in detail the History of GPS and Technical Specifications of its Orbits.	BT-6	Creating

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DEPARTMENT OF CIVIL ENGINEERING

CE 8351 SURVEYING

QUESTION BANK

S.No	Unit No.		BT1	BT2	BT3	BT4	BT5	BT6	Total Question
1	Unit-1	Part-A	6	4	3	3	2	2	20
		Part-B	4	3	2	3	1	1	14
		Part-C	-	2	1	-	-	1	4
2	Unit-2	Part-A	6	4	3	3	2	2	20
		Part-B	4	3	3	2	1	1	14
		Part-C	-	2	-	-	2	-	4
3	Unit-3	Part-A	6	4	3	3	2	2	20
		Part-B	4	3	2	3	1	1	14
		Part-C	-	2	-	-	-	2	4
4	Unit-4	Part-A	6	4	3	3	2	2	20
		Part-B	4	3	3	2	1	1	14
		Part-C	-	2	1	1	-	-	4
5	Unit-5	Part-A	6	4	3	3	2	2	20
		Part-B	4	3	2	3	1	1	14
		Part-C	-	1	1	-	1	1	4

TOTAL NO.OF QUESTIONS IN EACH PART

PART A	100
PART B	70
PART C	20
TOTAL	190

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UNIT 1

STRESS, STRAIN AND DEFORMATION OF SOLIDS

1. Define stress.

When an external force acts on a body, it undergoes deformation. At the same time the body resists deformation. The magnitude of the resisting force is numerically equal to the applied force. This internal resisting force per unit area is called stress.

Stress = Force/Area

= P/A unit is N/mm^2

2. Define strain

δ Strain = Change in length/Original length

e = δL/L

3. State Hooke's law.

It states that when a material is loaded, within its elastic limit, the stress is directly proportional to the strain.

Stress α Strain

σ α e

σ = Ee

Where,

E - Young's modulus

E = σ/e unit is N/mm^2

σ - Stress

e - Strain

4. Define shear stress and shear strain.

The two equal and opposite force act tangentially on any cross sectional plane of the body tending to slide one part of the body over the other part. The stress induced is called shear stress and the corresponding strain is known as shear strain.

5. Define Poisson's ratio.

When a body is stressed, within its elastic limit, the ratio of lateral strain to the longitudinal strain is constant for a given material.

Poisson' ratio (μ or 1/m) = Lateral strain /Longitudinal strain

6. State the relationship between Young's Modulus and Modulus of Rigidity.

$$E = 2G (1+1/m)$$

Where,

E - Young's Modulus

K - Bulk Modulus

1/m - Poisson's ratio

7. Define strain energy

Whenever a body is strained, some amount of energy is absorbed in the body.The energy which is absorbed in the body due to straining effect is known as strain energy.

8. Give the relationship between Bulk Modulus and Young's Modulus.

$$E = 3K (1-2/m)$$

Where,

E - Young's Modulus

K - Bulk Modulus

1/m - Poisson's ratio

9. What is compound bar?

A composite bar composed of two or more different materials joined together such that system is elongated or compressed in a single unit.

10. Define- elastic limit

Some external force is acting on the body, the body tends to deformation. If the force is released from the body its regain to the original position. This is called elastic limit .

11. Define – Young's modulus

The ratio of stress and strain is constant within the elastic limit.

$$E = \text{Stress} / \text{Strain}$$

12. Define Bulk-modulus

The ratio of direct stress to volumetric strain.

$$K = \text{Direct stress} / \text{Volumetric strain}$$

13. Define- lateral strain

When a body is subjected to axial load P. The length of the body is increased. The axial deformation of the length of the body is called lateral strain.

13. Define- longitudinal strain

The strain right angle to the direction of the applied load is called lateral strain.

14. What is principle of super position?

The resultant deformation of the body is equal to the algebric sum of the deformation of the individual section. Such principle is called as principle of super position

15. Define- Rigidity modulus

The shear stress is directly proportional to shear strain.

N = Shear stress / Shear strain

16. State principle plane.

The planes, which have no shear stress, are known as principal planes.

These planes carry only normal stresses.

17. Define principle stresses and principle plane.

Principle stress: The magnitude of normal stress, acting on a principal plane is known as principal stresses.

Principle plane: The planes which have no shear stress are known as principal planes.

18. What does the radius of Mohr's circle refer to?

Radius of Mohr's circle is equal to the maximum shear stress.

19. What is the use of Mohr's circle?

To find out the normal, resultant stresses and principle stress and their planes.

20. List the methods to find the stresses in oblique plane?

1. Analytical method
2. Graphical method

21. Define factor of safety.

The ratio of the ultimate strength to the actual working stress

UNIT –II

TRANSFER OF LOADS AND STRESSES IN BEAMS

1. Define beam?

Beam is a structural member which is supported along the length and subjected to external loads acting transversely (i.e) perpendicular to the center line of the beam.

2. What is mean by transverse loading on beam?

If a load is acting on the beam which perpendicular to the central line of it then it is called transverse loading.

3. What is Cantilever beam?

A beam one end free and the other end is fixed is called cantilever beam.

4. What is simply supported beam?

A beam supported or resting free on the support at its both ends.

5. What is mean by over hanging beam?

If one or both of the end portions are extended beyond the support then it is called over hanging beam.

6. What is mean by concentrated loads?

A load which is acting at a point is called point load.

7. What is uniformly distributed load.

If a load which is spread over a beam in such a manner that rate of loading 'w' is uniform through out the length then it is called as udl.

8. Define point of contra flexure? In which beam it occurs?

Point at which BM changes to zero is point of contra flexure. It occurs in overhanging beam.

9. What is mean by positive or sagging BM?

BM is said to positive if moment on left side of beam is clockwise or right side of the beam is counter clockwise.

10. What is mean by negative or hogging BM?

BM is said to negative if moment on left side of beam is counterclockwise or right side of the beam is clockwise.

11. Define shear force and bending moment?

SF at any cross section is defined as algebraic sum of all the forces acting either side of beam.

BM at any cross section is defined as algebraic sum of the moments of all the forces which are placed either side from that point.

12. When will bending moment is maximum?

BM will be maximum when shear force change its sign.

13. What is maximum bending moment in a simply supported beam of span 'L'subjected to UDL of 'w' over entire span?

Max BM =wL2/8

14. In a simply supported beam how will you locate point of maximum bending moment?

The bending moment is max.When SF is zero. Write SF equation at that point and equating to zero we can find out the distances 'x' from one end .then find maximum bending moment at that point by taking all moment on right or left hand side of beam.

15. What is shear force?

The algebric sum of the vertical forces at any section of the beam to the left or right of the section is called shear force.

16. What is shear force and bending moment diagram?

It shows the variation of the shear force and bending moment along the length of the beam.

17. What are the types of beams?

1. Cantilever beam
2. Simply supported beam
3. Fixed beam
4. Continuous beam
5. over hanging beam

18. What are the types of loads?

1. Concentrated load or point load
2. Uniform distributed load
3. Uniform varying load

19. In which point the bending moment is maximum?

When the shear force change of sign or the shear force is zero

20. Write the assumption in the theory of simple bending?

1. The material of the beam is homogeneous and isotropic.

2. The beam material is stressed within the elastic limit and thus obey hooke's law.

3. The transverse section which was plane before bending remains plains after bending also.

4. Each layer of the beam is free to expand or contract independently about the layer, above or below.

5. The value of E is the same in both compression and tension.

21. Write the theory of simple bending equation?

$$M/I = F/Y = E/R$$

M - Maximum bending moment

I - Moment of inertia

F - Maximum stress induced

Y - Distance from the neutral axis

E - Young's modulus

R - Constant.

22. What is meant by section modulus?

- Section modulus is a geometric property for a given cross-section used in the design of beams or flexural members.
- The elastic section modulus is defined as $S = I / y$,

Where,

I is the moment of inertia

y is the distance from the neutral axis

UNIT –III

DEFLECTION OF BEAMS

1. What are the methods for finding out the slope and deflection at a section?

The important methods used for finding out the slope and deflection at a section in a loaded beam are

1. Double integration method
2. Moment area method
3. Macaulay's method

The first two methods are suitable for a single load, where as the last one is suitable for several loads.

2. Why moment area method is more useful, when compared with double integration?

Moment area method is more useful, as compared with double integration method because many problems which do not have a simple mathematical solution can be simplified by the ending moment area method.

3. Explain the Theorem for conjugate beam method?

Theorem I : "The slope at any section of a loaded beam, relative to the original axis of the beam is equal to the shear in the conjugate beam at the corresponding section"

Theorem II: "The deflection at any given section of a loaded beam, relative to the original position is equal to the Bending moment at the corresponding section of the conjugate beam"

4. Define method of Singularity functions?

In Macaulay's method a single equation is formed for all loading on a beam, the equation is constructed in such away that the constant of Integration apply to all portions of the beam.

This method is also called method of singularity functions.

5. What are the points to be worth for conjugate beam method?

1. This method can be directly used for simply supported Beam

2. In this method for cantilevers and fixed beams, artificial constraints need to be supplied to the conjugate beam so that it is supported in a manner consistent with the constraints of the real beam.

6. What are the different sections in which the shear stress distribution is to be obtained?

- Rectangular section
- Circular section
- I- section
- T- section
- Miscellaneous section

7. What do you mean by shear stress in beams?

The stress produced in a beam, which is subjected to shear forces is know as stresses.

8. What is the formula to find a shear stress at a fiber in a section of a beam?

The shear stress at a fiber in a section of a beam is given by

$$q = F \times AY / I \times b$$

F = shear force acting at a section

A = Area of the section above the fiber

Y = Distance of C G of the Area A from Neutral axis

I = Moment of Inertia of whole section about N A

b = Actual width at the fiber

9. What is the shear stress distribution rectangular section?

The shear stress distribution rectangular section is parabolic and is given by

$$q = F/2I\ [d2\ /4 - y2]$$

d = Depth of the beam

y = Distance of the fiber from NA

10. What is the shear stress distribution Circular section?

$$q = F/3I\ [R2-y2]$$

11. State the main assumptions while deriving the general formula for shear stresses

- The material is homogeneous, isotropic and elastic
- The modulus of elasticity in tension and compression are same.

- The shear stress is constant along the beam width
- The presence of shear stress does not affect the distribution of bending stress.

12. Define: Shear stress distribution

The variation of shear stress along the depth of the beam is called shear stress distribution

13. What is the ratio of maximum shear stress to the average shear stress for the rectangular section?

Qmax is 1.5 times the Qavg.

14. What is the ratio of maximum shear stress to the average shear stress in the case of solid circular section?

Qmax is 4/3 times the Qave.

15. What is the shear stress distribution value of Flange portion of the I-section?

$$q= f/2I * (D2/4 - y)$$

D-depth

y- Distance from neutral axis

16. What is the value of maximum of minimum shear stress in a rectangular cross section?

$$Qmax=3/2 * F/ (bd)$$

17. What is the shear stress distribution for I-section?

The shear stress distribution I-section is parabolic, but at the junction of web and flange, the shear stress changes abruptly. It changes from F/8I [D2 –d2] to B/b x

$$F/8I [D2 –d2]$$

Where,

D = over all depth of the section

d = Depth of the web

b = Thickness of web

B = Over all width of the section.

18. How will you obtained shear stress distribution for unsymmetrical section?

The shear stress distribution for Unsymmetrical sections is obtained after calculating the position of N A.

19 Where the shear stress is max for Triangular section?

In the case of triangular section, the shear stress is not max at N A. The shear stress is max at a height of h/2

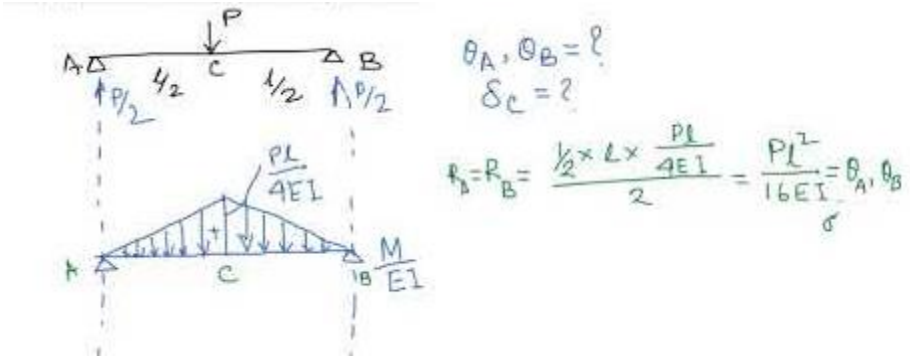
20. Where shear stress distribution diagram draw for composite section?

The shear stress distribution diagram for a composite section should be drawn by calculating the shear stress at important points.

21. What is the deflection at free end of a cantilever carrying central point load W

Deflection = $wl^3 / 3EI$

22. Draw the conjugate beam for cantilever carrying central point load W



UNIT –IV

TORSION

1. Define Torsion

When a pair of forces of equal magnitude but opposite directions acting on body, it tends to twist the body.

It is known as twisting moment or torsion moment or simply as torque.

Torque is equal to the product of the force applied and the distance between the point of application of the force and the axis of the shaft.

2. What are the assumptions made in Torsion equation

- The material of the shaft is homogeneous, perfectly elastic and obeys
- Hooke's law. o Twist is uniform along the length of the shaft
- The stress does not exceed the limit of proportionality
- The shaft circular in section remains circular after loading
- Strain and deformations are small.

3. Define polar modulus

It is the ratio between polar moment of inertia and radius of the shaft.

£ = polar moment of inertia = J

Radius R

4. Write the polar modulus for solid shaft and circular shaft.

£ = polar moment of inertia = J

Radius R

$$J = \pi D4 / 32$$

5. Why hollow circular shafts are preferred when compared to solid circular shafts?

- The torque transmitted by the hollow shaft is greater than the solid shaft.
- For same material, length and given torque, the weight of the hollow shaft will be less compared to solid shaft.

6. Write torsional equation (or)derive a relation for twisting moment carrying capacity of a solid circular shaft

$$T/J=C\theta/L=q/R$$

T-Torque

J- Polar moment of inertia

C-Modulus of rigidity

L- Length

q- Shear stress

R- Radius

7. Write down the expression for power transmitted by a shaft

$$P=2\pi NT/60$$

N-speed in rpm

T-torque

8. Write down the expression for torque transmitted by hollow shaft

$$T= (\pi/16)*Fs*((D4-d4)/d4$$

T-torque

q- Shear stress

D-outer diameter

d- Inner diameter

9. Write down the equation for maximum shear stress of a solid circular section in diameter 'D' when subjected to torque 'T' in a solid shaft.

$T=\pi/16 * Fs*D3$

T- Torque

q - Shear stress

D- Diameter

10. Define torsional rigidity

Product of rigidity modulus and polar moment of inertia is called torsional rigidity

11. What is composite shaft?

Some times a shaft is made up of composite section i.e. one type of shaft is sleeved over other types of shaft.

At the time of sleeving, the two shafts are joined together, that the composite shaft behaves like a single shaft.

12. What is a spring?

A spring is an elastic member, which deflects, or distorts under the action of load and regains its original shape after the load is removed.

13. State any two functions of springs.

1. To measure forces in spring balance, meters and engine indicators.

2. To store energy.

14. What are the various types of springs?

i. Helical springs

ii. Spiral springs

iii. Leaf springs

iv. Disc spring or Belleville springs

15. Classify the helical springs.

1. Close – coiled or tension helical spring.

2. Open –coiled or compression helical spring.

16. What is spring index (C)?

The ratio of mean or pitch diameter to the diameter of wire for the spring is called the spring index.

17. What is solid length?

The length of a spring under the maximum compression is called its solid length. It is the product of total number of coils and the diameter of wire.

$$Ls = nt \times d$$

Where, nt = total number of coils.

18. Define spring rate (stiffness).

The spring stiffness or spring constant is defined as the load required per unit deflection of the spring.

K= W/y

Where, W -load

Y – Deflection

19. Define helical springs.

The helical springs are made up of a wire coiled in the form of a helix and are primarily intended for compressive or tensile load.

20. What are the differences between closed coil & open coil helical springs?

The spring wires are coiled very the wires are coiled such that there closely, each turn is nearly at right is a gap between the two consecutive angles to the axis of helix turns.

Helix angle is less than 10o Helix angle is large (>100)

UNIT V

ANALYSIS OF TRUSSES

1. What is mean by perfect frame?

If a frame is composed of such members, which are just sufficient to keep the frame in equilibrium, when the frame is supporting the external load, then the frame is known as perfect frame.

2. What are the different types of frames?

The different types of frame are:

- Perfect frame and
- Imperfect frame.

3. What is mean by Imperfect frame?

A frame in which number of members and number of joints are not given by **n = 2j – 3** is known as imperfect frame. This means that number of members in an imperfect frame will be either more or less than (2j – 3).

4. What is mean by deficient frame?

If the number of member in a frame are less than (2j -3), then the frame is know as deficient frame

5. What is mean by redundant frame?

If the number of member in a frame are more than (2j -3), then the frame is know as deficient frame

6. What are the assumptions made in finding out the forces in a frame?

The assumptions made in finding out the forces in a frame are:

- The frame is a perfect frame
- The frame carries load at the joints
- All the members are pin-joined.

7. What are the reactions of supports of a frame?

The frame are generally supported

- on a roller support or
- On a hinged support.

8. How will you Analysis of a frame?

- Analysis of a frame consists of
- Determinations of the reactions at the supports and
- Determination of the forces in the members of the frame

9. What are the methods for Analysis the frame?

- Methods of joints,
- Methods of sections, and
- Graphical method.

10. How method of joints applied to Trusses carrying Horizontal loads.

If a truss carries horizontal loads (with or without vertical loads) hinged at one end supported on roller at the other end, the support reaction at the roller support end will be normal. Whereas the support reaction at the hinged end will consist of (i) horizontal reaction and (ii) vertical reaction

11. How method of joints applied to Trusses carrying inclined loads.

If a truss carries inclined loads hinged at one end supported on roller at the other end, the support reaction at the roller support end will be normal. Whereas the support reaction at the hinged end will consist of (i) horizontal reaction and (ii) vertical reaction

12. What is mean by compressive and tensile force?

The forces in the member will be compressive if the member pushes the joint to which it is connected whereas the force in the member will be tensile if the member pulls the joint to which it is connected.

13. How will you determine the forces in a member by method of joints?

While determining forces in a member by methods of joints, the joint should be selected in such a way that at any time there are only two members, in which the forces are unknown.

14. How method of joints applied to Trusses carrying Horizontal loads

If a truss carries horizontal loads (with or without vertical loads) hinged at one end and supported on roller at the other end, the support reaction at the roller support end will be normal, whereas the support reaction at the hinged end will consist of i)horizontal reaction and (ii) vertical reaction

15. How will you determine the forces in a member by method of joints?

In method of joint after determining the reactions at the supports, the equilibrium of every support is considered. This means the sum all vertical forces as well as the horizontal forces acting on a joint is equated to zero. The joint should be selected in such a way that at any time there are only two members, in which the forces are unknown

16. What are the benefits of method of sections compared with other methods?

- This method is very quick
- When the forces in few members of the truss are to be determined, then the method of section is mostly used.

17. What are the assumptions made in the analysis of a pin-jointed plane truss?

The two assumptions in the analysis of trusses are,

- All external forces act at the joints of the truss members and nowhere else.
- The truss members are acted upon either by axial tension or axial compression.

18. Define truss?

A truss is an assembly of beams or other elements that creates a rigid structure.

19. Define method of joints?

The method of joints is a way to find unknown forces in a truss structure. The principle behind this method is that all forces acting on a joint must add to zero. If there were a net force, the joint would move.

20. Define method of section?

The method of sections is a process used to solve for the unknown forces acting on members of a truss. The method involves breaking the truss down into individual sections and analyzing each section as a separate rigid body.

21. Define tension coefficient.

The tension co efficient for a member of a frame is defined as the pull or tension in that member is divided by its length. Analysis of space trusses using method of tension coefficients.

UNIT I

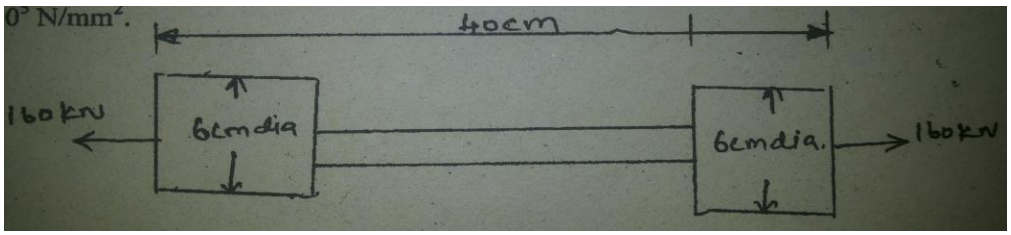
STRESS STRAIN DEFORMATION OF SOLIDS

PART- B

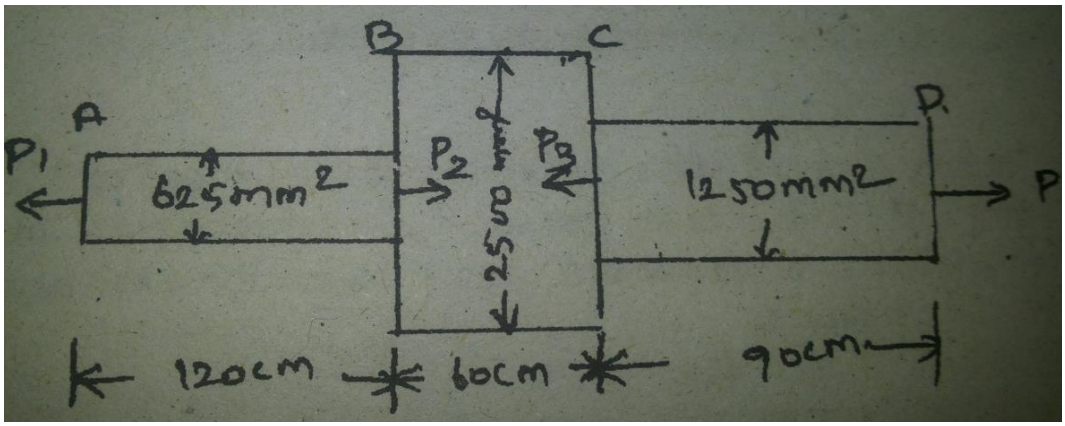
1. A Mild steel rod of 20 mm diameter and 300 mm long is enclosed centrally inside a hollow copper tube of external diameter 30 mm and internal diameter 25 mm. The ends of the rod and tube are brazed together, and the composite bar is subjected to an axial pull of 40 kN. If E for steel and copper is 200 GN/m2 and 100 GN/m2 respectively, find the stresses developed in the rod and the tube also find the extension of the rod.

2. A bar of 30 mm diameter is subjected to a pull of 60 kN. The measured extension on gauge length of 200 mm is 0.09 mm and the change in diameter is 0.0039 mm. calculate the Poisson's ratio and the values of the three moduli.

3. The bar shown in fig. is subjected to a tensile load of 160 KN. If the stress in the middle portion is limited to 150 N/mm^2, determine the diameter of the middle portion. Find also the length of the middle portion if the total elongation of the bar is to be 0.2mm. young's modulus is given as equal to 2.1 x 10^5 N/mm^2.



4. A member ABCD is subjected to point loads P1, P2, P3, P4 as shown in fig. calculate the force P2 necessary for equilibrium, if P1 = 45 KN, P3 = 450 KN and P4 = 139 KN. Determine the total elongation of the member, assuming the modulus of elasticity to be 2.1 x 10^5 N/mm^2.



5. A steel rod of 20mm diameter passes centrally through a copper tube of 50mm external diameter and 40mm internal diameter. The tube is closed at each end by rigid plates of negligible thickness. The nuts are tightened lightly home on the projecting parts of the rod. If the temperature of the assembly is raised by 50˚C, calculate the stress developed in copper and steel. Take E for steel and copper as 200 GN/m^2 and 100 GN/m^2 and α for steel and copper as 12 x 10^{-6} per ˚C and 18 x 10^{-6} per ˚C.

6. Two vertical rods one of steel and the other of copper are each rigidly fixed at the top and 50cm apart. Diameters and lengths of each rod are 2cm and 4m respectively. A cross bar fixed to the rods at the lower ends carries a load of 5000 N such that the cross bar remains horizontal even after loading. Find the stress in each rod and the position of the load on the bar. Take E for steel = 2 x 10^5 N/mm^2 and E for copper = 1x 10^5 N/mm^2.

7. The extension in a rectangular steel bar of length 400mm and thickness 3mm is found be 0.21mm .The bar tapers uniformly in width from 20mm to 60mm E for the bar is 2x 10^5 N/mm^2.Determine the axial load on the bar.

UNIT II

TRANSFER OF LOADS AND STRESSES IN BEAMS

PART- B

1. Three blanks of each 50 x200 mm timber are built up to a symmetrical I section for a beam. The maximum shear force over the beam is 4KN. Propose an alternate rectangular section of the same material so that the maximum shear stress developed is same in both sections. Assume then width of the section to be 2/3 of the depth.

2. A beam of uniform section 10 m long carries a udl of 2KN/m for the entire length and a concentrated load of 10 KN at right end. The beam is freely supported at the left end. Find the position of the second support so that the maximum bending moment in the beam is as minimum as possible. Also compute the maximum bending moment

3. A beam of size 150 mm wide, 250 mm deep carries a uniformly distributed load of w kN/m over entire span of 4 m. A concentrated load 1 kN is acting at a distance of 1.2 m from the left support. If the bending stress at a section 1.8 m from the left support is not to exceed 3.25 N/mm^2 find the load w.

4. A cantilever of 2m length carries a point load of 20 KN at 0.8 m from the fixed end and another point of 5 KN at the free end. In addition, a u.d.l. of 15 KN/m is spread over the entire length of the cantilever. Draw the S.F.D, and B.M.D.

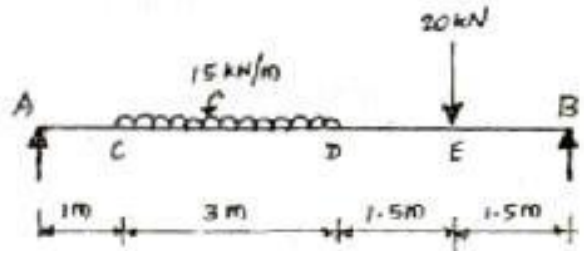
5. A Simply supported beam of effective span 6 m carries three point loads of 30 KN, 25 KN and 40 KN at 1m, 3m and 4.5m respectively from the left support. Draw the SFD and BMD. Indicating values at salient points.

6. A Simply supported beam of length 6 metres carries a udl of 20KN/m throughout its length and a point of 30 KN at 2 metres from the right support. Draw the shear force and bending moment diagram. Also find the position and magnitude of maximum Bending moment.

7. A Simply supported beam 6 metre span carries udl of 20 KN/m for left half of span and two point loads of 25 KN end 35 KN at 4 m and 5 m from left support. Find maximum SF and BM and their location drawing SF and BM diagrams.

8. A cantilever 1.5m long is loaded with a uniformly distribution load of 2 kN/m run over a length of 1.25m from the free end it also carries a point load of 3kn at a distance of 0.25m from the free end. Draw the shear force and bending moment diagram of the cantilever.

9. For the simply supported beam loaded as shown in Fig. , draw the shear force diagram and bending moment diagram. Also, obtain the maximum bending moment.



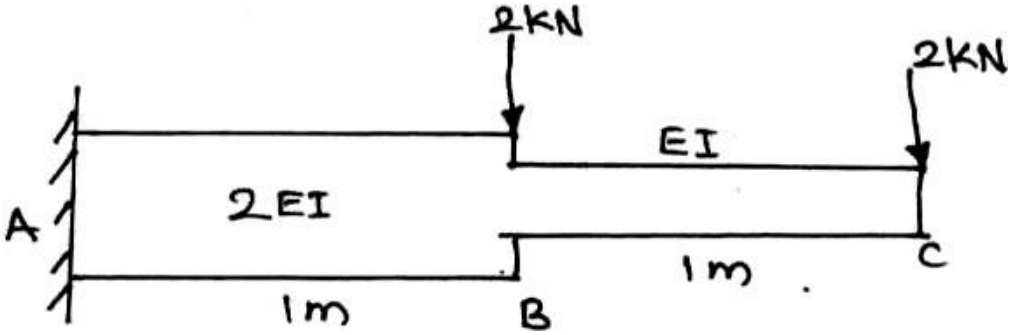
UNIT III
DEFLECTION OF BEAMS
PART-B

1. A beam AB of length 8 m is simply supported at its ends and carries two point loads of 50 kN and 40 kN at a distance of 2 m and 5 m respectively from left support A. Determine, deflection under each load, maximum deflection and the position at which maximum deflection occurs. Take E = 2 x 10^5 N/mm^2 and I = 8.5 $X10^6$ mm^4 .

2. A 1.2 m long column has a circular cross section of 45 mm diameter one of the ends of the column is fixed in direction and position and other ends are free. Taking factor of safety as 3, calculate the safe load using

(i) Rankine's formula, take yield stress = 560 N/mm^2 and a = 1/1600 for pinned ends.
(ii) Euler's formula, Young's modulus for cast iron = 1.2 x 105 N/mm^2.

3. For the cantilever beam shown in Fig.3. Find the deflection and slope at the free end. EI = 10000 kN/m^2.



4. A beam is simply supported at its ends over a span of 10 m and carries two concentrated loads of 100 KN and 60 KN at a distance of 2 m and 5 m respectively from the left support. Calculate (i) Slope at the left support (ii)slope and deflection under the 100 kN load. Assume EI = 36 x 104 kN-m^2.

5. Find the Euler critical load for a hollow cylindrical cast iron column 150 mm external diameter, 20 mm wall thickness if it is 6 m long with hinged at both ends. Assume Young's modulus of cast iron as 80 kN/mm^2. Compare this load with that given by Rankine formula. Using Rankine constants a =1/1600 and 567 N/mm^2.

6. A 3 m long cantilever of uniform rectangular cross–section 150 mm wide and 300 mm deep is loaded with a point load of 3 kN at the free end and a udl of 2 kN/m over the entire length. Find the maximum deflection. E = 210 kN/mm2. Use Macaulay's method.

7. A simply supported beam of span 6 m is subjected to an udl of 2 kN/m over the entire span and a point load of 3 kN at 4 m from the left support. Find the deflection under the point load in terms of EI. Use strain energy method.

8. A simply supported beam of uniform flexural rigidity EI and span *l*, carries two symmetrically placed loads *P* at one–third of the span from each end. Find the slope at the supports and the deflection at mid–span. Use moment area theorems.

9. Determine the section of a hollow C.I. cylindrical column 5 m long with ends firmly built in. The column has to carry an axial compressive load of 588.6 KN. The internal diameter of the column is 0.75 times the external diameter. Use Rankine's constants. a = 1 / 1600, σc = 57.58 KN/cm2 and F.O.S = 6.

UNIT IV
TORSION
PART- B

1. Determine the diameter of a solid shaft which will transmit 300 KN at 250 rpm. The maximum shear stress should not exceed 30 N/mm2 and twist should not be more than 10 in a shaft length 2m. Take modulus of rigidity = 1 x 10^5 N/mm^2.

2. The stiffness of the closed coil helical spring at mean diameter 20 cm is made of 3 cm diameter rod and has 16 turns. A weight of 3 KN is dropped on this spring. Find the height by which the weight should be dropped before striking the spring so that the spring may be compressed by 18 cm. Take C= 8x10^4 N/mm^2.

3. It is required to design a closed coiled helical spring which shall deflect 1mm under an axial load of 100 N at a shear stress of 90 Mpa. The spring is to be made of round wire having shear modulus of 0.8 x 10^5 Mpa. The mean diameter of the coil is 10 times that of the coil wire. Find the diameter and length of the wire.

4. A steel shaft ABCD having a total length of 2400 mm is contributed by three different sections as follows. The portion AB is hollow having outside and inside diameters 80 mm and 50 mm respectively, BC is solid and 80 mm diameter. CD is also solid and 70 mm diameter. If the angle of twist is same for each section, determine the length of each portion and the total angle of twist. Maximum permissible shear stress is 50 Mpa and shear modulus 0.82 x 10^5 MPa

5. The stiffness of close coiled helical spring is 1.5 N/mm of compression under a maximum load of 60 N. The maximum shear stress in the wire of the spring is 125 N/mm^2. The solid length of the spring (when the coils are touching) is 50 mm. Find the diameter of coil, diameter of wire and number of coils. C = 4.5.

6. Calculate the power that can be transmitted at a 300 r.p.m. by a hollow steel shaft of 75 mm external diameter and 50 mm internal diameter when the permissible shear stress for the steel is 70 N/mm2 and the maximum torque is 1.3 times the mean. Compare the strength of this hollow shaft with that of an solid shaft. The same material, weight and length of both the shafts are the same.

7. A solid cylindrical shaft is to transmit 300 kN power at 100 rpm. If the shear stress is not to exceed 60 N/mm^2, find its diameter. What percent saving in weight would be obtained if this

shaft is replaced by a hollow one whose internal diameter equals to 0.6 of the external diameter, the length, the material and maximum shear stress being the same?

8. A helical spring of circular cross-section wire 18 mm in diameter is loaded by a force of 500 N. The mean coil diameter of the spring is 125mm. The modulus of rigidity is 80 kN/mm^2. Determine the maximum shear stress in the material of the spring. What number of coils must the spring have for its deflection to be 6 mm?

9. A closely coiled helical spring of round steel wire 10 mm in diameter having 10 complete turns with a mean diameter of 12 cm is subjected to an axial load of 250 N. Determine

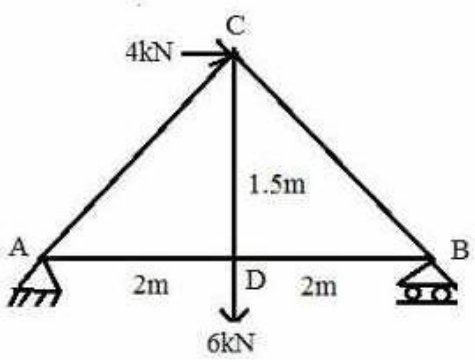
I. the deflection of the spring
II. maximum shear stress in the wire and
III. stiffness of the spring and
IV. Frequency of vibration. Take C = 0.8 x 10^5 N/mm^2.

UNIT 5

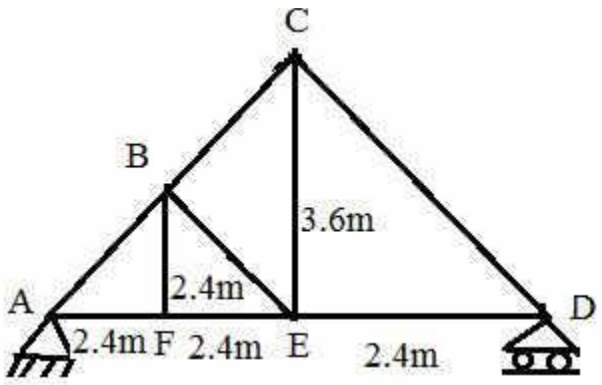
ANALYSIS OF TRUSSES

PART-B

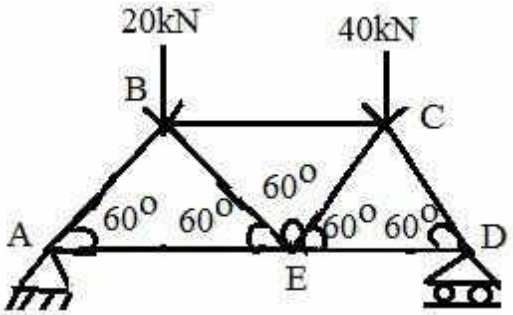
1. Find the forces in the members of the truss shown in Fig.



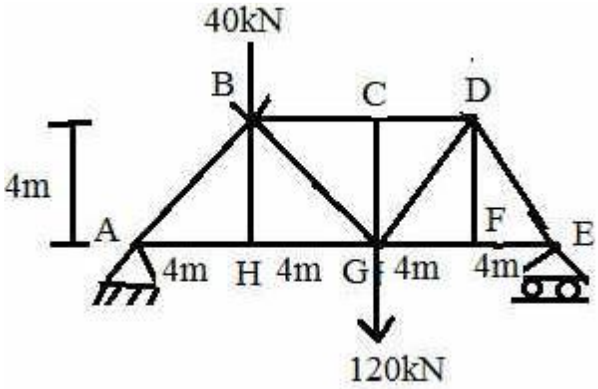
2. Find the forces in the member of the truss shown in fig. by method of joints.



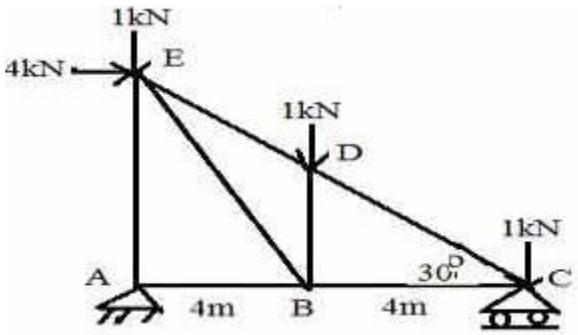
3. Find the forces in all the members of the girder shown in Fig. by the method of joints, indicating whether the force is compressive or tensile.



4. Determine the forces in all members of a truss as shown in fig.



5. For the truss shown in fig find the forces in members CD, CB, BD and AE by method of Joints.



CE6301 ENGINEERING GEOLOGY

UNIT I PHYSICAL GEOLOGY

Geology in civil engineering – branches of geology – structure of earth and its composition – weathering of rocks – scale of weathering – soils - landforms and processes associated with river, wind, groundwater and sea – relevance to civil engineering. Plate tectonics – Earth quakes – Seismic zones in India.

Part – A

1. Define engineering geology.
2. Describe briefly the layers of interior of earth.
3. Define weathering.
4. What is meant by seismic zone?
5. What is physical weathering?
6. What is chemical weathering?
7. What is spheroidal weathering?
8. Define mohorovicic and Guttenburg discontinuity.
9. Define plate tectonics.
10. Define exfoliation.
11. Differentiate between water table and perched water table.
12. Define aquifer and mention its types.
13. What is meant by subduction zone? Mention its importance.
14. Differentiate aquifer and aquiclude.
15. Write about Mercalli scale.
16. Describe the interior of the earth.
17. Define confined aquifer.
18. Define unconfined aquifer.
19. Mention the seismic zones of India.

Part – B

1. Write in detail about the structure of the earth and its composition with a neat diagram.
2. Give an account on mode of occurrence and prospecting of ground water.
3. Describe in detail about plate tectonics.
4. Explain in detail about weathering of rocks and its engineering importance.
5. Explain in detail about the earthquake belts of India.
6. Write in detail about the scope of geology and importance of geology in Civil Engineering.
7. Explain physical and chemical weathering process in detail. Add a note on weathering grade and its engineering significance.
8. Explain the process associated with river. Write their engineering significance.
9. Explain the process associated with winds. Write their engineering significance.
10. Explain the process associated with sea. Write their engineering significance.

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UNIT II MINEROLOGY

Physical properties of minerals – Quartz group, Feldspar group, Pyroxene - hypersthene and augite, Amphibole – hornblende, Mica – muscovite and biotite, Calcite, Gypsum and Clay minerals.

Part – A

1. Define mineralogy.
2. Define hexagonal, isometric, tetragonal, orthorhombic, triclinic, monoclinic system of crystals.
3. Write the physical properties of mica.
4. Define mineral.
5. What are the different physical properties of minerals?
6. Give the physical properties and uses of quartz, augite.
7. What is moh's scale of hardness?
8. Give the physical properties and uses of hornblende, biotite.
9. Name at least 4 clay minerals and their important engineering properties.
10. Give the physical properties of muscovite, calcite.
11. Define ore minerals.
12. Define isometric system of crystals.
13. Define tetragonal system of crystals.
14. Define orthorhombic system of crystals.
15. Define triclinic system of crystals.
16. Define monoclinic system of crystals.
17. Define colour.
18. Define lusture.
19. Define form,streak.
20. Define hardness, fracture and specific gravity.

Part – B

1. Elaborate the various physical properties which help in identification of minerals.
2. Explain the physical properties of Feldspar group of minerals.
3. Explain the physical properties of Quartz group of minerals.
4. Explain the physical properties of Augite, hornblende.
5. Explain the physical properties of Mica group of minerals.
6. Give a detailed account on chemical composition, physical properties, origin, occurrence, engineering behaviour and uses of clay minerals.
7. Explain the physical properties biotite, calcite.

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UNIT III PETROLOGY

Classification of rocks, distinction between Igneous, Sedimentary and Metamorphic rocks. Engineering properties of rocks. Description, occurrence, engineering properties, distribution and uses of Granite, Dolerite, Basalt, Sandstone, Limestone, Laterite, Shale, Quartzite, Marble, Slate, Gneiss and Schist.

Part – A

1. Define petrology.
2. Write the classification of petrology.
3. What are metamorphic rocks?
4. What are igneous rocks?
5. What are sedimentary rocks?
6. Write about the occurrence of granite.
7. Write about the occurrence of basalt.
8. Classify igneous rocks.
9. Classify metamorphic rocks.
10. Write the composition uses of shale, slate and marble.
11. Write about the structure of igneous rocks.
12. Write about the occurrence of sandstone.
13. What are the factors controlling the specific gravity, porosity and strength of rocks?
14. List out the tests to be carried out to determine the strength of building stones.
15. What is black granite? List its uses.
16. Bring out the differences between granite and slate.
17. What is ductility of rocks?
18. Describe the uses of black granite.
19. How igneous rocks are classified according to their occurrences?
20. What is ductility of rocks?

Part – B

1. What are sedimentary rocks? Explain the properties of any 4 sedimentary rocks.
2. What are metamorphic rocks? Explain the properties of any 4 metamorphic rocks.
3. What are igneous rocks? Explain the properties of any 4 igneous rocks.
4. Describe the engineering properties of rocks.
5. Describe the different types of rocks. Give the classification, texture and structure of igneous, sedimentary and metamorphic rocks.
6. List the various field and laboratory tests to determine the engineering properties of rocks.
7. Describe in detail about the mineral composition, structure, texture, origin, engineering properties and uses of basalt, dolerite, sandstone and slate.
8. Describe the composition, texture, characteristics, occurrence and uses of black granite and basalt
9. Describe the composition, texture, characteristics, occurrence and uses of limestone and slate.
10. Describe the engineering properties of igneous rocks.

UNIT IV STRUCTURAL GEOLOGY AND GEOPHYSICAL METHODS

Geological maps – attitude of beds, study of structures – folds, faults and joints – relevance to civil engineering. Geophysical methods – Seismic and electrical methods for subsurface investigations.

Part – A

1. Define structural geology.
2. Define dip and strike.
3. Define recumbent fold and overturn fold.
4. Differentiate anticline and syncline.
5. Describe wenner's configuration of electrodes.
6. What are joints? Explain their significance.
7. What are folds? Explain their significance.
8. What are faults? Explain their significance.
9. Differentiate Wenner and berger methods.
10. Define outcrops.
11. What are true and apparent dips?
12. What are the types of dips?
13. What is a geological map?
14. Draw and describe the parts of folds.
15. Classify folds.
16. Write the causes of foldings?
17. What are the engineering considerations of a fold?
18. What are the engineering considerations of a fault?
19. What are the mechanisms of faulting?
20. Write about the origin of joints.

Part – B

1. What is a fault? Discuss the various types of faults and write about the engineering applications.
2. What is a fold? Discuss the various types of faults and write about the engineering applications.
3. What is a joint? Discuss the various types of faults and write about the engineering applications.
4. Explain in detail the role of electrical methods of subsurface investigation in civil engineering practice.
5. Describe seismic refraction survey to de conducted for determining the depth of bed rock.
6. Discuss in detail electrical method of investigation for ground water exploration.
7. Classify folds and faults in rocks and explain how they influence the design of dams.
8. Classify and describe joint structures with neat sketches and also write their role in dam and tunnel construction.
9. Give a detailed account of the various geological structures and their role in selection of sites for engineering projects.
10. Describe fault structures with neat sketches and also write their role in dam and tunnel construction.

UNIT V APPLICATION OF GEOLOGICAL INVESTIGATIONS

Remote sensing for civil engineering applications; Geological conditions necessary for design and construction of Dams, Reservoirs, Tunnels, and Road cuttings - Hydrogeological investigations and mining - Coastal protection structures. Investigation of Landslides, causes and mitigation.

Part – A

1. List a few measures of coastal protection.
2. What is the function of groynes in coastal protection?
3. Enumerate the coastal protection structures.
4. What is the application of satellite imagery?
5. Define overlap.
6. Define overbreak.
7. Describe sea wall and jetties in coastal protection structures.
8. List any 4 methods to prevent landslides.
9. Define stand up time in dam construction.
10. Define pay line in dam construction.
11. Differentiate swelling ground and running ground in construction site.
12. Define the term parallax in aerial photograph.
13. What are the various types of aerial photographs?
14. Explain how the study of bed rocks is essential before the construction of tunnels.
15. Define overbreak and pay line in tunnelling operations.
16. Write the applications of remote sensing techniques.
17. What are multipurpose dams?
18. Give the functions of breakwater.

Part – B

1. Write in detail about landslides and their causative effects. Explain about the measures to prevent them.
2. What are the various geological factors to be considered for the construction of dams? Explain with examples.
3. Explain in detail the role of aerial photographs and satellite images in planning and execution of civil Engineering projects.
4. Write in detail about sea erosion and coastal protection structures.
5. Using case studies, describe the various aspects of coastal erosion and the various methods of coastal protection.
6. What are the various geological factors to be considered for the construction of tunnels? Explain in detail with examples.
7. What are the various geological factors to be considered for the construction of, road cuttings? Explain in detail with examples.
8. What are the various geological factors to be considered for the construction of buildings? Explain in detail with examples.

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Reg. No. : | | | | | | | | | | | | |

Question Paper Code : 80072

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Third Semester

Civil Engineering

CE 8392 — ENGINEERING GEOLOGY

(Regulation 2017)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Draw neat sketch and show the internal structure of the earth.
2. Distinguish – Stalagmites and stalactites.
3. Define: Mineral with few examples.
4. Write the uses of Gypsum in Civil engineering.
5. What are quasi elastic rocks? Give example.
6. How does a slate form?
7. Draw a sketch and show Joint sets and Joint system.
8. What are passive geophysical techniques?
9. List the photo recognition elements in Remote Sensing technique.
10. Relate the angle of repose with landslides.

PART B — (5 × 13 = 65 marks)

11. (a) Explain on erosional and depositional landforms produced by river and its significance in Civil Engineering.

Or

(b) Discuss in detail on physical and chemical weathering of rocks, its effects and relate to soil profile.

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12. (a) Enumerate with neat sketches on the important physical properties of minerals which are essential for identification.

Or

(b) Write in detail on the following minerals:

(i) Feldspars

(ii) Quartz

(iii) Calcite and

(iv) Hornblende.

13. (a) Explain on the classification of igneous rocks, their engineering properties and their uses in Civil Engineering.

Or

(b) Describe the following rocks:

(i) Sandstone

(ii) Marble

(iii) Slate and

(iv) Dolerite.

14. (a) Discuss on Faults, their types and their effects on Civil Engineering structures.

Or

(b) Explain electrical resistivity methods and their applications in Civil Engineering.

15. (a) Describe the importance of geological investigations and the effects of geology on Dams.

Or

(b) How remote sensing is done? Elaborate on the applications of Remote Sensing in the field of Civil Engineering.

PART C — (1 × 15 = 15 marks)

16. (a) "The success or failure of an engineering project depends largely upon the physical conditions with in the province of geology"- Professor Boyd Dawkins.

Justify the above sentence by illustrating the importance of subsurface geophysical investigations prior to major Civil Engineering constructions.

Or

(b) Every landslide starts with a few pebbles moving in the same direction- Robin Lings.

Relate the above quote and explain the importance of knowledge of geology in landslide mitigation and management.

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Question Paper Code : 90112

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019

Third/Fifth Semester

Civil Engineering

CE 8392 – ENGINEERING GEOLOGY

(Common to Environmental Engineering)

(Regulations 2017)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A (10×2=20 Marks)

1. Write short notes on the landforms of Sea.
2. Write short note on the interior of the earth.
3. Briefly write the physical properties of Quartz group of minerals.
4. Write notes on the physical properties of clay minerals.
5. Distinguish between igneous and sedimentary rocks.
6. Give the engineering properties of Limestones and Marble.
7. Write short notes on the attitude of beds.
8. Discuss the various coastal protection structures.
9. Distinguish between geological map and topographic map.
10. Write few lines on applications of Remote Sensing in civil engineering.

PART – B (5×13=65 Marks)

11. a) Explain how the Earthquakes occur. Write notes on seismic zones in India.

(OR)

b) Describe the different types of physical and chemical weathering of rocks.

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12. a) Give a detailed account on the physical properties of Pyroxene group of minerals.

(OR)

b) Enumerate the physical properties of Feldspar group of minerals and their suitability in civil engineering applications.

13. a) Write an essay on the engineering properties, distribution and uses of Granite and Laterites.

(OR)

b) Describe the classification of rocks and their brief engineering properties.

14. a) Explain in detail on the seismic methods for subsurface investigations.

(OR)

b) Describe the various types of Faults and their control on geological structures.

15. a) Discuss in detail on the geological conditions essential for construction of dams.

(OR)

b) Write an essay on causes and mitigation of Landslides.

PART – C **(1×15=15 Marks)**

16. a) Write an essay and critically analyze the formations of various landforms of river and wind.

(OR)

b) Discuss in detail on the geological conditions required for the design and construction of Tunnels.

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Question Paper Code : 57141

30/5/16 AN

B.E./B.Tech. DEGREE EXAMINATION, MAY/JUNE 2016

Third Semester

Civil Engineering

CE 6301 – ENGINEERING GEOLOGY

(Regulations 2013)

Time : Three Hours **Maximum : 100 Marks**

Answer ALL questions.

PART – A (10 × 2 = 20 Marks)

1. What is meant by soil profile ? List the informations that are to be provided in soil profile.
2. Write a note on Sea cliff with neat diagram.
3. What is meant by Moh's scale of hardness ?
4. Differentiate between cleavage and fracture.
5. Difference between concordant bodies and discordant bodies.
6. What do you mean by granulation in metamorphic petrology ?
7. What is meant by a Recumbent folds ? Draw a neat diagram.
8. What is meant by Wenner's Arrangement ? Draw a neat diagram.
9. Define the term Dead storage in reservoir.
10. What is meant by "Rock bolting" ? Draw a neat diagram.

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PART – B (5 × 16 = 80 Marks)

11. (a) Write an essay on weathering of rocks and its significance in engineering point of view.

OR

(b) Give a general view of Internal Structure of the Earth as revealed by the seismological evidence.

12. (a) Give salient features and mention important properties of Feldspar group of minerals.

OR

(b) Explain Amphibole group of minerals.

13. (a) Write an essay on engineering properties distribution and uses of granite.

OR

(b) How would you differentiate between igneous rock, metamorphic rock and sedimentary rock on the basis of structure and texture ?

14. (a) Explain gravity method in geophysics.

OR

(b) Write a note on faults, their causes and effects on the engineering quality of rocks.

15. (a) Give an account of causes of inherent weakness in rocks. How rock qualities could be improved by artificial treatment.

OR

(b) What do you understand by a 'dam disaster' ? Discuss these geological situations which when ignored could be the cause of the possible dam disaster.

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Question Paper Code : 71543

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Third Semester

Civil Engineering

CE 6301 — ENGINEERING GEOLOGY

(Regulations 2013)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Draw the internal structure of earth.
2. What is Exfoliation?
3. List the name of clay group of minerals.
4. Bring out the differences between muscovite and biotite.
5. How do you classify rocks? Give examples.
6. Differentiate between Gneiss and Schist.
7. Differentiate between True dip and apparent dip of rock formation.
8. What are joints and joint sets?
9. Define remote sensing.
10. List at least four measures to prevent coastal erosion.

PART B — (5 × 13 = 65 marks)

11. (a) Explain the concept of plate tectonics and describe how earthquakes occur. Add a note on the distribution of earthquake in the world and in India.

Or

(b) Illustrate and explain the erosion and depositional features formed by flow of river and add a note on the significance in civil engineering projects.

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12. (a) List the physical properties of minerals and describe each property with examples from the mineral kingdom.

Or

(b) Describe the composition, properties, varieties and uses of Gypsum, Quartz, and Feldspar.

13. (a) What are the engineering properties of rocks to be tested for constructions of dams and tunnels and how will you determine the engineering properties of rocks at site and laboratory?

Or

(b) Write briefly about the classification of rocks? Describe the origin, texture, structure and occurrence of granite, dolerite, marble and sandstone rocks.

14. (a) Explain folding in rocks and describe the various types of folds.

Or

(b) Explain how the geophysical methods help in knowing about sub-surface features during civil engineering investigations.

15. (a) What are landslides? How do they occur? Describe the types of landslides with a sketch and enumerate the geological investigation required for mitigation of landslides.

Or

(b) Explain in detail the various causes and effects of sea erosion. Add a detailed note on coastal protection measures.

PART C — (1 × 15 = 15 marks)

16. (a) Using case study write a detailed account of the application of remote sensing in civil engineering?

Or

(b) Write the geological considerations to be taken into account during tunneling.

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Question Paper Code : 40770

26/04/18 AN

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2018
Third Semester
Civil Engineering
CE 6301 – ENGINEERING GEOLOGY
(Regulations 2013)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A (10×2=20 Marks)

1. Define the term soil pedogenesis.
2. Explain in short the erosional landforms associated with ground water flow and depletion.
3. Define various varieties of quartz group of minerals.
4. Briefly, describe the important clay minerals and their use in modern industries.
5. Describe some distinguishing prime physical properties of the metamorphic rocks.
6. Explain briefly the term metamorphic facies and mineral paragenesis.
7. Describe the use of geological maps in understanding structural geology of a tectonically active area.
8. What are the different geological structures associated with convergent plate regimes ?
9. Differentiate between Arch and Gravity dams.
10. What are the various coastal protection structures ?

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PART – B (5×13=65 Marks)

11. a) Explain in detail the internal composition of earth with emphasis on earth's upper mantle.

(OR)

b) Discuss in detail the theory of plate tectonics. Provide some evidences that support this concept.

12. a) Discuss in detail the physical properties of feldspar and Pyroxene group of minerals.

(OR)

b) List and explain various physical properties of Gypsum and Clay group of minerals with emphasis of their use in engineering aspects.

13. a) Describe how the mineral textures of an igneous rock be used to infer its origin.

(OR)

b) What is a Dolerite ? Describe its composition, origin and distribution.

14. a) Write a detailed note on the mechanics and classification of folds and faults.

(OR)

b) Explain in detail the principle, procedure and applicability of seismic methods for subsurface investigations.

15. a) Write a detailed note on the foundation evaluation techniques and influence of geological conditions on foundation and design of Dams.

(OR)

b) Discuss the use of remote sensing applications in hydrogeological and mining investigations studies.

PART – C (1×15=15 Marks)

16. Discuss the use of geospatial techniques for disaster management. Enumerate your answer with case studies on landslide mitigation adopted in the Himalayan region.

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25/4/19 AN

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Question Paper Code : 52753

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Third Semester

Civil Engineering

CE 6301 — ENGINEERING GEOLOGY

(Regulation 2013)

(Common to : PTCE 6301 – Engineering Geology for B.E. (Part – Time) – Second Semester – Civil Engineering – (Regulation – 2014))

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Draw the internal structure of earth.

2. What is Exfoliation?

3. Differentiate between colour and streak of minerals.

4. Differentiate between muscovite and biotite.

5. Define crushing strength of a rock.

6. Write short note on the importance of texture and structure of a building stone.

7. What are Joints and Joint sets?

8. Define the term 'Rock Quality Designation' (RQD).

9. What is meant by the term "overlap" in remote sensing?

10. List the causative factors of landslides.

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PART B — (5 × 13 = 65 marks)

11. (a) Write elaborately on physical and chemical weathering of rocks. What is the significance of weathering of rocks in Civil Engineering?

Or

(b) Describe the geological work of sea and its engineering considerations.

12. (a) List the physical properties of minerals and describe each property with examples from the mineral kingdom.

Or

(b) Describe the composition, properties, varieties and uses of Gypsum, Quartz and Feldspar.

13. (a) Outline the various engineering properties of rocks, and give a detailed account of the laboratory and field test to be carried out to estimate such properties.

Or

(b) Write detailed notes on the mineral composition, texture, origin, engineering properties and uses of

(i) Granite

(ii) Dolerite

(iii) Sandstone and

(iv) Marble.

14. (a) Explain folding in rocks and describe the various types of folds.

Or

(b) Explain how the geophysical methods help in knowing about sub–surface features during civil engineering investigations.

15. (a) Illustrate with neat sketches about land slides and their types. What are the various measures to control landslides?

Or

(b) Explain how remote sensing is done. Discuss in detail on remote sensing application in Civil Engineering.

STUCOR APP

PART C — (1 × 15 = 15 marks)

16. (a) Using case studies of structural failures, discuss the importance of geological investigations for the design and construction of large civil structures.

Or

(b) Natural disasters in India can be understood better and controlled well, if geology is understood well. Give your opinion about this statement using appropriate case studies.

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Question Paper Code : 80193

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2016.

Third Semester

Civil Engineering

CE 6301 — ENGINEERING GEOLOGY

(Regulations 2013)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Differentiate between Physical and chemical weathering.
2. Mention about the seismic zones of India.
3. Differentiate between colour and streak of minerals.
4. Differentiate between muscovite and biotite.
5. Write briefly about attrition test.
6. Compare the relative strengths of shale, sandstone and quartzite.
7. What are joints in rocks? How do they influence the strength of rocks?
8. What is meant by Wenner Configuration?
9. What is over break in tunelling? How can it he controlled?
10. Mention a few coastal protection structures.

PART B — (5 × 13 = 65 marks)

11. (a) Using diagrams and explanations, describe the internal structure and composition of the Earth.

Or

(b) Write in detail about the tectonic plates and their role in generation of earthquakes.

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12. (a) Using appropriate examples from the mineral kingdom, describe the physical properties of minerals.

Or

(b) Describe in detail, the properties, composition and uses of Feldspar and Calcite.

13. (a) Outline the various engineering properties of rocks, and give a detailed account of the laboratory and field tests to be carried out to estimate such properties.

Or

(b) Write detailed notes on the mineral composition, texture, origin, engineering properties and uses of (i) Granite (ii) Dolerite, (iii) Sandstone and (iv) Marble.

14. (a) Write in detail about the types of faults and their influence on dams and tunnels.

Or

(b) Discuss the operating principle of the seismic methods of subsurface investigation.

15. (a) What are the effects of the action of sea waves on the coastal zones? Add a note on the various coastal protection structures.

Or

(b) List the causes of landslides. Also, classify landslides and give a detailed account of the methods of preventing them.

PART C — (1 × 15 = 15 marks)

16. (a) Using case studies of structural failures, discuss the importance of geological investigations for the design and construction of large civil structures.

Or

(b) Natural disasters in India can be understood better and controlled well, if geology is understood well. Give your opinion about this statement using appropriate case studies.

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Question Paper Code : 50252

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2017
Third Semester
Civil Engineering
CE 6301 – ENGINEERING GEOLOGY
(Regulations 2013)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A (10×2=20 Marks)

1. Draw the internal structure of Earth.
2. Mention the characteristics of levees and floodplains.
3. Write Mohs' Scale of Hardness.
4. Mention the properties and uses of Gypsum.
5. Write the significance of determining RQD in Engineering Constructions.
6. What is contact metamorphism ?
7. Using a diagram, explain the terms "Dip" and "Strike".
8. Briefly explain the principle involved in electrical resistivity survey for sub-surface investigation.
9. What is Richter's scale ?
10. What are the elements of aerial photographs ?

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PART – B (5×13=65 Marks)

11. a) Explain the process of weathering in detail and comment on the role of weathering in influencing strength of rocks.

(OR)

b) Explain in detail on the geological actions of streams and rivers. Write a note on its significance in Civil Engineering constructions.

12. a) Write in detail on the following properties of minerals :

(i) Specific Gravity (ii) Hardness (iii) Fracture (iv) Crystalline Structure.

(OR)

b) Explain the structure and physical properties of clay minerals.

13. a) List the various engineering properties of rocks and give a detailed account of the field and laboratory tests to be conducted to determine such properties.

(OR)

b) Mention the mineralogical composition, texture, origin and uses of (i) Limestone (ii) Slate (iii) Marble.

14. a) Illustrate with neat sketches on folds, classification of folds, folding processes and their Civil Engineering significance.

(OR)

b) Give an account of the procedure for seismic refraction survey methods.

15. a) Enumerate with appropriate figures on the types, causes of landslides and their mitigation measures.

(OR)

b) Explain the important geological factors governing coastal process and give various coastal protection structures.

PART – C (1×15=15 Marks)

16. Lithological and structural aspects are the primary considerations for the selection of suitable sites and design of tunnels. Using appropriate case studies or examples, explain the validity of this statement.

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Reg. No. :

Question Paper Code : 91288

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019

Third Semester

Civil Engineering

CE 6301 – ENGINEERING GEOLOGY

(Common to PTCE 6301 : Engineering Geology for B.E. Part-Time for – Civil Engineering – Second Semester – Regulations 2014)

(Regulations 2013)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A **(10×2=20 Marks)**

1. Mention the thickness of Earth's crust.
2. How are rocks classified according to the scale of weathering ?
3. Write the important minerals from Feldspar group and their uses.
4. Write the name of the minerals belonging to clay group.
5. Write briefly about attrition test.
6. Compare the relative strengths of shale, sandstone, and quartzite.
7. What are Joints and Joint sets ?
8. Define the term 'Rock Quality Designation' (RQD).
9. Define remote sensing.
10. List at least four measures to prevent coastal erosion.

PART – B **(5×13=65 Marks)**

11. a) Write elaborately on Physical and chemical weathering of rocks. What is the significance of weathering of rocks in Civil Engineering ?

(OR)

b) Describe the geological work of sea and its engineering considerations.

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12. a) Using appropriate examples from the mineral kingdom, describe the physical properties of minerals.

(OR)

b) Describe in detail, the properties, composition and uses of Feldspar and Calcite.

13. a) List the various engineering properties of rocks. Describe the instrumentation required and the procedure for tests to determine these properties.

(OR)

b) Describe the mineralogical composition, texture, engineering properties and uses of Dolerite, Laterite, Sandstone and Limestone.

14. a) What are faults ? Explain in detail with sketches on (i) Normal faults (ii) Reverse faults (iii) Strike slip fault (iv) Oblique fault ?

(OR)

b) Explain in detail on electrical resistivity method and Wenner's configuration. Add a note on their application in civil engineering.

15. a) What are landslides ? How do they occur ? Describe the types of landslides with a sketch and enumerate the geological investigation required for mitigation of landslides.

(OR)

b) Explain in detail the various causes and effects of sea erosion, Add a detailed note on coastal protection measures.

PART – C

(1×15=15 Marks)

16. a) Explain with a neat sketch on how Remote Sensing is done. Add a note on applications of Remote Sensing in civil Engineering.

(OR)

b) Explain on various geological factors to be considered for Dam construction.

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Question Paper Code : 80070

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Third and Fourth Semester

Civil Engineering

CE 8391 — CONSTRUCTION MATERIALS

(Common to B.E. Environmental Engineering)

(Regulation 2017)

10-5-19

AN

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Define "natural bed of stone".
2. Differentiate first and second class bricks.
3. Define "slaking of lime".
4. Mention the function of coarse aggregate in concrete.
5. List the various ingredients involved in Ready Mixed concrete.
6. Explain the purpose of curing.
7. Write the different types of Industrial timber.
8. Write the various constituents of enamel paint
9. Mention the application of fibre glass reinforced plastic.
10. Define "Composite Materials".

PART B — (5 × 13 = 65 marks)

11. (a) Explain the properties of a good brick.

Or

(b) Discuss the test procedure for two types of test on stones.

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12. (a) Explain field tests on river sand and cement.

Or

(b) Describe the various properties of coarse aggregate.

13. (a) Discuss the functions of cement, fine aggregate and water in the concrete.

Or

(b) Explain various means of compaction of concrete.

14. (a) Write the properties and uses of timber.

Or

(b) Explain the different types of paints.

15. (a) List few modern materials and explain the uses of any two materials.

Or

(b) Explain the properties and uses of glass.

PART C — (1 × 15 = 15 marks)

16. (a) Discuss the criteria to recommend stone as a good building material.

Or

(b) As a civil Engineer recommend various building Materials involved in the construction of a building with their criteria with respect to quality.

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Question Paper Code : 25048

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2018.

Third Semester

Civil Engineering

CE 8351 — SURVEYING

(Common to Environmental Engineering)

(Regulations 2017)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Write the Principles of surveying.
2. Convert the whole circle bearing to quadrantal bearing
   (a) 170° 12'
   (b) 211° 54'.
3. Write the advantages of measuring horizontal angle by repetition method.
4. What are the advantages of tacheometric surveying over other methods?
5. What do you understand by horizontal control? What are different methods of establishing a horizontal control?
6. What are different types of errors in surveying measurements? Give one example of each.
7. What are the various operations conducted in Hydrographic surveying?
8. Determine the local mean time at a longitude 60°E if the local hour angle of the mean sun is 5h30m20s.
9. Write about the maintenance of Total Station equipment.
10. Give the importance of Selective Availability in GPS.

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PART B — (5 × 13 = 65 marks)

11. (a) In a closed traverse ABCDE, the bearing of the line AB was measured as 150°30'. The included angles were measured as under, Angle A = 130°10'; Angle B = 89°45'; Angle C = 125°22'; Angle D = 135°34': Angle E = 59°9'. Calculate the bearings of all other lines. (13)

Or

(b) The following readings were observed successively with a levelling instrument The instrument was shifted after fifth and eleventh readings.

(i) 0.585
(ii) 1.010
(iii) 1.735
(iv) 3.295
(v) 3.775
(vi) 0.350
(vii) 1.300
(viii) 1.795
(ix) 2.575
(x) 3.375
(xi) 3.865
(xii) 1.735
(xiii) 0.635
(xiv) 1.605 m.

Draw up a page of level book and determine the R.L of various point on which the first reading was taken is 136.440. Use Rise and Fall method. (13)

12. (a) A tacheometer is used to obtain the difference of levels between two points A and B. The instrument is set up at another station C, and the following observations were taken

Staff at	Vertical Angle	Stadia readings
A	–6°30'	3.500, 2.815, 2.130
B	–8°30'	1.870, 0.990, 0.110

If the R.L. of A is 100.000, determine the R.L. of B. Also determine the horizontal distance of A from C. Take k = 50.0 and C = 0.50. (13)

Or

(b) What is contouring? Describe about direct contouring and various methods of indirect Contouring. (13)

13. (a) Calculate the heights and distances by trigonometrical levelling. (13)

Or

(b) Find the most propable values of the following angles closing the horizon at a station by methods of correlate (13)

P	=	45°23'37"	weight 1
Q	=	75°37'15"	weight 2
R	=	125°21'21"	weight 3
S	=	113°37'59"	weight 3

14. (a) What do you understand by hydrographic surveying? Explain about the various operations conducted in hydrographic surveying. (13)

Or

(b) Determine the azimuth and altitude of a star from the following data

Declination of star	=	10°40'S
Hour angle of star	=	325°
Latitude of the observer	=	48°N

(13)

15. (a) Draw a neat sketch and explain the working principle of a Microwave Total Station equipment. (13)

Or

(b) Explain about the concept and segments of GPS with neat sketch. (13)

PART C — (1 × 15 = 15 marks)

16. Elaborate the procedure to be followed to conduct the survey of a village using modern survey equipments. (15)

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Question Paper Code : 90110

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019

Third/Fourth Semester

Civil Engineering

CE 8391 : CONSTRUCTION MATERIALS

(Common to Environmental Engineering)

(Regulations 2017)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A **(10×2=20 Marks)**

1. Write the any four characteristics of good building stones.
2. What is the meaning of preservation of stones.
3. List the uses of lime.
4. Mention the properties of cement.
5. Reproduce the methods of proportioning concrete.
6. Write the different types of concrete mixers.
7. Name the characteristics of good timber.
8. What are the types of varnishes.
9. Name the constituents of glass.
10. What are composite materials.

PART – B **(5×13=65 Marks)**

11. a) Describe the various tests to determine the suitability of a building stone.

(OR)

b) Describe any four tests to find the suitability of brick for the construction work.

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12. a) Describe the following with neat sketches.

i) Consistency test.

ii) Soundness test.

(OR)

b) Reproduce the following test procedure on coarse aggregate.

i) Crushing strength test.

ii) Impact strength test.

13. a) i) List the advantages of compaction by vibration.

ii) Labels the methods of curing concrete.

(OR)

b) List the advantages of ready mix concrete and mention its types.

14. a) Describe the following wood based products with neat sketch.

i) Veneers

ii) Plywood

(OR)

b) Recall the physical and mechanical properties of (i) Aluminum (ii) Copper and (iii) Lead.

15. a) i) List the types of glass used in construction.

ii) Mention the properties and applications of Fiber glass reinforced plastic.

(OR)

b) Explain the types of ceramics and their applications.

PART – C **(1×15=15 Marks)**

16. a) Explain the functions and application of geo textiles in construction.

(OR)

b) i) List the defects in painting.

ii) Explain the forms of bitumen.

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Reg. No. | | | | | | | | | | | | |

Question Paper Code : 57152

24/05/2016 FN

B.E/B.Tech. DEGREE EXAMINATION, MAY/JUNE 2016

Fourth Semester

Civil Engineering

CE 6401T – CONSTRUCTION MATERIALS

(Common to Environmental Engineering)

(Regulations 2013)

Time : Three Hours **Maximum : 100 Marks**

Answer ALL questions.

PART – A (10 × 2 = 20 Marks)

1. Write down the characteristics of good stone.
2. List the different types of refractory bricks.
3. What do you understand by Transition Zone ?
4. What is bulking of sand ? Why is it important ?
5. Distinguish between HSC and HPC.
6. Why do we need compaction of concrete ?
7. Give the application of Aluminium composites.
8. Enumerate the various market forms of steel.
9. What is a sealant ? State its applications.
10. Differentiate between geo membranes and geo textile.

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PART – B (5 × 16 = 80 Marks)

11. (a) Explain the various types of stones which are used for building works and give in brief the specifications for a good building stone. **(16)**

OR

(b) With sketches explain in detail about the manufacturing process of conventional bricks.

12. (a) (i) Add a note on the process of hydration of cement. **(8)**

(ii) How do you perform the soundness test of cement ? Why it is important ? **(8)**

OR

(b) What are tests to be conducted for conventional course aggregates ? Explain any four tests in detail. **(16)**

13. (a) With a flow chart explain the step by step process of manufacturing of concrete. **(16)**

OR

(b) Write down the complete procedure of concrete mix design as per BIS. Assume necessary data.

14. (a) (i) Discuss the possible sources of defects in timber. **(8)**

(ii) Briefly explain the following terms :

(1) Veneer **(4)**

(2) Thermacole **(4)**

OR

(b) Explain in detail about :

(i) Preparation of paint **(8)**

(ii) Distemper **(4)**

(iii) Bitumen **(4)**

15. (a) (i) Add a note on types of Glass and its properties. **(8)**

(ii) Discuss about the manufacture and properties of various clay products. **(8)**

OR

(b) Explain various applications of

(i) Geomembranes **(8)**

(ii) Geo textiles **(8)**

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Reg. No. : | | | | | | | | | | | | |

Question Paper Code : 71553

23/05/20 VJ

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Fourth Semester

Civil Engineering

CE 6401 – CONSTRUCTION MATERIALS

(Common to Environmental Engineering)

(Regulations 2013)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Write any four tests on stones.
2. Define light weight clay Bricks.
3. What do you understand by 53 grade cement?
4. What do you mean by graded aggregate?
5. Distinguish between bleeding and segregation.
6. List out the methods of transportation of concrete.
7. What are the Advantages of using veneer?
8. Name any four merits of aluminum in construction.
9. What are the forms of sealants?
10. Define the term Refractories.

PART B — (5 × 13 = 65 marks)

11. (a) Explain the various types of common building stones and their uses. (13)

Or

(b) Describe various tests to be conducted for testing of conventional bricks. (13)

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12. (a) How do you perform the tensile strength of cement? Also explain the effect of fineness of cement. (13)

Or

(b) Enumerate the various methods for testing of traditional coarse aggregates. Discuss any two of them in brief. (13)

13. (a) Explain any two tests for testing of conventional fresh concrete. (13)

Or

(b) With sketches explain any two tests for testing of workability for self-compacting concrete. (13)

14. (a) (i) Describe the characteristics of good timber.

(ii) Explain in detail about:

(1) Plywood

(2) Thermacole.

Or

(b) Explain in detail about:

(i) Paint. (5)

(ii) Distemper (4)

(iii) Bitumen. (4)

15. (a) (i) Discuss the various forms of commercial glass. (7)

(ii) Write an explanatory note on Mechanical properties of ceramics. (6)

Or

(b) Brief explains the following terms:

(i) Fibre glass reinforced plastic (7)

(ii) Composite materials (6)

PART C — (1 × 15 = 15 marks)

16. (a) Describe the various applications of geosynthetic in civil engineering construction works. (15)

Or

(b) Explain any three types of special concrete and their applications. (15)

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Question Paper Code : 40780

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2018

Fourth Semester

Civil Engineering

CE 6401 – CONSTRUCTION MATERIALS

(Regulations 2013)

(Common to : Environmental Engineering)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A (10×2=20 Marks)

1. Define efflorescence in bricks. How can it be removed ?
2. What is the significance of light weight concrete blocks ?
3. What is meant by hydration of cement ? State its importance.
4. Write down the chemical and physical characteristics of flyash.
5. Mention the properties of fresh concrete.
6. Give the relationship between slump value and degree of workability.
7. List out the defects in timber.
8. State the important consideration to be made in choosing paints.
9. Write down the uses Fibre glass reinforced plastic.
10. What is Geo Membrane ? List out the uses of Geo membrane.

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PART – B (5×13=65 Marks)

11. a) Describe the characteristics to be considered for selection of stones for various civil engineering works.

(OR)

b) List out the types of special bricks. Briefly explain any four of them.

12. a) What are the various ingredients required for manufacturing cement ? State their functions.

(OR)

b) Compare and contrast the advantages and disadvantages of using lime and cement in engineering works.

13. a) Explain the test on workability of concrete.

(OR)

b) Write short notes on :

i) High Strength Concrete. (6)

ii) Self Compacting Concrete. (7)

14. a) List out and explain the various methods of seasoning of timber.

(OR)

b) Investigate the various applications of aluminium.

15. a) Describe the process of manufacturing of Glass ? What are the uses of glass in construction industry ?

(OR)

b) Discuss the uses of Clay Products.

PART – C (1×15=15 Marks)

16. a) Explain the application of geotextiles for earth reinforcement.

(OR)

b) Explain the application of Bitumen in road construction.

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Question Paper Code : 52758

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Fourth Semester

30-4-19 FN

Civil Engineering

CE 6401 — CONSTRUCTION MATERIALS

(Common to: Environmental Engineering)

(Regulation 2013)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. List out the raw materials used for the manufacture of Fire bricks.
2. Write the classification of concrete blocks.
3. What do you mean by hydration of cement?
4. What is meant by bulking of sand?
5. Distinguish between nominal mix and design mix.
6. What do you understand by consolidation of concrete?
7. State any four characteristics of good timber.
8. Write any four tests on bitumen.
9. What are the forms of sealants?
10. Define the term composite materials.

PART B — (5 × 13 = 65 marks)

11. (a) Describe the various Qualities of a good building stone. (13)

Or

(b) Discuss the reasons for the causes of defects in bricks. (13)

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12. (a) Explain with a flow chart the step by step dry process of manufacturing of cement. (13)

Or

(b) Enumerate the various methods for physical testing of coarse aggregate. Discuss any two tests in detail. (5 + 8)

13. (a) Explain in detail about the manufacture process of concrete. (13)

Or

(b) Explain the fresh and hardened properties of concrete. (13)

14. (a) Explain the various types of heat treatment of steel and their purpose. (8 + 5)

Or

(b) (i) Enumerate the advantages of using aluminium as building material. (5)

(ii) What are Varnishes? Describe the various types of varnishes. (2 + 6)

15. (a) (i) Describe the various classification of glass. (6)

(ii) Describe the various applications of clay products in construction works. (7)

Or

(b) Explains the following terms:

(i) Refractories

(ii) laminar Composite.

PART C — (1 × 15 = 15 marks)

16. (a) Discuss in detail about the factors affecting the durability of concrete and state protective measures established for durable concrete. (8 + 7)

Or

(b) Describe the various classifications of geosynthetics and also explain various applications of geosynthetic in construction works.

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Question Paper Code : 80198

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2016.

Fourth Semester

Civil Engineering

CE 6401 — CONSTRUCTION MATERIALS

(Common to Environmental Engineering)

(Regulations 2013)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. What is meant by dressing of stones?
2. Write down the uses of bricks.
3. Define the term slaked lime.
4. What are the grades of cement?
5. Define RMC.
6. What are the factors to be considered in mix proportioning?
7. State any four characteristic of good sealant.
8. Write down the functions of solvent.
9. What are the necessity of bituminous paint?
10. Define geomembranes.

PART B — (5 × 16 = 80 marks)

11. (a) (i) Briefly discuss about the various tests conducted on stone. (12)

(ii) Why stone is called as a building material? (4)

Or

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(b) (i) Write a short note on refractory bricks. (12)

(ii) List down the application of lightweight concrete blocks. (4)

12. (a) (i) Explain the procedure for consistency as per IS specification. (8)

(ii) What is meant by hydration of cement? (8)

Or

(b) (i) Write down the importance of industrial by products. (8)

(ii) Explain the shape test for aggregate with the neat sketches. (8)

13. (a) With neat sketches explain the manufacturing process of concrete. (16)

Or

(b) Write in detail : (4 + 8 + 4)

(i) Modulus of rupture

(ii) Self compacting concrete

(iii) High performance concrete.

14. (a) Explain in detail about the principal process involved in heat treatment of steel. (16)

Or

(b) Write a short note on : (4 + 4 + 4 + 4)

(i) Forms of aluminium

(ii) Varnish

(iii) Thermocole

(iv) Classification of timber.

15. (a) (i) Explain in detail about fibre glass reinforced plastic. (8)

(ii) Explain in detail about ceramics. (8)

Or

(b) Discuss the various applications of

(i) fibre textiles

(ii) geotextiles.

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31/10/17 AN

Reg. No. :

Question Paper Code : 50262

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2017

Fourth Semester

Civil Engineering

CE 6401 – CONSTRUCTION MATERIALS

(Common to : Environmental Engineering)

(Regulations 2013)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A (10×2=20 Marks)

1. What are the uses of stone in construction ?
2. List out the types of bricks for special use.
3. Write the grade of cement.
4. What do you understand by bulking of sand ?
5. State the functions of coarse aggregate in a concrete.
6. Write the formula for flow percent.
7. Enumerate the various market forms of timber.
8. List out the various test on bitumen.
9. What is meant by sealant ?
10. Write the properties of refractories.

PART – B (5×13=65 Marks)

11. a) Describe various characteristics of good building stones. (13)

(OR)

b) Explain in detail about the manufacture process of conventional bricks with neat sketches. (13)

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12. a) What are tests to be conducted for cement ? Explain any two tests in detail. **(13)**

(OR)

b) Describe various methods for testing of conventional coarse aggregate. Discuss any two tests in detail. **(13)**

13. a) Explain in detail about manufacturing process of concrete. **(13)**

(OR)

b) Explain the procedure of compressive and split tensile strength of hardened concrete with neat sketch. **(13)**

14. a) Discuss the various types of heat treatment of steel and their purpose. **(13)**

(OR)

b) i) Discuss the reasons for the causes of defects in painting work. **(8)**

ii) Write a short note on varnishes. **(5)**

15. a) i) Explain the classification of glass. **(6)**

ii) Discuss the various factors you considered for choosing floor tiles. **(7)**

(OR)

b) Explain the following :

i) Properties of plastics **7**

ii) Composite materials. **6**

PART – C **(1×15=15 Marks)**

16. a) Describe the various applications of geotextiles in geotechnical engineering works. And also explain the properties of geotextiles. **(15)**

(OR)

b) Explain the following :

i) Light weight concrete

ii) Fibre reinforced concrete

iii) Self-compacting concrete. **(3×5=15 Marks)**

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Question Paper Code : 20259

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2018.

Fourth Semester

Civil Engineering

CE 6401 – CONSTRUCTION MATERIALS

(Common to : Environmental Engineering)

(Regulations 2013)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. What are the causes for deterioration of stones?
2. Reason out for efflorescence in brick?
3. Why hydration of cement takes place?
4. List any four industrial by-products.
5. Infer the factors affecting workability.
6. Define RMC.
7. Identify the various grade's of Bitumen.
8. List out the properties of mild steel.
9. What are the major constituents of glass?
10. Where geotextiles are used?

PART B — (5 × 13 = 65 marks)

11. (a) Discuss in details the properties of good brick.

Or

(b) Discuss in detail about various characteristics of good building stones.

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12. (a) Briefly discuss test procedures to determine any three physical properties on cement with suitable sketch.

Or

(b) Explain in detail about wet process of manufacturing of cement.

13. (a) Briefly discuss about the manufacturing process of concrete.

Or

(b) Explain the various test on hardened concrete with neat sketch.

14. (a) Categorize and explain the various defects in timber in detail with neat sketches.

Or

(b) Summarize the uses and properties of distemper and aluminium.

15. (a) Explain about geomembranes for earth reinforcement.

Or

(b) Outline the role of laminar composites in building construction.

PART C — (1 × 15 = 15 marks)

16. (a) Explain in detail about the applications of various types of industrial timbers in building construction.

Or

(b) Relate the application of light weight concrete blocks with first class bricks.

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20/11/19/AN

Reg. No. :

Question Paper Code : 91293

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019
Fourth Semester
Civil Engineering
CE 6401 – CONSTRUCTION MATERIALS
(Common to Environmental Engineering)
(Regulations 2013)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A **(10×2=20 Marks)**

1. State any four characteristics of good building stone.
2. What do you mean by lightweight concrete blocks ?
3. What are the normal ingredients of cement ?
4. Define Elongation index.
5. Define RMC.
6. What are the factors to be considered in mix proportioning ?
7. Give the application of Aluminium composites.
8. Enumerate the various market forms of steel.
9. What are the forms of sealants ?
10. How glass is manufactured ?

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PART – B (5×13=65 Marks)

11. a) i) Briefly discuss about the various tests conducted on stone.

ii) Why stone is called as a building material ?

(OR)

b) i) Write a short note on refractory bricks.

ii) List down the application of lightweight concrete blocks.

12. a) How do you assess the tensile strength of cement ? Also explain the effect of fineness of cement.

(OR)

b) Enumerate the various methods for testing of traditional coarse aggregates. Discuss any two of them in brief.

13. a) With sketches explain in detail about the different flow properties of concrete.

(OR)

b) With sketches explain the various specifications for self compacting concrete.

14. a) i) Enumerate the advantages of aluminium in construction. (5)

ii) Explain the defects in timber. (8)

(OR)

b) Write short notes on :

i) Thermocole. (4)

ii) Steel (3)

iii) Distemper (3)

iv) Varnish (3)

15. a) i) Add a note on types of Glass and its properties.

ii) Discuss about the manufacture and properties of various clay products.

(OR)

b) Explain various applications of :

i) Geomembranes. (7)

ii) Geo textiles. (6)

PART – C (1×15=15 Marks)

16. a) List and explain properties of hardened concrete along with tests required.

(OR)

b) Explain the various types of sealants used in construction industry and their suitability.

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14/05/19 AN

Reg. No. : | | | | | | | | | | | | |

Question Paper Code : 80068

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Third Semester

Civil Engineering

CE 8351 — SURVEYING

(Common to Environmental Engineering)

(Regulation 2017)

Time : Three hours | Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Define an "agonic and isogonic lines".
2. Find the combined correction for curvature and refraction for a distance of a
   (a) 3400 m
   (b) 1 .29 km.
3. What is the basic principle followed in stadia method?
4. What are the factors on which the choice of contour interval depends?
5. Differentiate between Laplace station from satellite station.
6. Find the most probable value of the angle A from the following observation equations
   A = 40° 20' 12"
   2A = 80° 40' 20"
   6A = 40° 20' 12"
7. Name the factors to be considered in the selection of a discharge site.
8. What is the relation between the Right ascension and Hour angle?
9. Bring out the temporary adjustments of a total station.
10. List the advantages of GPS surveying.

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PART B — (5 × 13 = 65 marks)

11. (a) (i) A chain line PQ intersects a pond. Two points A and B are taken on the chain line on opposite sides of the pond. A line AC, 250 m long, is set out on the left of AB and other line AD, 300 m long is set out on the right of AB. Points C, B and D are in the same straight line. CB and BD are 100 m and 150 m long respectively. Calculate the length of AB. (7)

(ii) A traverse ABCDA is made in the form of a square taking in clockwise order. If the bearing of AB is 120° 30', find the bearing of the other sides. (6)

Or

(b) The following consecutive readings were taken with a levelling instrument at intervals of 20 m. The readings are 2.375, 1.730, 0.615, 3.450, 2.835, 2.070, 1.835, 0.985, 0.435, 1.630 2.255, and 3.630 m. The instrument was shifted after fourth and eighth readings. The last reading was taken on a BM of RL 110.200 m. Find the RL of all the points. (13)

12. (a) The top (Q) of a chimney was sighted from two stations P and R at very different levels, the stations P and R being in line with the top of the chimney. The angle of elevation from P to the top of the chimney was 38° 21' and that from R to the top of the chimney was 21° 18'. The angle of the elevation from R to a vane 2m above the foot of the staff held at P was 15° 11'. The heights of instrument at P and R were 1 .87m and 1.64m respectively. The horizontal distance between P and R was 127 m and the reduced level of R was 112.78 m. Find the RL of the top of the chimney and the horizontal distance from P to the chimney. (13)

Or

(b) The following observations were made using a tacheometer fitted with an anallactic lens.

Instrument station	Height of instrumentation	Staff station	WCB	Vertical angle	Stadia Hair readings	Remarks
O	1.550	A	30°30'	4° 30'	1.155, 1.755, 2.355	RL of O = 150. 000 m
	1.550	B	75° 30'	10° 15'	1.250, 2.000, 2.750	

Calculate the distance AB, RL of A and B. Find also the gradient of line AB. (13)

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13. (a) A satellite station S is 6.5 m from the main station A and the following observations were taken A = 0° 0'; B = 102°48' ; C = 256°12'; D = 324°6'. The length AB, AC and AD were computed to be 1895 m, 2277 m, 2522 m respectively. Determine the direction of the line AB, AC and AD. (13)

Or

(b) The following angles were measured at a station O so as to close the horizon:

AOB	= 83° 42' 28.75"	weight 3
BOC	= 102° 15' 43.26"	weight 2
COD	= 94° 38' 27.22"	weight 4
DOA	= 79° 23' 23.77"	weight 2

Adjust the angle by method of correlates. (13)

14. (a) Explain the different coordinates systems by which the position of heavenly body can be specified. (13)

Or

(b) Explain various sounding methods in detail. (13)

15. (a) (i) Explain the working principle of a total station. (8)

(ii) Discuss the different sources of errors in a total station. (5)

Or

(b) Explain in detail about the different segments of GPS. (13)

PART C — (1 × 15 = 15 marks)

16. (a) Discuss the field procedures involved in preparing a map of a proposed hospital building. (15)

Or

(b) With a neat sketch, explain the types of surveying for the construction of highway include all possible measurements required.

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Question Paper Code : 90108

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019

Third Semester

Civil Engineering

CE 8351 : SURVEYING

(Common to Environmental Engineering)

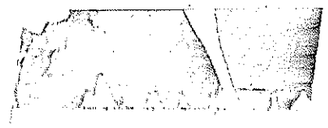
(Regulations 2017)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A **(10×2=20 Marks)**

1. Differentiate between true bearing and magnetic bearing.
2. What is meant by balancing of sights ?
3. What is the need for providing anallatic lens ?
4. Define contour interval.
5. What do you mean by reduction to centre ?
6. Find the most probable value and the probable error of the area of a circle whose radius is 15.40 ± 0.02 m.
7. What is celestial sphere ?
8. List the different solutions for a three-point problem.
9. What is a total station ?
10. What is the need for anti-spoofing in GPS ?



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PART – B (5×13=65 Marks)

11. a) Describe the different equipment required for ranging and chaining and explain the different methods of ranging.

(OR)

b) Explain the temporary adjustments of a level. How is the reduction of levels and booking of staff readings done using the rise and fall system ?

12. a) A tacheometer is set up at an intermediate point on a traverse course AB and the following observations are taken on a staff held vertically. The instrument is fitted with an anallatic lens and the multiplying constant is 100. The reduced level of A being given as 350.75 m, calculate the length of AB and the reduced level B.

Staff station	Bearing	Vertical angle	Intercept	Axial hair reading
A	40°35'	– 4°24'	2.172	1.962
B	220°35'	– 5°12'	1.986	1.866

(OR)

b) To determine the elevation of the top of the aerial pole, the following observations were made :

Instrument Station	Reading on BM	Angle of elevation	Remarks
A	1.377	11°53'	RL of BM =
B	1.263	8°5'	30.150 m

Station A and B and the top of the aerial pole are in the same vertical plane. Find the elevation of the top of the aerial pole, if the distance between A and B was 30 m.

13. a) Find the most probable values of the following angles closing the horizontal at a station.

P = 45°23'37" weight = 1

Q = 75°37'15" weight = 2

R = 125°21'21" weight = 3

S = 113°37'59" weight = 3

(OR)

b) In measuring angles at a triangulation station C, it was found necessary to set the transit over another station P south west of C and 3 m from C, so that the angle APB is approximately bisected by the line PC. The angles APC and CPB were found to be 28°20'35" and 31°26'45" respectively. The side AB was computed to be 975 m in the adjacent triangle, and when the station C was observed, the mean values of the angles CAB and CBA were recorded as 61°30'25" and 58°34'20" respectively. Determine the angle ABC.

14. a) What is meant by soundings ? Describe briefly any four methods of locating soundings.

(OR)

b) Describe the different types of celestial coordinate systems.

15. a) With neat sketches explain the working of a modern total station.

(OR)

b) Explain the various components of a GPS and its working principle.

PART – C (1×15=15 Marks)

16. a) What are the sources of error in surveying ? Explain the different precautions and correction procedures that can be adopted to eliminate the errors.

(OR)

b) List the characteristics of contour lines and uses of contouring. Also explain the different methods of locating and interpolating contours.

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07/05/19

Reg. No. :

Question Paper Code : 80064

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Third Semester

Civil Engineering

CE 8302 – FLUID MECHANICS

(Regulation 2017)

Time : Three hours Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Differentiate solid and fluid.
2. Define buoyancy.
3. Describe stream function.
4. Recall the application of Orifice-meter.
5. Describe dimensional homogeneity.
6. Describe distorted models
7. Recall the types of pipe flow based on viscosity.
8. Define major and minor losses.
9. Describe Boundary layer.
10. Define drag force.

PART B — (5 × 13 = 65 marks)

11. (a) (i) Describe the following properties of the fluid with the values of water at standard temperature and pressure: (1) Mass density, (2) Specific weight, (3) Specific gravity and (4) Viscosity. (7)

(ii) Explain surface tension and capillarity and derive an expression for capillarity. (6)

Or

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(b) (i) Compute the pressure of water in the pipe, if U tube mercury manometer is connected to a pipe line conveying water at 0.7m below the centre of pipe and the other leg (right leg) is open to atmosphere. The level of mercury in the right leg is 0.2m below the centre of pipe and the space above mercury in the right leg contains oil of specific gravity 0.9 to a height of 0.3m. (6)

(ii) Explain centre of pressure and total pressure, also derive an expression for it. (7)

12. (a) Compute the form of velocity potential if exists with proof and also find stream function in a two dimensional incompressible flow if the fluid velocity components are given by $u=x-4y$ and $v=-y-4x$.

Or

(b) Calculate the discharge of the oil and the pressure difference between the entrance section and throat section of a venturi-meter of size 30 cm × 15 cm fixed in a vertical pipe line carrying oil of specific gravity 0.9 flowing upwards. The difference in elevation of the throat section and entrance section of the venturimeter is 30 cm. The differential U tube mercury manometer shows a deflection of 25 cm. Take the co-efficient of meter as 0.98.

13. (a) Explain the procedure for dimensional analysis by Buckingham's law method.

Or

(b) Compute the velocity and rate of flow in the model, if a pipe of diameter 1.2 m is required to transport an oil of specific gravity 0.9 and viscosity of 0.03 poise flowing at the rate of 3000 l/s. Tests were conducted on a 15cm diameter pipe using water at 20°C. Find the velocity and rate of flow in the model. Take Viscosity of water at 20°C=0.01 poise.

14. (a) Derive Hagen — Poiseuille's equation for viscous flow through a circular pipe.

Or

(b) Determine (i) Reynolds number of flow, (ii) Centre line velocity, (iii) Wall shear stress and (iv) Power required to maintain the flow, for an oil of viscosity 1 poise and specific gravity 0.8 is flowing through 50 mm diameter pipe of length 500 m at the rate of 1.9 litres/sec.

15. (a) Calculate (i) The displacement thickness, (ii) The momentum thickness and (iii) The energy thickness in a boundary layer over the face of a high spillway for which the velocity distribution is $u/U=(y/\delta)$.

Or

(b) Describe boundary layer and the methods of preventing the separation of boundary layer.

PART C — (1 × 15 = 15 marks)

16. (a) Calculate the increase in discharge by neglecting minor losses, if a pipe line of 50 cm diameter of 1.5 km long is laid parallel to the second half of the existing line of 60 cm diameter. Take frictional coefficient of the pipe as 0.01 and head at inlet is 0.3 m.

Or

(b) Compute the forces required to drag a thin plate of surface area 0.75m² length 1 m, between the plane surfaces with a velocity of 0.3m/s, having 4 cm wide gap and the gap is filled with an oil of specific gravity 0.80 and dynamic viscosity 8.5 poise, if

(i) the thin plate is in the middle of the two plane surfaces

(ii) the thin plate is at a distance of 2.5 cm from one of the surfaces

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Reg. No. :

Question Paper Code : 90104

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019
Third Semester
Civil Engineering
CE 8302 – FLUID MECHANICS
(Regulations 2017)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions.

PART – A **(10×2=20 Marks)**

1. Write the unit and dimension for dynamic viscosity.
2. Write the unit and dimension for surface tension.
3. What are the advantages of orifice meter ?
4. A pitot tube was inserted in a river to measure the velocity of water in it. If the water rises in the tube above free surface of water is 300 mm, find the velocity of water (Cv = 0.98).
5. Illustrate the principle of dimensional homogeneity.
6. Write the fundamental dimensions and base units.
7. Illustrate the Moody's diagram.
8. What are major and minor losses of flow in pipes ?
9. Define the term boundary layer.
10. How do you classify boundary layer ?

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PART – B (5×13=65 Marks)

11. a) i) Explain the difference between the solid and fluid. (7)

ii) A liquid has a mass density of 1550 kg/m^3. Calculate its specific weight, specific gravity and specific volume. (6)

(OR)

b) i) Explain briefly about the Newtonian Fluids and Non Newtonian Fluids with neat sketch. (6)

ii) Derive an expression for centre of pressure and hydrostatic pressure force for a plane submerged vertically in static liquid. (7)

12. a) i) Explain in detail about the various types of fluid flow. (6)

ii) Derive three dimensional continuity equation in Cartesian co-ordinate system. (7)

(OR)

b) i) A pitot static tube placed in the centre of a 250 mm pipe line has one leg pointing up stream and other perpendicular to that. The mean velocity in the pipe is 75% of the central velocity. Find the discharge through the pipe if the pressure difference between the two legs in 80 mm of water and take $C_v = 0.99$. (5)

ii) A venturimeter is used for measurement of discharge water in horizontal pipeline. If the ratio of upstream pipe diameter to that of throat in 2:1, upstream diameter is 300 mm the difference in pressure between the throat and upstream is equal to 3 m head of water and loss of head through meter is one-eight of the throat velocity head, calculate the discharge in the pipe. (8)

13. a) Drag force (F) of an partially submerged body is a function of relative velocity (v), linear dimension (L), surface roughness (k), fluid density (ρ), viscosity (μ), and acceleration due to gravity (g). Using Buckingham pi theorem method of dimensional analysis obtain an expressions for the drag in terms of dimensionless number. (13)

(OR)

b) The lift force 'F' on a missile is a function of its length L, velocity V, Diameter D, Angle of attack α, density ρ, viscosity μ and speed of sound 'C' of the air. Find the functional relationship in dimensionless form. (13)

14. a) Discuss in detail about the test procedure, apparatus and observation of Reynold's experiment with neat sketch. (13)

(OR)

b) Derive the Hagen-Poiseuille's equation for laminar flow of fluid in straight and circular pipe with proper assumptions and neat sketch. (13)

15. a) Describe briefly about the general characteristics of boundary layer on a flat plate and illustrate the definition for boundary layer thickness, displacement thickness and momentum thickness with neat sketch. (13)

(OR)

b) Explain in detail about the boundary layer separation phenomena and methods of controlling boundary layer separation with neat sketch. (13)

PART – C (1×15=15 Marks)

16. a) Explain in detail about the construction details and working principle of venturimeter with neat sketch and also derive the equation to determine volumetric flow rate. (15)

(OR)

b) Using Buckingham's π Theorem, show that the velocity through a circular orifice is given by $V = \sqrt{2gH}\,\phi\left[\frac{D}{H}, \frac{\mu}{\rho VH}\right]$ where H is the head causing flow, D is the diameter of the orifice, μ is the coefficient of viscosity, ρ is mass density and g is the acceleration due to gravity. (15)

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PART C — (1 × 15 = 15 marks)

16. (a) A beam of rectangular cross section is to be cut from a circular log of diameter D. What should be ratio of the depth of the beam to its width to resist maximum bending moment?

Or

(b) Determine the slope at the supports and deflection under the load for the beam given in Q 16 b. Use conjugate beam method.

$I = 1 \times 10^8 \text{ mm}^4$. $E = 200 \text{ GN/m}^2$.

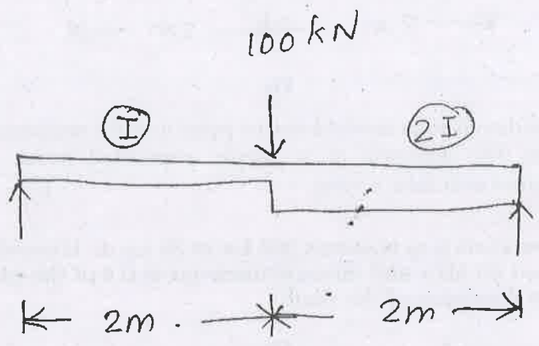


Figure – Q 16 (b)

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B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Third Semester

Civil Engineering

CE 8301 — STRENGTH OF MATERIALS — I

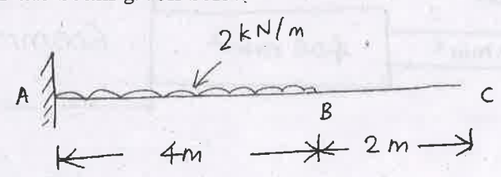
(Regulation 2017)

Time : Three hours Maximum : 100 marks

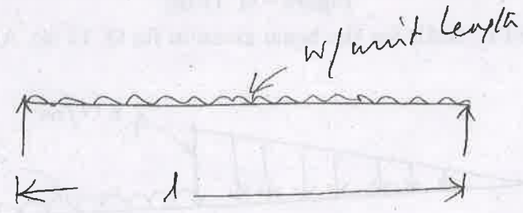
Answer ALL questions.

PART A — (10 × 2 = 20 marks)

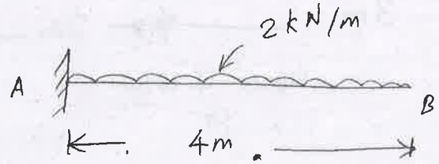
1. Define Poisson's ratio.
2. What is Mohr's Circle?
3. Draw SFD for the beam given below



4. Draw qualitative shear stress distribution for a T-section.
5. What is the maximum deflection for the beam given below



6. Draw Conjugate beam for the beam given below



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7. Define stiffness.

8. What is the bending stress induced in a close coiled helical spring subject to axial load?

9. What is redundant frame?

10. Write the advantages of method of section for calculating member forces in a truss.

PART B — (5 × 13 = 65 marks)

11. (a) (i) Derive a relation for change in length of a low hanging freely under its own weight. (6)

(ii) Derive a relation for change in length of a bar with uniformly varying diameter and subjected to an axial tensile load 'P' (7)

Or

(b) The rod given in fig Q. 11 (b) is held between rigid supports. Find the stress developed in each material when the temperature is raised by 55°C. Use

$E_s = 2 \times 10^5\ N/mm^2$, $\alpha_s = 1.2 \times 10^{-5}/^\circ c$

$E_c = 1.0 \times 10^5\ N/mm^2$, $\alpha_c = 1.75 \times 10^{-5}/^\circ c$

$E_A = 0.7 \times 10^5\ N/mm^2$, $\alpha_A = 2.2 \times 10^{-5}/^\circ c$.

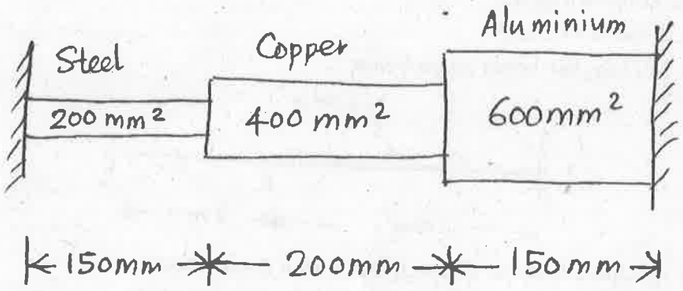


Figure – Q. 11 (b)

12. (a) Draw SFD, BMD for the beam given in fig Q. 12 (a). Also find Maximum BM

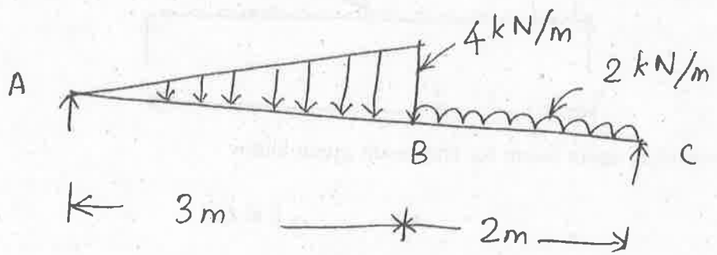


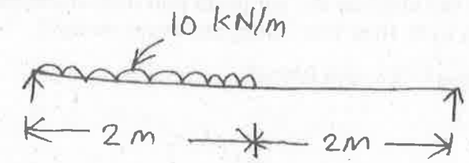
Figure – Q. 12 (a)

Or

(b) A T-beam of flange size 250× 30 *mm* (thickness), web 280 × 40 mm (thickness) is subjects to shear force of 20 kN. Sketch the shear stress distribution.

13. (a) Find the maximum deflection of the beam given in Q. 13 (a). It is of circular cross section with 200 mm as diameter.

Take E = 2 × 10^4 N/mm² use Macaulay's method.



Or

(b) Using moment area method derive equations for maximum deflection and slope at the supports of a simply supported beam carrying U.D.L. distributed over entire span.

14. (a) A hollow shaft is to transmit 200 kw at 80 r.p.m. If the shear stress is not to exceed 60 MPa and internal diameter is 0.6 of the external diameter, find the diameters of the shaft.

Or

(b) Derive the relations for deflection, stiffness of a close coiled helical spring subjected to axial load.

15. (a) Determine the forces in all the members of the truss given in Fig Q. 15 (a). Use method of joints

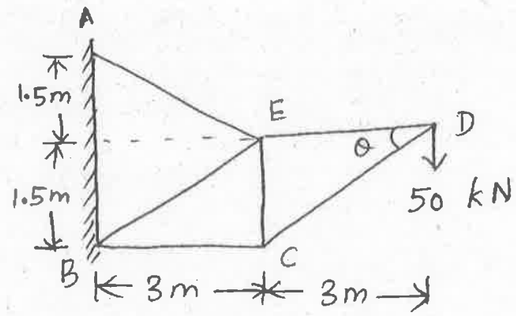


Figure – Q 15 (a)

Or

(b) Using method of sections solve the truss given in Fig Q. 15 (a).

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Reg. No. :

Question Paper Code : 90102

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019

Third Semester

Civil Engineering

CE 8301 – STRENGTH OF MATERIALS – I

(Regulations 2017)

Time : Three Hours | Maximum : 100 Marks

Answer ALL questions

PART – A **(10×2=20 Marks)**

1. Recall Poisson's Ratio.
2. Show that in a strained material subjected to two-dimensional stress, the sum of the normal components of stresses on any two mutually perpendicular planes is constant.
3. A beam of triangular section having base width 20 cm and height of 30 cm is subjected to a shear force of 3 kN. Calculate the value of maximum shear stress.
4. State the relationship between intensity of load, shear force and bending moment.
5. In what situation, conjugate beam method is preferred ?
6. A beam 5 m span of 200 mm width and 300 mm depth is simply supported, carries a uniformly distributed load of 9 kN/m. If $E = 10^4$ N/mm^2 find slope at support and maximum deflection.
7. List four assumptions made in determination of torsion of shafts.
8. Compare open coiled and closed coiled helical springs.
9. Outline determinate and indeterminate structures.
10. Define : Tension Coefficient.

PART – B **(5×13=65 Marks)**

(Restrict to a maximum of 2 sub-divisions)

11. a) A tie bar 300 mm long has enlarged ends of square section 60 mm × 60 mm. If the middle portion of the bar is also of square cross section, find the size and length of the middle portion, if the stress there is 140 MN/m^2. The bar is subjected to an axial pull of 87.5 kN and the total extension of the bar is 0.14 mm. Take E = 200 GN/m^2.

(OR)

b) A rectangular block of material is subjected to a tensile stress of 100 MN/m^2 on one plane and a tensile stress of 50 MN/m^2 at right angle, together with the shear stresses of 60 MN/m^2 on the same planes. Find

i) the magnitude of the principal stresses,

ii) the direction of the principal planes and

iii) the magnitude of the greatest shear stress.

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12. a) i) Enlist the assumptions made in the theory of pure bending. **(5)**

ii) Obtain the ratio of section modulii of a plane solid circular section of 100 mm to a hollow circular section of external diameter 100 mm and internal diameter 25 mm. **(8)**

(OR)

b) i) Classify beams depending upon the type of supports with suitable sketches. **(5)**

ii) A hollow circular bar having outside diameter twice the inside diameter is used as a beam. The bar is subjected to a bending moment of 40 kNm. If the allowable bending stress in the beam is limited to 100 MN/m², find the inside diameter of the bar. **(8)**

13. a) A simply supported beam 6 m long carries a concentrated moment of 150 kNm (clockwise) and concentrated load of 300 kN, both at 2 m from the right support. If the flexural rigidity of the beam is 8×10^4 kNm², find the maximum deflection and slope at the loaded point using MaCaulay's method.

(OR)

b) Using conjugate beam method, find the slopes and deflections at A, B, C and D in the beam shown in Fig. 13 (b). Given $E = 200 \times 10^6$ kN/m² and $I = 300 \times 10^{-4}$ m⁴.

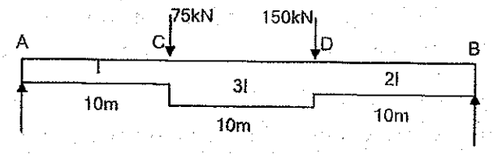


Fig. 13 (b)

14. a) A solid cylindrical shaft is to transmit 300 kW at 100 r.p.m. (i) If the shear stress is not to exceed 80 MN/m², find it diameter. (ii) What percentage saving in weight would be obtained if this shaft is replaced by a hollow one whose internal diameter equals 0.6 of the external diameter, the length, the material and maximum shear stress being the same ?

(OR)

b) A closely coiled helical spring is made out of 10 mm diameter steel rod. The coil consists of 10 complete turns with a mean diameter of 120 mm. The spring carries an axial pull of 200 N. Find the maximum shear stress induced in the section of the rod. If C = 80 GN/m², find the deflection in the spring, the stiffness and strain energy stored in the spring.

15. a) Analyse the pin jointed plane determinate truss shown in Fig. 15 (a) by the method of joints.

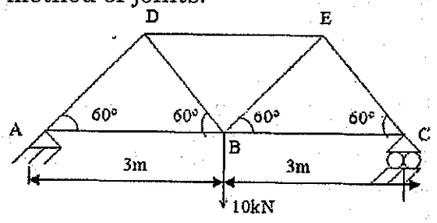


Fig. 15 (a)

(OR)

b) Determine the forces in the members of the space truss shown in the Fig. 15 (b) by tension coefficient method.

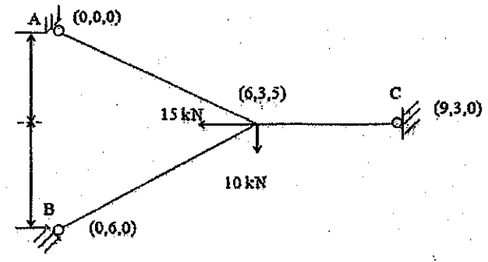


Fig. 15 (b)

PART – C **(1×15=15 Marks)**

16. a) A horizontal beam AB 8 m long is supported at A and C, 6 m from A. The beam supports a uniformly distributed load of 1.5 kN/m over its entire length and also concentrated loads of 3 kN and 1.5 kN at D and B respectively, D being 2 m from A. Plot shear force and bending moment diagrams. Determine the magnitude and position of maximum bending moment.

(OR)

b) A composite bar made up of aluminium and steel is held between two supports as shown in Fig. 16 (b). The bars are stress free at a temperature of 40°C. What will be the stresses in the two bars when the temperature is 20°C if (i) the supports are non-yielding, and (ii) the supports come nearer to each other by 0.1 mm. The change of temperature is uniform along the length of the bar. Take $E_s = 210$ GN/m², $E_a = 74$ GN/m², $\alpha_s = 11.7 \times 10^{-6}$ per °C and $\alpha_a = 23.4 \times 10^{-6}$ per °C.

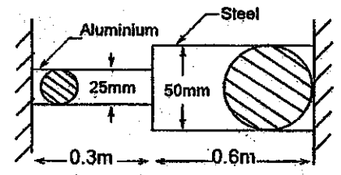


Fig. 16 (b)

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